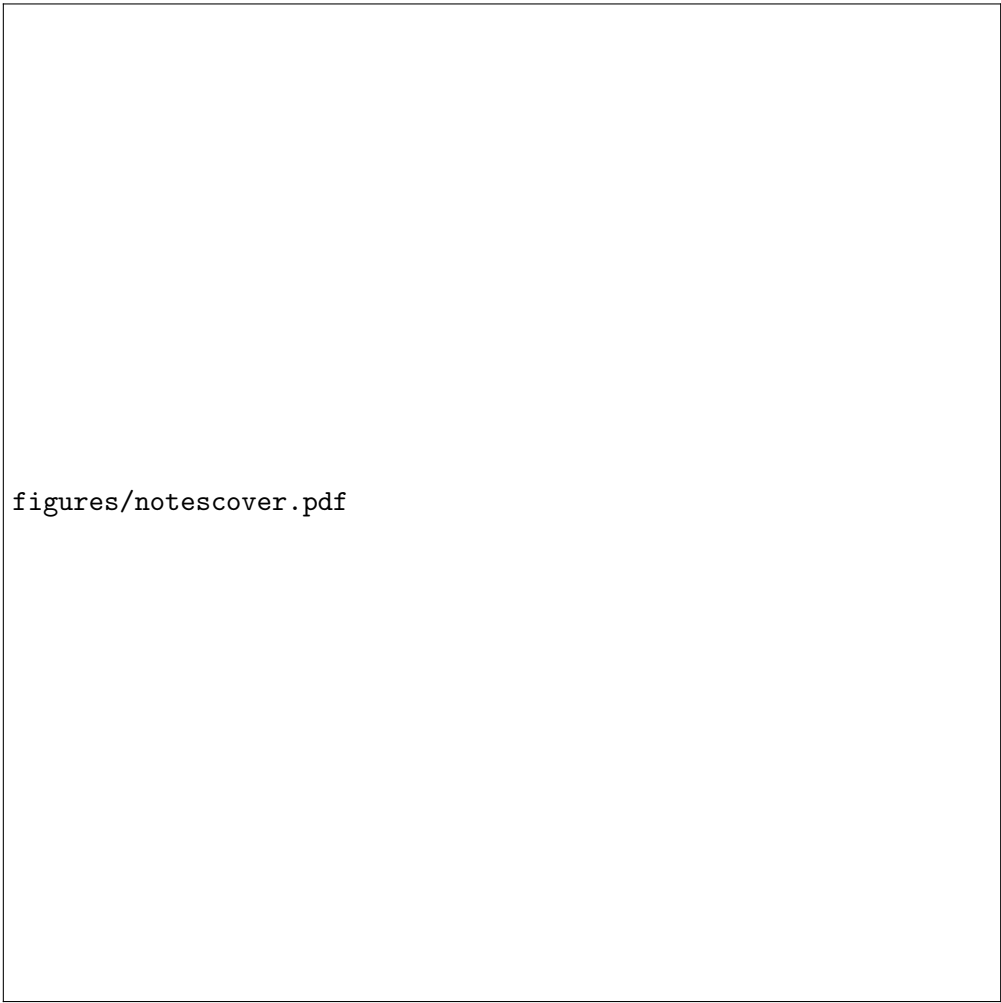

Microwave Cavity Warm Atoms Theory Notes

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figures/notescovers.pdf

Typeset by Aiden Diaz

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1 Atomic Structure

1.1 The Hydrogen Atom

We will begin our study of atomic physics by considering the simplest possible object: the hydrogen atom. The basic structure of a single electron and a nucleus can be well understood, and despite the simplicity, these results are broadly applicable to many forms of matter.

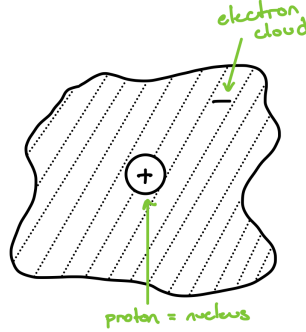


Figure 1: Simple proton and nucleus diagram for the hydrogen atom.

Consider the simple picture: an atom consists of a single proton and electron. We may ask the general question:

"What is the quantum state of the electron in the Coulomb potential?"

To start, we will look at the time-independent Schrödinger equation (TISE),

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(r) \right] \psi(\mathbf{r}) = E \psi(\mathbf{r}).$$

Here, $-\hbar^2/2m$ is the usual kinetic energy term, and $V(r)$ is the typical potential energy. In this case, it is the Coulomb potential assuming radial symmetry so that $V(\mathbf{r}) = V(r)$. For this analysis, we will work in the position representation, since we can easily represent $V(\mathbf{r})$ here.

Due to radial symmetry (we use the physics convention), we will work in spherical coordinates. Here, the Laplacian is of the form

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) - \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{1}{r^2} \hat{\mathbf{I}}^2.$$

We construct $\hat{\mathbf{I}}^2$ in such a way that it defines the angular momentum operator $\hbar \hat{\mathbf{I}}$, so that

$$\hat{\mathbf{I}}^2 = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) - \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2}.$$

The angular components of the gradient $\hat{\mathbf{I}}^2$ are implicitly related to angular momentum. That is, we can say that angular momentum is the generator of rotations.

Returning to the Schrödinger equation (SE), we consider the solutions

$$\begin{aligned} -\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{1}{r^2} \hat{\mathbf{I}}^2 \right] \psi(\mathbf{r}) &= [E - V(r)] \psi(\mathbf{r}) \\ \left\{ -\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + [V(r) - E] \right\} \psi(\mathbf{r}) &= -\frac{\hbar^2}{2mr^2} \hat{\mathbf{I}}^2 \psi(\mathbf{r}) \\ \left\{ \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{2mr^2}{\hbar^2} [E - V(r)] \right\} \psi(\mathbf{r}) &= \hat{\mathbf{I}}^2 \psi(\mathbf{r}). \end{aligned}$$

Note that the left-hand side (LHS) is of a functional form: it depends only on r , and not θ or ϕ . On the right-hand side (RHS), the term $\hat{\mathbf{I}}^2$ depends only on θ and ϕ . This suggests that we have a "separable" solution,

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$$

where $R(r)$ contains all radial dependence of the wave function, and $Y(\theta, \phi)$ contains all angular dependence of the wave function. Separating variables, we find

$$\begin{aligned} Y(\theta, \phi) \left\{ \frac{\partial}{\partial r} \left[r^2 \frac{\partial R(r)}{\partial r} \right] + \frac{2mr^2}{\hbar^2} [E - V(r)] R(r) \right\} &= R(r) \hat{\mathbf{I}}^2 Y(\theta, \phi) \\ \frac{1}{R(r)} \frac{\partial}{\partial r} \left[r^2 \frac{\partial R(r)}{\partial r} \right] + \frac{2mr^2}{\hbar^2} [E - V(r)] &= \frac{1}{Y(\theta, \phi)} \hat{\mathbf{I}}^2 Y(\theta, \phi). \end{aligned}$$

For a solution $\psi(r, \theta, \phi)$ to hold for all (r, θ, ϕ) , we must have a constant solution. We will call this constant A to get two equations:

$$\frac{\partial}{\partial r} \left[r^2 \frac{\partial R(r)}{\partial r} \right] + \frac{2mr^2}{\hbar^2} [E - V(r)] R(r) = AR(r). \quad (\text{radial equation})$$

$$\hat{\mathbf{I}}^2 Y(\theta, \phi) = AY(\theta, \phi). \quad (\text{angular equation})$$

Our new goal is to determine A and the solutions associated with it. To proceed, we take the angular equation and further separate variables:

$$\hat{\mathbf{I}}^2 Y(\theta, \phi) = \left[-\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) - \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] Y(\theta, \phi) = AY(\theta, \phi).$$

Supposing $Y(\theta, \phi) = \Theta(\theta)\Phi(\phi)$, we find

$$\begin{aligned} \Phi(\phi) \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left[\sin \theta \frac{\partial \Theta(\theta)}{\partial \theta} \right] + \Theta(\theta) \frac{1}{\sin^2 \theta} \frac{\partial^2 \Phi}{\partial \phi^2} &= -A \Theta(\theta) \Phi(\phi) \\ \Phi(\phi) \sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Theta}{\partial \theta} \right) + A \sin^2 \theta \Theta(\theta) \Phi(\phi) &= -\Phi(\phi) \frac{\partial^2 \Phi(\phi)}{\partial \phi^2} \\ \frac{1}{\Theta(\theta)} \sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Theta}{\partial \theta} \right) + A^2 \sin^2 \theta &= -\frac{1}{\Phi(\phi)} \frac{\partial^2 \Phi}{\partial \phi^2}. \end{aligned}$$

Now, this equation must be equal to a different constant. We shall call this constant m^2 , so that we find the first solution

$$-\frac{1}{\Phi(\phi)} \frac{\partial^2 \Phi}{\partial \phi^2} = m^2 \implies \frac{\partial^2 \Phi}{\partial \phi^2} = -m^2 \Phi(\phi) \implies \boxed{\Phi(\phi) = e^{im\phi}}.$$

This solution tells us that the azimuthal dependence of the final wavefunction is a phase factor (no ϕ dependence of amplitude).

The next step is to solve for the polar angle dependence, θ . Here, we use ladder operators to reach a solution. The Cartesian representation of $\hat{\mathbf{I}}^2$ is

$$\begin{aligned}\hat{l}_x &= i \left[\sin \phi \frac{\partial}{\partial \theta} + \frac{\cos \theta}{\sin \theta} \cos \phi \frac{\partial}{\partial \phi} \right] \\ \hat{l}_y &= i \left[-\cos \phi \frac{\partial}{\partial \theta} + \frac{\cos \theta}{\sin \theta} \sin \phi \frac{\partial}{\partial \phi} \right] \\ \hat{l}_z &= -i \frac{\partial}{\partial \phi}\end{aligned}$$

and the raising and lowering operators may be defined by

$$\hat{\mathbf{I}}_{\pm}^2 = \hat{l}_x \pm i \hat{l}_y = e^{\pm i \phi} \left[\frac{\partial}{\partial \theta} \pm i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \phi} \right].$$

We evaluate the action of these operators on our wave function, starting with l_z ,

$$\hat{l}_z Y(\theta, \phi) = \Theta(\theta) \left[-i \frac{\partial}{\partial \phi} \Phi(\phi) \right] = \Theta(\theta) \left[-i \frac{\partial}{\partial \phi} e^{im\phi} \right] = -i(im) \Theta(\theta) \Phi(\phi) = m Y(\theta, \phi).$$

Notice that m is the eigenvalue of the $\hat{l}_z$ operator, and so the states $Y(\theta, \phi)$ have angular momentum m along with z . As a sidenote, we may confirm this by taking the expectation value of $\hat{l}_z$,

$$\langle \hat{l}_z \rangle = \langle Y(\theta, \phi) | \hat{l}_z | \psi(\theta, \phi) \rangle = \langle Y(\theta, \phi) | m | \psi(\theta, \phi) \rangle = m \langle Y(\theta, \phi) | Y(\theta, \phi) \rangle = m.$$

In the same way, we determine the angular momentum along z , on states to which $\hat{l}_{\pm}$ have been applied,

$$\hat{l}_+ \hat{l}_z Y(\theta, \phi) \text{ or } \hat{l}_- \hat{l}_z Y(\theta, \phi).$$

We use the commutation relationship $[\hat{l}_z, \hat{l}_{\pm}] = \pm \hat{l}_{\pm}$ to determine that

$$\begin{aligned}\hat{l}_z \hat{l}_{\pm} &= \pm \hat{l}_{\pm} + \hat{l}_{\pm} \hat{l}_z \\ \hat{l}_z \hat{l}_{\pm} Y(\theta, \phi) &= \pm \hat{l}_{\pm} Y(\theta, \phi) + \hat{l}_{\pm} \hat{l}_z Y(\theta, \phi) \\ &= \pm \hat{l}_{\pm} Y(\theta, \phi) + l_{\pm} m Y(\theta, \phi) \\ &= (m \pm 1) \hat{l}_{\pm} Y(\theta, \phi) \\ \hat{l}_z [\hat{l}_{\pm} Y(\theta, \phi)] &= (m \pm 1) [\hat{l}_{\pm} Y(\theta, \phi)].\end{aligned}$$

This, of course, tells us that the state $[\hat{l}_{\pm} Y(\theta, \phi)]$ has angular momentum along z of $m \pm 1$, which justifies the names of the raising or lowering operators $\hat{l}_{\pm}$.

Another important property of the operators $\hat{l}_z$ and $\hat{l}_{\pm}$ is that they commute with $\hat{\mathbf{I}}^2$:

$$[\hat{\mathbf{I}}^2, \hat{l}_z] = 0, \quad [\hat{\mathbf{I}}^2, \hat{l}_{\pm}] = 0$$

which dictate that any eigenstate of $\hat{l}_z$ or $\hat{l}_{\pm}$ is also a (simultaneous) eigenstate of $\hat{\mathbf{I}}^2$. We can thus express that $\hat{\mathbf{I}}^2$ operator as

$$\hat{\mathbf{I}}^2 = \hat{l}_- \hat{l}_+ + \hat{\mathbf{I}}_z^2 + \hat{l}_z = \hat{l}_+ \hat{l}_- + \hat{\mathbf{I}}_z^2 - \hat{l}_z.$$

With this, we can make an assumption about the possible states in the system: for a set of states that have the eigenvalue of $\hat{\mathbf{I}}^2$, several values of the $\hat{l}_z$ eigenvalue are possible, i.e., the two expressions for $\hat{\mathbf{I}}^2$ shown above are composed of different terms that sum to the same value.

From this, we can argue that there are states of minimum and maximum m values. If this were the case, the raising and lowering operators could generate $\langle \hat{\mathbf{I}}_z^2 \rangle > \langle \hat{\mathbf{I}}^2 \rangle$. We will now show this. Consider an eigenvalue of $\hat{\mathbf{I}}^2$,

$$\hat{\mathbf{I}}^2 Y(\theta, \phi) = AY(\theta, \phi).$$

We will use the following results:

$$\hat{\mathbf{I}}^2 - \hat{\mathbf{I}}_z^2 = \hat{l}_+ \hat{l}_- - \hat{l}_z = \hat{l}_- \hat{l}_+ + \hat{l}_z = \frac{1}{2}(\hat{l}_+ \hat{l}_- + \hat{l}_- \hat{l}_+)$$

and,

$$\hat{l}_+ = \hat{l}_-^\dagger \implies \hat{\mathbf{I}}^2 - \hat{\mathbf{I}}_z^2 = \frac{1}{2}(\hat{l}_+ \hat{l}_+^\dagger + \hat{l}_+^\dagger \hat{l}_+) > 0.$$

Now, we will take a generic eigenstate of $\hat{\mathbf{I}}^2$ and $\hat{l}_z$, labelled by the two quantum numbers. The two eigenvalues of these operators should show the following:

$$\begin{aligned} \hat{\mathbf{I}}^2 |A, m\rangle &= A |A, m\rangle & \langle A, m | \hat{\mathbf{I}}^2 - \hat{\mathbf{I}}_z^2 | A, m \rangle &\geq 0 \\ \hat{\mathbf{I}}_z^2 |A, m\rangle &= m^2 |A, m\rangle & \implies & \\ & & A - m^2 &\geq 0 \\ & & m^2 &\leq A. \end{aligned}$$

Indeed, we should find that

$$-A \leq m \leq A$$

showing that $|m|$ is bounded by a maximum value. Now we might ask: what is this maximum value, say $m_{\max}$? If we apply the raising operator to the maximum value, it must "fail" and return zero. So,

$$\begin{aligned} \hat{l}_+ |A, m_{\max}\rangle &= 0 \\ \hat{l}_- \hat{l}_+ |A, m_{\max}\rangle &= 0 \\ (\hat{\mathbf{I}}^2 - \hat{\mathbf{I}}_z^2 - \hat{l}_z) |A, m_{\max}\rangle &= 0 \\ (A - m_{\max}^2 - m_{\max}) |A, m_{\max}\rangle &= 0. \end{aligned}$$

Thus, we must have

$$m_{\max}^2 + m_{\max} = A \implies m_{\max}(m_{\max} + 1) = A.$$

We repeat the same process for the "bottom" state, with $m = m_{\min}$, to which we should find that

$$m_{\min}(m_{\min} - 1) = A.$$

Note that this is the same A eigenvalue as found above. Since these states have the same eigenvalue of $\hat{\mathbf{I}}^2$, we should expect that

$$m_{\min}(m_{\min} - 1) = m_{\max}(m_{\max} + 1) \implies m_{\max} = -m_{\min}.$$

By convention, we give this maximum value the name " l ", and call it the eigenvalue of $\hat{\mathbf{I}}^2$. So,

$$A = l(l + 1) \implies \begin{cases} m_{\max} = l \\ m_{\min} = -l. \end{cases}$$

We saw earlier that the eigenstates of $\hat{\mathbf{l}}^2$ are eigenstates of $\hat{l}_\pm$ and $\hat{l}_z$, so the possible values of m for these eigenstates must vary by 1. Indeed,

$$m = \{-l, -l+1, \dots, l-1, l\}.$$

Connecting this all back to the solutions of the angular part of the hydrogen atom, we find that

$$\hat{\mathbf{l}}^2 Y(\theta, \phi) = l(l+1)Y(\theta, \phi)$$

such that we can also say about the radial part

$$\frac{\partial}{\partial r} \left[r^2 \frac{\partial R(r)}{\partial r} \right] + \frac{2mr^2}{\hbar^2} [E - V(r)] R(r) = l(l+1)R(r).$$

1.1.1 Angular Wavefunctions

To interpret the angular part of the wave function, we will observe specifically the solution $Y(\theta, \phi)$. Notice that each solution is defined by two eigenvalues, and thus two quantum numbers:

$$Y_{lm}(\theta, \phi) \rightarrow |lm\rangle$$

which is the quantum state of total angular momentum l and z -angular momentum, m . We know these quantum numbers can take discrete values only, which shows that angular momentum is quantized, and the quantization is along an particular axis (which we call here "z", by convention).

This quantization can be thought of in terms of the requirement of single-valuedness of the phase. We found previously that $\Phi(\phi) = e^{im\phi}$ which is indiscernible from $e^{im(\phi+2\pi)}$. However, the value of m must give the same Φ whether $\phi = \phi_0$ or $\phi = \phi_0 + 2\pi$, ϕ_0 being some arbitrary value. Thus, we require

$$e^{im2\pi} = 1 \implies 2\pi m = \{0, \pm 2\pi, \pm 4\pi, \dots\} = \pm 2\pi n \implies m = \pm n$$

where m, n are integers. Note also that we must have $e^{im2\pi} = +1$ for positive single-values wavefunctions (negative is problematic!).

Recalling that "velocity is the gradient of the phase", observe that m is a measure of how much the phase changes as the z -axis is circumnavigated, indicating the angular velocity about z .¹

Consider now the position-space representation of the wave function. For the $\Theta(\theta)$ solution, we can see that

$$\begin{aligned} \frac{1}{\Theta(\theta)} \sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Theta}{\partial \theta} \right) + A \sin^2 \theta &= -\frac{1}{\Phi(\phi)} \frac{\partial^2 \Phi(\phi)}{\partial \phi^2} \\ \sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Theta}{\partial \theta} \right) + [l(l+1) \sin^2 \theta - m^2] \Theta(\theta) &= 0. \end{aligned}$$

This differential equation has known solutions.² The solution is of the form

$$\Theta(\theta) = AP_l^m(\cos \theta)$$

¹This interpretation is simplified, but will get the job done.

²See Griffiths p. 126.

which are the "associated Legendre polynomials." In particular, $P_l^m(x)$ are of the form

$$P_l^m(x) = (1-x^2)^{|m|/2} \left(\frac{\partial}{\partial x} \right)^{|m|} P_l(x)$$

$$P_l(x) = \frac{1}{2^l l!} \left(\frac{\partial}{\partial x} \right)^l (x^2-1)^l.$$

Together, the solutions $\Phi(\phi)$ and $\Theta(\theta)$ given $Y_{lm}(\theta, \phi)$ solutions known as the *spherical harmonics*. The first few results are shown here:

$$Y_{00} = \sqrt{\frac{1}{4\pi}}$$

$$Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta$$

$$Y_{1(\pm 1)} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi}$$

$$Y_{20} = \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1).$$

Another useful interpretation of the wave functions is the *parity* of a system. We define the parity operator as

$$\hat{P} |r\rangle = \pm |-r\rangle \implies \hat{P} \psi(\mathbf{r}) = \pm \psi(\mathbf{r})$$

where "+" refers to even and "-" is odd. For the spherical coordinates we work with, we have

$$\mathbf{r} = -\mathbf{r} \implies \begin{cases} r \rightarrow r \\ \theta \rightarrow \pi - \theta \\ \phi \rightarrow \pi + \phi. \end{cases}$$

So, if $\mathbf{r} \rightarrow -\mathbf{r}$, we have $\cos \theta \rightarrow \cos(\pi - \theta) = -\cos \theta$, so that we can evaluate the action of parity on each

$$\hat{P}[e^{im\phi}] = e^{im(\phi+\pi)} = e^{im\pi} e^{im\phi} = (e^{i\pi})^m e^{im\phi} = (-1)^m e^{im\phi}$$

which is even if m is even and odd if m is odd. Similarly, we have the results

$$\hat{P}[P_l^m(\cos \theta)] = P_l^m(-\cos \theta) = (-1)^m (-1)^l P_l^m(\cos \theta)$$

$$\hat{P}[Y_{lm}(\theta, \phi)] = (-1)^m (-1)^m (-1)^l P_l^m(\cos \theta) = (-1)^l P_l^m(\cos \theta).$$

Observe that the parity of the $Y_{lm}(\theta, \phi)$ depends only on the l quantum number; the function is even if l is even, and odd if l is odd.

We state another property without proof, but one which follows from the fact that these are eigenfunctions. That is, the spherical harmonics are orthogonal to each other (and orthonormal when properly normalized). We write the normalization conditions as

$$\int_0^{2\pi} \int_0^\pi d\theta d\phi \sin \theta = 1$$

$$\int_0^{2\pi} \int_0^\pi d\theta d\phi Y_{l'm'}^*(\theta, \phi) Y_{lm}(\theta, \phi) \sin \theta = \delta_{ll'} \delta_{mm'}$$

$$\langle l'm' | lm \rangle = \delta_{ll'} \delta_{mm'}.$$

1.1.2 Radial Wavefunctions

Now consider the radial part of the wave function,

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) - \frac{2mr^2}{\hbar^2} [V(r) - E] R(r) = l(l+1)R(r).$$

Here, we will make the substitution $\rho(r) = rR(r)$ with the Coulomb potential $V(r) = -e^2/4\pi\epsilon_0 r$. Citing the solution,³ we find that

$$\frac{\partial \rho(r)}{\partial r} = R(r) + r \frac{\partial R}{\partial r}$$

so

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) = \frac{\partial}{\partial r} \left[r \left(\frac{\partial \rho}{\partial r} - R(r) \right) \right] = \frac{\partial}{\partial r} \left[r \frac{\partial \rho}{\partial r} - \rho(r) \right] = \frac{\partial \rho}{\partial r} + r \frac{\partial^2 \rho}{\partial r^2} - \frac{\partial \rho}{\partial r} = r \frac{\partial^2 \rho}{\partial r^2}.$$

Thus,

$$r \frac{\partial^2 \rho}{\partial r^2} - \frac{2mr^2}{\hbar^2} \left[-\frac{e^2}{4\pi\epsilon_0 r} - E \right] = l(l+1) \frac{\partial \rho(r)}{\partial r}.$$

The solutions to this differential equation yield energy eigenvalues

$$E_n = - \left[\frac{m_e}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \right] \frac{1}{n^2}, \quad n = 1, 2, \dots$$

and radial wavefunctions of the form

$$R_{nl}(r) = \left\{ \left(\frac{2}{na} \right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3} \right\}^{1/2} e^{-r/na} \left(\frac{2r}{na} \right)^l L_{n-l-1}^{2l+1} \left(\frac{2r}{na} \right)$$

where $a = 4\pi\epsilon_0 \hbar^2 / m_e e^2 = 0.529 \text{Å}$ is the Bohr radius and

$$L_{q-p}^\rho(x) = (-1)^p \left(\frac{\partial}{\partial x} \right)^p L_q(x)$$

are the "associated Laguerre polynomials" with $n = 1, 2, \dots$ and $l = 0, 1, \dots$. There are a couple notes to make about these solutions:

- Energy depends only on "n", not on "l" or "m."
- Radial wavefunctions with $n = 0$ have non-zero probability at the origin, meaning the electron can overlap the nucleus.

For historical reasons, spectroscopic notation refers to the different energies by letters, and are still commonly used.

l	0	1	2	3	4	5	...
letter	s	p	d	f	g	n	... alphabetical

Table 1: Spectroscopic labelling for the first five indexes of l . After the fifth, the lettering becomes alphabetical.

³See Griffiths pg. 134 for a full derivation.

1.2 Transitions

One of the primary ways we interact with matter is through electromagnetic (EM) radiation. Since EM radiation couples to charge, it is through the electronic wavefunction that these interactions are often important (and strongest). The relevant Hamiltonian describing this interaction is the electric dipole Hamiltonian:

$$\hat{H}_{\text{int}} = -e\mathbf{r} \cdot \mathbf{E}(t).$$

Physically, this describes the interaction of an electric dipole with an electric field; and, often, the dipole is induced by the field, which pulls electrons away from the nucleus.

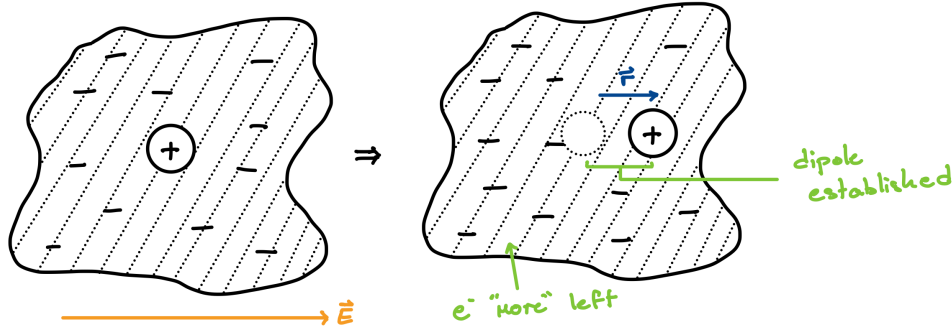


Figure 2: External $\mathbf{E}$ field applied onto a nucleus, shifting it by a distance $\mathbf{r}$.

For EM radiation, the electric field is oscillating by

$$\mathbf{E}(t) = |\mathbf{E}_0| \text{Re}(e^{i\omega_0 t} \mathbf{e}_{\text{rad}})$$

where $|\mathbf{E}_0|$ is the constant amplitude and $\mathbf{e}_{\text{rad}}$ is the polarization vector of radiation.⁴ One way to understand the rate of transitions from one state to the other is by Fermi's Golden Rule:

$$\text{Rate} \propto \frac{2\pi}{\hbar^2} \left| \langle \psi_f | \hat{H}_{\text{int}} | \psi_i \rangle \right|^2$$

where $\langle \psi_f | \hat{H}_{\text{int}} | \psi_i \rangle$ is the *transition matrix element*. Expanding further, we find that

$$\text{Rate} \propto \frac{2\pi}{\hbar^2} |eE_0|^2 |\langle \psi_f | \mathbf{r} \cdot \mathbf{e}_{\text{rad}} | \psi_i \rangle|^2 = \frac{2\pi}{\hbar} |eE_0|^2 \left| \int d^3\mathbf{r} \psi_f^*(\mathbf{r}) (\mathbf{r} \cdot \mathbf{e}_{\text{rad}}) \psi_i(\mathbf{r}) \right|^2.$$

Note that

$$\hat{H}_{\text{int}} | \psi_i \rangle = | \psi_{\text{new}} \rangle$$

so some inner product $\langle \psi_f | \psi_{\text{new}} \rangle$ quantifies "how much" this new state is like $|\psi_f\rangle$, a particular final state.

In spherical coordinates, the matrix element can be expressed as

$$\begin{aligned} \langle \psi_f | \mathbf{r} \cdot \mathbf{e}_{\text{rad}} | \psi_i \rangle &= \langle R_{nl,f} | r | R_{nl,i} \rangle \times \langle Y_{l_f m_f} | \mathbf{e}_{\mathbf{r}} \cdot \mathbf{e}_{\text{rad}} | Y_{l_i m_i} \rangle \\ &= \langle n_f l_f | r | n_i l_i \rangle \langle l_f m_f | \mathbf{e}_{\mathbf{r}} \cdot \mathbf{e}_{\text{rad}} | l_i m_i \rangle \\ &= D_{12} \mathcal{I}_{\text{ang}} \end{aligned}$$

⁴We make several assumptions here. For one, we assume that the field is spatially uniform across the size of the atom.

where

$$D_{12} = \int_0^\infty dr R_{n_f l_f}(r) |\mathbf{r}| R_{n_i l_i}(r) r^3 = \int_0^\infty dr R_{n_f l_f}(r) r R_{n_i l_i}(r) r^3$$

is the radial integral, involving the radial hydrogenic wavefunctions $R_{nl}(r)$. The magnitude of D_{12} is not generally zero for $n_f l_f \neq n_i l_i$, but can be quite small if they differ by a lot. Likewise, we may write

$$\mathcal{I}_{\text{ang}} = \int_0^{2\pi} \int_0^\pi d\theta d\phi Y_{l_f m_f}^*(\theta, \phi) \mathbf{e}_r \cdot \mathbf{e}_{\text{rad}} Y_{l_i m_i}(\theta, \phi) \sin \theta$$

for the angular integral, which ends up being zero much of the time, due to the symmetries of the spherical harmonics.

For an intuitive understanding of the angular integral, we will recast the operator part in the same basis as the spherical harmonics. The position vector may be represented as

$$\begin{aligned} \mathbf{e}_r &= \frac{1}{r}(x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z) \\ &= \sin \theta \cos \phi \mathbf{e}_x + \sin \theta \sin \phi \mathbf{e}_y + \cos \theta \mathbf{e}_z \\ &= \left[\sqrt{\frac{2\pi}{3}}(Y_{1(-1)} - Y_{11})\mathbf{e}_x + i\sqrt{\frac{2\pi}{3}}(Y_{1(-1)} + Y_{11})\mathbf{e}_y + \sqrt{\frac{4\pi}{3}}Y_{10}\mathbf{e}_z \right] \\ &= \sqrt{\frac{2\pi}{3}} \left[\frac{(\mathbf{e}_x + i\mathbf{e}_y)}{\sqrt{2}} Y_{1(-1)} + \frac{(-\mathbf{e}_x + i\mathbf{e}_y)}{\sqrt{2}} Y_{11} + \mathbf{e}_z Y_{10} \right]. \end{aligned}$$

Notice that the direction of $\mathbf{r}$, $\mathbf{e}_r$ are represented as a linear combination of spherical harmonics with $l = 1$, one unit of angular momentum. This new "basis" has its own unit vectors, which represent a polarization basis.

The unit vectors in this polarization basis are

$$\begin{aligned} \mathbf{e}_{\sigma+} &= \frac{1}{\sqrt{2}}(\mathbf{e}_x + i\mathbf{e}_y) \\ \mathbf{e}_{\sigma-} &= \frac{1}{\sqrt{2}}(\mathbf{e}_x - i\mathbf{e}_y) \\ \mathbf{e}_\pi &= \mathbf{e}_z \end{aligned}$$

such that the polarization of the light has the unit vector

$$\mathbf{e}_{\text{rad}} = A_{\sigma+}\mathbf{e}_{\sigma+} + A_{\sigma-}\mathbf{e}_{\sigma-} + A_\pi\mathbf{e}_\pi.$$

Here, the A are the amplitude coefficients which quantify how much of each polarization is there.

For normalization, we require

$$\sqrt{A_{\sigma+}^2 + A_{\sigma-}^2 + A_\pi^2} = 1.$$

This polarization convention for EM light is quite useful. We use the names,

$$\begin{aligned} \sigma^+ &: \text{"sigma-plus"} \rightarrow \text{right-circularly polarized} \\ \sigma^- &: \text{"sigma-minus"} \rightarrow \text{left-circularly polarized} \\ \pi &: \text{"pi"} \rightarrow \text{parallel-linear polarized.} \end{aligned}$$

The corresponding transformation back to the Cartesian basis may be helpful, and is

$$\mathbf{e}_{\text{rad}} = A_x \mathbf{e}_x + A_y \mathbf{e}_y + A_z \mathbf{e}_z$$

with

$$A_x = \frac{(A_{\sigma^-} - A_{\sigma^+})}{\sqrt{2}}, \quad A_y = \frac{(A_{\sigma^-} + A_{\sigma^+})}{\sqrt{2}i}, \quad A_z = A_{\pi}.$$

To return to the angular integral for the transition matrix element, note that with the complex numbers

$$\left(\frac{\mathbf{e}_x - i\mathbf{e}_y}{\sqrt{2}} \right) \cdot \left(\frac{\mathbf{e}_x + i\mathbf{e}_y}{\sqrt{2}} \right) = 1, \quad \left(\frac{\mathbf{e}_x + i\mathbf{e}_y}{\sqrt{2}} \right) \cdot \left(\frac{\mathbf{e}_x + i\mathbf{e}_y}{\sqrt{2}} \right) = 0.$$

So, we yield the term in our integral

$$\mathbf{e}_{\mathbf{r}} \cdot \mathbf{e}_{\text{rad}} \propto A_{\sigma^+} Y_{11} + A_{\sigma^-} Y_{1(-1)} + A_{\pi} Y_{10} \implies e\mathbf{E} \cdot \mathbf{r} = eE_0 r [A_{\sigma^+} Y_{11} + A_{\sigma^-} Y_{1(-1)} + A_{\pi} Y_{10}]$$

and the matrix element for the angular part of the integral is

$$\begin{aligned} \langle l_f m_f | \mathbf{e}_{\mathbf{r}} \cdot \mathbf{e}_{\text{rad}} | l_i m_i \rangle &\propto \langle l_f m_f | A_{\sigma^+} Y_{11} + A_{\sigma^-} Y_{1(-1)} + A_{\pi} Y_{10} | l_i m_i \rangle \\ &= \langle l_f m_f | A_{\sigma^+} Y_{11} | l_i m_i \rangle + \langle l_f m_f | A_{\sigma^-} Y_{1(-1)} | l_i m_i \rangle \langle l_f m_f | A_{\pi} Y_{10} | l_i m_i \rangle. \end{aligned}$$

Each term is thus the result of an integral over three spherical harmonic functions:

$$\begin{aligned} \langle l_3 m_3 | Y_{l_2 m_2} | l_1 m_1 \rangle &= \int_0^{2\pi} d\theta \sin \theta \int_0^{\pi} d\phi Y_{l_3 m_3}^*(\theta, \phi) Y_{l_2 m_2} Y_{l_1 m_1} \\ &= \left[\frac{(2l_3 + 1)(2l_2 + 1)(2l_1 + 1)}{4\pi} \right]^{1/2} \begin{pmatrix} l_3 & l_2 & l_1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_3 & l_2 & l_1 \\ -m_3 & m_2 & m_1 \end{pmatrix} \end{aligned}$$

where we have used the shorthand of the *Wigner 3-j symbol*,

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}.$$

There are two things to note here:

- (1) we have placed a negative $-m_3$ within the second Wigner 3-j symbol since we have used the complex conjugate $Y_{lm}^* = Y_{l(-m)}$;
- (2) the Wigner 3-j symbol itself is **not** a matrix, it just uses a matrix-like representation.

This symbol may be familiar as it shows up in the addition of angular momentum, and is closely related to the *Clebsch-Gordon coefficients*. Specifically, the Clebsch-Gordon coefficients evaluates to a single number.

These Wigner 3-j's have certain rules baked into them that tell us whether the symbols are zero or not. In fact, *the Wigner 3-j symbol is zero unless **all** of the following are true:*

- (1) all input values are integers or half-integers;
- (2) the m -values are in the manifold of the j -values in the same column:

$$m_i \in \{-j_i, -j_i + 1, \dots, j_i - 1, j_i\};$$

(3) m -values must sum to zero:

$$m_1 + m_2 + m_3 = 0;$$

(4) the triangle rule holds:

$$|j_1 - j_2| \leq j_3 \leq |j_1 + j_2|;$$

(5) j -values must sum to an integer, even if j -values are half-integer. In the special case $m_1 = m_2 = m_3 = 0$, the j -sum must be even.

We will now apply these rules to the radiative transition matrix elements.

1.2.1 Sigma-plus (σ^+) Transitions

These transitions are of the form:

$$\langle l_f m_f | A_{\sigma^+} Y_{11} | l_i m_i \rangle \propto \begin{pmatrix} l_f & 1 & l_i \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_f & 1 & l_i \\ -m_f & 1 & m_i \end{pmatrix}.$$

We must show that each rule is satisfied.

- (1) Rules 1 and 2 are both satisfied from our solutions.
- (2) Rule 3 gives $m_f = m_i + 1$. Observe that this result shows that the m quantum number *increases by one* for a σ^+ transition.
- (3) Rule 4 gives $|l_f - 1| \leq l_i \leq |l_f + 1|$.
- (4) Rule 5 gives, for the first 3-j, $l_i + l_f + 1 = \text{even}$. Thus, combining this and the previous rule, we find that

$$|l_f - l_i| = 1.$$

Observe that the l quantum number *must change* (increase or decrease) *by one*.

1.2.2 Sigma-minus (σ^-) Transitions

These transitions are of the form:

$$\langle l_f m_f | A_{\sigma^-} Y_{1(-1)} | l_i m_i \rangle \propto \begin{pmatrix} l_f & 1 & l_i \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_f & 1 & l_i \\ -m_f & -1 & m_i \end{pmatrix}.$$

Following all five rules once more, we should observe the conditions

$$m_f = m_i - 1, \quad |l_f - l_i| = 1.$$

Again, we have the result that l must change by one, however the σ^- transitions decrease m by one.

1.2.3 Pi (π) Transitions

These transitions are of the form

$$\langle l_f m_f | A_\pi Y_{10} | l_i m_i \rangle \propto \begin{pmatrix} l_f & 1 & l_i \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_f & 1 & l_i \\ -m_f & 0 & m_i \end{pmatrix}.$$

We find from our rules the conditions

$$m_f = m_i, \quad |l_f - l_i| = 1.$$

Again, l must change by one, however, the π -transitions keep the same m -value.

To interpret these rules, which are known as *selection rules* (for electric dipole transitions), we have in summary:

Quantum Number	l -rule	σ^+	π	σ^-
l	must change by ± 1	—	—	—
m	—	+1	0	-1

Table 2: Summarized selection rules for electric dipole transitions. As before, the l -value changes by one, while the m -value increases, decreases, or remains constant depending on the transition.

Observe that the requirement that l changes is a statement about the parity of the operator and of the wavefunctions: the electric dipole operator has odd parity, and so for the dipole matrix element to be non-zero, the two states must be of different parity so that the final integrand is even.

$$e\mathbf{E} \cdot \mathbf{r} : \text{ odd since } r \text{ is odd.}$$

In spherical coordinates, the parity is carried by the angular coordinates, so the angular integrals must reflect this, thus showing on l .

Further, the change of l by a single unit is a statement about the angular momentum of light: it carries a single unit of angular momentum.

Finally, we can think about the change of m that depends on polarization as the polarization transferring angular momentum to the electron upon transition.

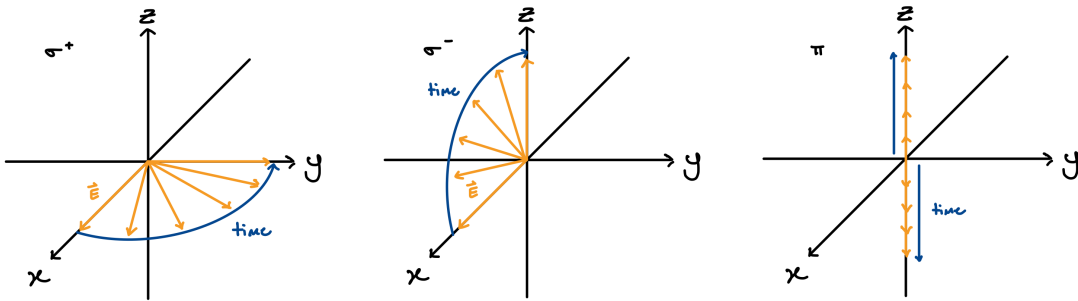


Figure 3: Time evolution for an arbitrary $\mathbf{E}$ vector for each transition.

For the σ^+ transition, we expect $\mathbf{E}$ to evolve under the form

$$\mathbf{E}_{\sigma^+} = E_0 \text{Re} \left[e^{-i\omega t} \left(\frac{\mathbf{e}_x + i\mathbf{e}_y}{\sqrt{2}} \right) \right] = \frac{E_0}{\sqrt{2}} [\cos(\omega t)\mathbf{e}_x + \sin(\omega t)\mathbf{e}_y]$$

which is the field "stirring" the electron to have more angular momentum along $+z$.

For the σ^- transition, we expect $\mathbf{E}$ to evolve under the form

$$\mathbf{E}_{\sigma^-} = E_0 \text{Re} \left[e^{-i\omega t} \left(\frac{\mathbf{e}_x - i\mathbf{e}_y}{\sqrt{2}} \right) \right] = E_0 [\cos(\omega t)\mathbf{e}_x - \sin(\omega t)\mathbf{e}_y]$$

which is the field "stirring" the electron to have less angular momentum along $+z$. Equivalently, we may think of this as the field generating more angular momentum along $-z$.

For the π transition, we expect $\mathbf{E}$ to evolve under the form

$$\mathbf{E}_{\pi} = E_0 \text{Re} [e^{-i\omega t} \mathbf{e}_z] = E_0 \cos(\omega t) \mathbf{e}_z$$

which shows no change to angular momentum along $\pm z$, since the force is parallel to z .

1.3 Zeeman Effect

The Zeeman effect is the effect of a magnetic field on a system. We will begin its analysis by considering an electron with angular momentum

$$\hbar|\mathbf{l}| = \sqrt{l(l+1)}\hbar.$$

The classical equivalence of this value is $\hbar\mathbf{l} = \mathbf{r} \times \mathbf{p} = m_e v r \mathbf{e}_z$, where m_e is the electron mass. Note also that we define l to be the average momentum divided by $\hbar$, assuming all momentum is along $\mathbf{e}_z$.

The electric current associate with the classical orbit of this circular motion is given by

$$I = -e \left(\frac{v}{2\pi r} \right), \quad \left(\frac{[\text{J}]}{[\text{s}]} = [\text{J}] \frac{[\text{m/s}]}{[\text{m}]} \right).$$

The associated magnetic moment is given by

$$\boldsymbol{\mu} = I\mathbf{A} = -e \frac{v}{2\pi r} \pi r^2 \mathbf{e}_z = -\frac{e}{2} \frac{\hbar\mathbf{l}}{m_e} = -\frac{e\hbar}{2m_e} \mathbf{l}.$$

We determine the potential energy associated with a magnetic dipole momentum through the Hamiltonian

$$\hat{H}_z = -\boldsymbol{\mu} \cdot \mathbf{B} = \frac{e\hbar}{2m_e} \mathbf{l} \cdot \mathbf{B}.$$

Notice that when $\mathbf{B}$ points along $\mathbf{e}_z$, we have that $\mathbf{l} \cdot \mathbf{B} = l_z B$; and, in the quantum picture, for hydrogenic states like the ones we have been considering, we may have

$$l_z B Y_{lm}(\theta, \phi) = m B Y_{lm}(\theta, \phi) \implies \mathbf{l} \cdot \mathbf{B} \rightarrow l_z B \rightarrow m B.$$

Thus, the potential energy is

$$E_z = \frac{e\hbar}{2m_e} m B = \mu_B m B, \quad \mu_B \equiv \frac{e\hbar}{2m_e} \approx 9.27 \times 10^{-24} \text{ J/T}.$$

where μ_B is the *Bohr magneton*.

In an external magnetic field, the energies of the otherwise degenerate $(2l + 1)$ levels with eigenvalues (quantum numbers) $m = -l, \dots, l$ are split⁵ due to the orbital magnetic moment. This is called the *Normal Zeeman Effect*, and we can calculate an energy level spacing given by

$$\Delta E_z = E_z(m) - E_z(m - 1) = \mu_B B.$$

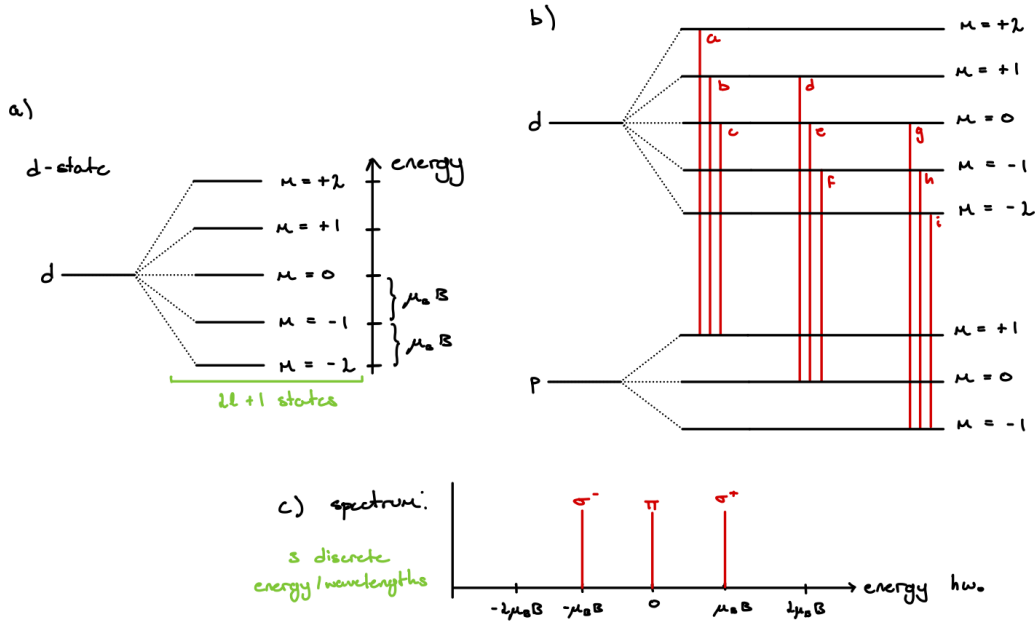


Figure 4: a) Example *d*-state showing the energy level spacings and $2l + 1 = 5$ states. b) Absorption/emission spectrum for transitions between *p* and *d* states separated by $E_0 = \hbar\omega_0$. c) Three discrete energy spectra for each transition, separated by an energy unit $\hbar\omega_0$.

Observe from Fig. 4 that each transition in b) corresponds to the following:

$$a, d, g \rightarrow \sigma^+ \text{ transitions: } m \rightarrow m + 1$$

$$b, e, h \rightarrow \pi \text{ transitions: } m \rightarrow m$$

$$c, f, i \rightarrow \sigma^- \text{ transitions: } m \rightarrow m - 1.$$

1.4 Spin-Orbit Coupling

We have considered, so far, the electron only as a charged particle; however, we must also include its implicitly quantum mechanical character in its *spin*. This intrinsic angular momentum has no spatial structure, so we cannot include it in our position-space wavefunction, but it does have its own quantum number that defines it in its Hilbert space.

⁵We say that the "degeneracy is lifted" in this case.

While the Dirac equation⁶ is the more complete description of incorporating spin (and relativity) into quantum mechanics (QM), we will use a simpler picture to develop intuition using classical concepts.

We begin by defining spin via its own operator, describing and intrinsic angular momentum.

$$\hbar \mathbf{s} \rightarrow |\hbar \mathbf{s}| = \frac{\hbar}{2} \quad (\text{for a single electron})$$

Notice that the electron has a "half-unit" of angular momentum ($\hbar/2$). Two quantum numbers will define this (analogous to l and m_l for orbital angular momentum). We write:

(1) The eigenstates of spin are: $|s, m_s\rangle$;

(2) For a single electron: $s = 1/2$, $m_s = \pm 1/2$

$$\begin{aligned} \hat{\mathbf{S}}^2 |s, m_s\rangle &= s(s+1) |s, m_s\rangle = \frac{3}{4} |s, m_s\rangle \\ \hat{S}_z |s, m_s\rangle &= m_s |s, m_s\rangle = \pm \frac{1}{2} |s, m_s\rangle. \end{aligned}$$

The total hydrogenic wavefunction, in Dirac notation, is given by its quantum numbers

$$|\psi\rangle = |nlm_lsm_s\rangle$$

where we have redefined our previous $m \rightarrow m_l$ to distinguish from m_s .

Electron spin is associated with a magnetic moment, akin to the orbital magnetic moment, but the relationship needs a "g-factor." Here, $g_s \approx 2$ is the Landé g-factor, given by

$$\boldsymbol{\mu}_s = -g_s \mu_B \mathbf{S}.$$

In considering the effect of spin on the spectrum of electrons in atoms, we will consider the consequence of the spin's motion. That is, the electron's orbit around a charged nucleus. In the frame of the electron, the rotating charged nucleus "creates" an effective magnetic field given by

$$\begin{aligned} \mathbf{B}_{\text{eff}} &= \mathbf{B}_0 - \frac{1}{c^2} \mathbf{v} \times \mathbf{E} \quad (\text{for non-relativistic speeds}) \\ \mathbf{B}_{\text{eff}} &= -\frac{1}{c^2} \mathbf{v} \times \mathbf{E}. \end{aligned}$$

We assume that $\mathbf{B}_0$ is zero for now so that we can consider only the second term. The electric field is due to a Coulomb potential, so by spherical symmetry we may write

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}, \quad \mathbf{E}(r) = \frac{1}{e} \frac{\partial V}{\partial r} \mathbf{e}_r = \frac{1}{er} \frac{\partial V}{\partial r} \mathbf{r} = \frac{e}{4\pi\epsilon_0 r^3} \mathbf{r}.$$

Thus, our effective magnetic field becomes

$$\mathbf{B}_{\text{eff}} = -\frac{e}{4\pi\epsilon_0 r^3 c^2} \mathbf{v} \times \mathbf{r} = \frac{e}{4\pi\epsilon_0 r^3} \frac{1}{mc^2} \mathbf{r} \times \mathbf{p} = \frac{e}{4\pi\epsilon_0 r^3 mc^2} \hbar \mathbf{l}.$$

⁶See Sakurai; Shankar chp. 20.

Given this field, and together with the electron spin, a term in the atomic Hamiltonian arises and is

$$\hat{H}_{\text{so}} = -\boldsymbol{\mu}_s \cdot \mathbf{B}_{\text{eff}} = \frac{1}{2}g_s\mu_B \left(\frac{\hbar e}{4\pi\epsilon_0 m_e c^2 r^3} \right) \mathbf{s} \cdot \mathbf{l}.$$

Here, $\mathbf{s} \cdot \mathbf{l}$ is the *spin-orbit coupling*, and there is an extra $1/2$ Thomas correction factor.⁷ So, with $g_s = 2$ and $\mu_B = e\hbar/2m_e$, we find the spin-orbit Hamiltonian

$$\hat{H}_{\text{so}} = \frac{e^2 \hbar^2}{8\pi\epsilon_0 m_e^2 c^2 r^3} (\hat{\mathbf{s}} \cdot \hat{\mathbf{l}}).$$

The expectation value of this spin-orbit energy is

$$\langle E_{\text{so}} \rangle = \frac{e^2 \hbar^2}{8\pi\epsilon_0 m_e^2 c^2} \left\langle \frac{1}{r^3} \right\rangle \langle \hat{\mathbf{s}} \cdot \hat{\mathbf{l}} \rangle.$$

The $\langle 1/r^3 \rangle$ term is used for position space wavefunctions to evaluate; additionally, we define a special name for a constant factor pulled out front. Namely, the *fine structure constant* α ,

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \implies \frac{\hbar^3}{2m_e^2 c} \frac{e^2}{4\pi\epsilon_0 \hbar c} = \frac{\hbar^3 \alpha}{2m_e^2 c}.$$

For hydrogenic wavefunctions, we can calculate this position dependence⁸ to be

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{1}{l(l+1/2)(l+1)} \left(\frac{Z}{na_0} \right)^3$$

where Z is the atomic number and a_0 is the Bohr radius. Now that we have a new term in the Hamiltonian, we must consider the eigenstates of the entire Hamiltonian, namely

$$\hat{H}_{\text{tot}} = \hat{H}_0 + \hat{H}_{\text{so}}$$

where $\hat{H}_0$ is the Hamiltonian we have already solved. We must be careful to note that the eigenstates of $\hat{H}_0$ are not necessarily eigenstates of $\hat{H}_{\text{so}}$, and thus, not necessarily eigenstates of $\hat{H}_{\text{tot}}$.

To find eigenstates of the full Hamiltonian, we will consider a new angular momentum, called the *total angular momentum*

$$\mathbf{j} = \mathbf{l} + \mathbf{s} \rightarrow \hat{\mathbf{j}} = \hat{\mathbf{l}} + \hat{\mathbf{s}}.$$

This new quantity $\mathbf{j}$ must obey the "usual" rules for adding angular momentum in QM. That is, it must follow

$$|l - s| \leq j \leq |l + s| \quad (\text{triangle rule})$$

where j are eigenvalues of $\hat{\mathbf{j}}$,

$$m_l + m_s = m_j \quad (\text{zero-sum})$$

and,

$$\langle ls; m_l m_s | ls; j m_j \rangle = (-1)^{l-s+m_j} \sqrt{2j+1} \begin{pmatrix} l & s & j \\ m_l & m_s & -m_j \end{pmatrix}. \quad (\text{Clebsch-Gordan})$$

⁷This also arises from the Dirac equation, since electron orbits are not actually circles.

⁸See eq. 2.23 in Foot

We now pay attention to $\hat{\mathbf{j}}^2$.

$$\begin{aligned}\hat{\mathbf{j}}^2 &= (\hat{\mathbf{l}}^2 + \hat{\mathbf{s}}^2 + \hat{\mathbf{s}} \cdot \hat{\mathbf{l}} + \hat{\mathbf{l}} \cdot \hat{\mathbf{s}}). \\ \hat{\mathbf{s}} \cdot \hat{\mathbf{l}} &= \frac{1}{2}(\hat{\mathbf{j}}^2 - \hat{\mathbf{l}}^2 - \hat{\mathbf{s}}^2).\end{aligned}$$

We note that the spin-orbit operator is related to the squares of the individual angular momenta, and the sum of their square. We note also that $\hat{\mathbf{s}}$ and $\hat{\mathbf{l}}$ commute, since they act on different, separable parts of the wavefunction.

The eigenstates of spin and of $\hat{H}_0$ were $|\psi_0\rangle = |nlm_lsm_s\rangle$, and in our old scenario, these states were separable. Indeed, we could've written

$$|nlm_lsm_s\rangle = |nlm_s\rangle \otimes |sm_s\rangle.$$

However, with the coupling between spin and orbital momentum, these states are no longer separable, and we require a new basis. In this case, there is a basis in which $\mathbf{s} \cdot \mathbf{l}$ is diagonal:

$$|nlsjm_j\rangle = \sum |nls; m_sm_l\rangle \langle nls; m_sm_l | nlsjm_j\rangle.$$

In this step we have incorporated a *resolution of unity* on $|nls; m_sm_l\rangle \langle nls; m_sm_l|$, where there is a sum over a complete basis set. Note also that in this notation, we have

$$|nls; m_lm_s\rangle = |nlm_l\rangle \otimes |sm_s\rangle.$$

Further, we note that the inner product $\langle nls; m_sm_l | nlsjm_j\rangle$ are the Clebsch-Gordon coefficients. We may not wish to choose m_l and m_s anymore for our primary quantum numbers, since in this new picture we have $[\hat{l}_z, \hat{j}_z] \neq 0$ and $[\hat{s}_z, \hat{j}_z] \neq 0$.

We might ask: what are the eigenstates of $\hat{\mathbf{j}}^2$? We have

$$\hat{\mathbf{j}}^2 |nlsjm_j\rangle = j(j+1) |nlsjm_j\rangle$$

together with

$$\begin{aligned}\hat{\mathbf{l}}^2 |nlsjm_j\rangle &= l(l+1) |nlsjm_j\rangle \\ \hat{\mathbf{s}}^2 |nlsjm_j\rangle &= s(s+1) |nlsjm_j\rangle.\end{aligned}$$

So, the spin-orbit term gives

$$\langle \hat{\mathbf{s}} \cdot \hat{\mathbf{l}} \rangle = \left\langle \frac{1}{2}(\hat{\mathbf{j}}^2 - \hat{\mathbf{l}}^2 - \hat{\mathbf{s}}^2) \right\rangle = \frac{1}{2}[j(j+1) - l(l+1) - s(s+1)].$$

The energy associated with this coupling is

$$E_{\text{so}} = \frac{\beta}{2}[j(j+1) - l(l+1) - s(s+1)]$$

where

$$\beta = \frac{e^2 \hbar^2}{8\pi\epsilon_0 m_e^2 c^2} \frac{1}{l(l+1/2)(l+1)} \left(\frac{Z}{na_0} \right)^3.$$

For hydrogen's ground state, we have $l = 0$ and $s = 1/2$, so that $j = 1/2$ and

$$\langle \hat{\mathbf{s}} \cdot \hat{\mathbf{l}} \rangle = \frac{1}{2} \left[\frac{1}{2} \left(\frac{3}{2} \right) - 0 - \frac{1}{2} \left(\frac{3}{2} \right) \right] = 0$$

suggesting there is no spin-orbit coupling of the ground state. For hydrogen's first excited state, we have $l = 1$ and $s = 1/2$, so that $j = 1 - 1/2 = 1/2$ and $j = 1 + 1/2 = 3/2$. So,

$$j = \frac{1}{2} : \langle \hat{\mathbf{s}} \cdot \hat{\mathbf{l}} \rangle = \frac{1}{2} \left[\frac{1}{2} \left(\frac{3}{2} \right) - 1(2) - \frac{1}{2} \left(\frac{3}{2} \right) \right] = -1$$

$$j = \frac{3}{2} : \langle \hat{\mathbf{s}} \cdot \hat{\mathbf{l}} \rangle = \frac{1}{2} \left[\frac{3}{2} \left(\frac{5}{2} \right) - 1(2) - \frac{1}{2} \left(\frac{3}{2} \right) \right] = \frac{1}{2}.$$

We use the common spectroscopic notation which is of the form nL_j or $n^{2s+1}L_j$.

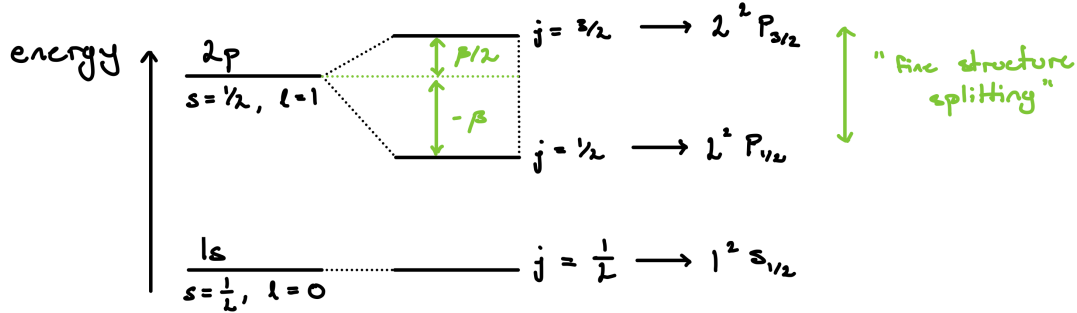


Figure 5: Hydrogen spin-orbit spectrum.

The spin-orbit energy split is known as *fine structure splitting*. In alkali atoms specifically, fine structure results in *D-lines*, after the observation of the sodium "doublet." We list the D-lines for the alkali atoms below.

Atom, $\rightarrow$ Ground State	D1 (nm)	D2 (nm)
H, $1S \rightarrow 1S_{1/2} \rightarrow$	$2P_{1/2} : 656.277$	$2P_{3/2} : 656.272$
Li, $1S^2 2S \rightarrow 2S_{1/2} \rightarrow$	$2P_{1/2} : 670.979$	$2P_{3/2} : 670.977$
Na, $1S^2 2S^2 2P^6 3S \rightarrow 3S_{1/2} \rightarrow$	$3P_{1/2} : 589.755$	$3P_{3/2} : 589.159$
K, $\dots 4S \rightarrow 4S_{1/2} \rightarrow$	$4P_{1/2} : 770.108$	$4P_{3/2} : 767.701$
Rb, $\dots 5S \rightarrow 5S_{1/2} \rightarrow$	$5P_{1/2} : 794.978$	$5P_{3/2} : 780.241$
Cs, $\dots 6S \rightarrow 6S_{1/2} \rightarrow$	$6P_{1/2} : 894.593$	$6P_{3/2} : 852.347$

Table 3: D1 and D2 lines for the alkali atoms including their spectroscopic notation.

We note also that, on the same order of correction, a shift in energies due to a relativistic correct to the momentum should also be considered.

1.4.1 Notes on Change of Basis

[Placeholder]

1.5 Hyperfine Interaction

So far, we have ignored details about the nucleus, but for our next step we must consider its effect on our atomic spectrum. The nucleus has a non-zero magnetic moment due to the combined spin

of the nucleons, which add to give the "nuclear spin", $\mathbf{I}$. We write the nuclear magnetic moment as

$$\boldsymbol{\mu}_I = g_I \mu_N \mathbf{I}, \quad \mu_N = \frac{m_e \mu_B}{m_n} \simeq \frac{\mu_B}{1836}$$

where m_n is the neutron mass.

Like our discussion of spin-orbit coupling, the neutron, with its own spin, "sees" the motion of the electron as an effective magnetic field, and the interaction of the nuclear spin with this effective field gives rise to the hyperfine term in the Hamiltonian,

$$\hat{H}_{hfs} = -\boldsymbol{\mu}_I \cdot \mathbf{B}_e.$$

To determine the effective magnetic field due to the electron, we can use the concept of a magnetization distribution

$$\mathbf{M} = -g_s \mu_B \mathbf{s} |\psi(\mathbf{r})|^2$$

and, by classical EM,

$$\mathbf{B}_e = \frac{2}{3} \mu_0 \mathbf{M}.$$

For s -electrons ($l = 0$), any contributions right at $r = 0$ matter, since contributions from "shells" at each radius cancel when averaged at $r = 0$. Indeed,

$$\mathbf{B}_e = -\frac{2}{3} \mu_0 g_s \mu_B |\psi_{ns}(0)|^2 \mathbf{s}$$

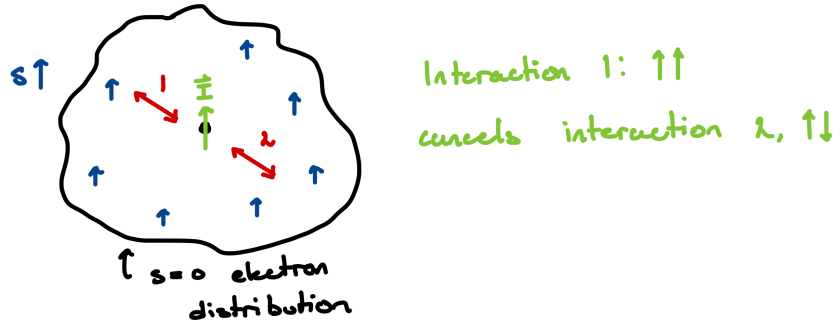


Figure 6: s -electron interaction at $r = 0$ showing how there is a net effect on the magnetic field.

For higher l , the orbital motion of the electrons also contributes an effective $\mathbf{B}$ -field at the nucleus, also from the dipole moment of a spin some distance from the nucleus. We can determine this field as

$$\mathbf{B}_e = \frac{\mu_0}{4\pi} \left[-\frac{3\mathbf{v} \times (-\mathbf{r})}{r^3} - \frac{\boldsymbol{\mu}_e - 3(\boldsymbol{\mu}_e \cdot \mathbf{e}_r)\mathbf{e}_r}{r^3} \right].$$

Using $-e\mathbf{v} \times \mathbf{r} = -2\mu_B \mathbf{l}$ and $\boldsymbol{\mu}_e = -2\mu_B \mathbf{s}$, we determine the effective $\mathbf{B}$ field

$$\mathbf{B}_e = -\frac{2\mu_0 \mu_B}{4\pi r^3} \left[\mathbf{l} - \mathbf{s} + \frac{3(\mathbf{s} \cdot \mathbf{r})\mathbf{r}}{r^2} \right].$$

This is a rather complicated operator, so we will use an approximation to it through the *Wigner-Eckhart* theorem (also known as the *Projection Theorem*). This lets us help describe off-diagonal elements of angular momentum operators.⁹

⁹Foot describes this briefly in page 83, and there is an extensive description in Sakurai, Shankar, and other grad-level quantum mechanics textbooks.

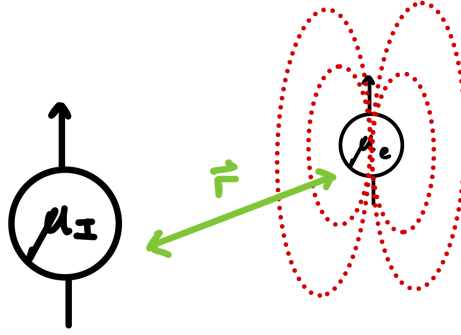


Figure 7: Magnetic moment of the nucleus and electron. The nucleus observes an effective magnetic field from the magnetic moment of the electron.

In our case, the eigenvalues of the electrons' angular momentum are, to a good approximation, the eigenvalues of $\mathbf{j}$ (so long as the hyperfine interactions are much less than the fine-structure interactions, i.e., $\langle \hat{H}_{\text{hfs}} \rangle \ll \langle \hat{H}_{\text{soc}} \rangle$).

If this is the case, we can assume that the matrix element of any vector operator will be proportional to one for $\mathbf{j}$, so

$$\langle jm_j | \mathbf{V} | jm_j \rangle = c \langle jm_j | \mathbf{j} | jm_j \rangle$$

where c is a small coefficient. Further, we project $\mathbf{V}$ onto $\mathbf{j}$. Taking the inner product of the operator with $\mathbf{j}$ on both sides, we should find that

$$\langle jm_j | \mathbf{V} \cdot \mathbf{j} | jm_j \rangle = c \langle jm_j | \mathbf{j}^2 | jm_j \rangle = c[j(j+1)] \implies c = \frac{\langle jm_j | \mathbf{V} \cdot \mathbf{j} | jm_j \rangle}{j(j+1)}.$$

Observe that this is exactly analogous to the case for vectors where

$$\mathbf{V} \cong \frac{\mathbf{V} \cdot \mathbf{j}}{|\mathbf{j}|^2} \mathbf{j} \quad \text{for } |\mathbf{V}| \ll |\mathbf{j}|.$$

Here, we implement this and gather all our coefficients into a "hyperfine constant," while defining $\mathbf{B}_e$ in terms of $\mathbf{j}$,

$$\mathbf{B}_e = -\frac{2\mu_0\mu_B}{4\pi r^3} \left[1 - \mathbf{s} + \frac{3(\mathbf{s} \cdot \mathbf{r})\mathbf{r}}{r^2} \right] \approx \frac{\mathbf{B}_e \cdot \mathbf{j}}{j(j+1)} \mathbf{j}.$$

So,

$$\hat{H}_{\text{hfs}} = -\mu_I \mathbf{I} \cdot \left[\frac{\mathbf{B}_e \cdot \mathbf{j}}{j(j+1)} \right] \mathbf{j} \equiv A_{\text{hfs}} \mathbf{I} \cdot \mathbf{j}.$$

We will generalize this case to a multi-electron case, so rather than using $\mathbf{j}$ we shall use $\mathbf{J}$. So,

$$\hat{H}_{\text{hfs}} \equiv A_{\text{hfs}} \mathbf{I} \cdot \mathbf{J}.$$

The eigenstates of the new total Hamiltonian will thus be the eigenstates of the "total angular momentum operator of the atom". We define

$$\mathbf{F} = \mathbf{I} + \mathbf{J}$$

with the new eigenstates

$$|nlsjfm_f\rangle \text{ or } |nLSJFm_F\rangle$$

where capital letters can refer to single or multiple electrons. These eigenstates are diagonal in $\hat{\mathbf{L}}^2$, $\hat{\mathbf{S}}^2$, $\hat{\mathbf{J}}^2$, $\hat{\mathbf{F}}^2$ and $\hat{\mathbf{F}}_z$, but not $\hat{\mathbf{L}}_z$, $\hat{\mathbf{S}}_z$ or $\hat{\mathbf{J}}_z$. Our newest eigenvalues are

$$\begin{aligned}\hat{\mathbf{F}}^2 |nLSJFm_F\rangle &\rightarrow \hat{\mathbf{F}}^2 |F, m_F\rangle = F(F+1) |F, m_F\rangle \\ \hat{\mathbf{F}}_z |F, m_F\rangle &= m_F |F, m_F\rangle\end{aligned}$$

where we drop the $nLSJ$ in the ket as a shorthand. As a sum of angular momenta, $\mathbf{F}$ is subject to the usual rules:

$$|I - J| \leq F \leq |I + J|, \quad m_F = -F, -F+1, \dots, F-1, F$$

so that m_F has $2F+1$ total possible values. We determine the energy of the hyperfine splitting to be

$$E_{\text{hfs}} = \langle \hat{H}_{\text{hfs}} \rangle = A_{\text{hfs}} \langle \mathbf{I} \cdot \mathbf{J} \rangle$$

and, like before with $\mathbf{L} \cdot \mathbf{S}$, determine

$$E_{\text{hfs}} = \frac{A_{\text{hfs}}}{2} [F(F+1) - I(I+1) - J(J+1)].$$

As an example, consider the hydrogen ground ($L=0$) state. The nuclear spin of a single proton is $I=1/2$, so that we have

$$\begin{aligned}L=0 \\ S=\frac{1}{2} &\implies \begin{cases} J=\frac{1}{2} \\ I=\frac{1}{2} \end{cases} \implies \begin{aligned} F &= \frac{1}{2} + \frac{1}{2} = 1 \\ F &= \frac{1}{2} - \frac{1}{2} = 0. \end{aligned}\end{aligned}$$

So, we have that $F=0, 1$, and so the energy of the hyperfine splitting are

$$\begin{aligned}E_{\text{hfs}}^{F=1} &= \frac{A_{\text{hfs}}}{2} \left[1(2) - \frac{1}{2} \left(\frac{3}{2} \right) - \frac{1}{2} \left(\frac{3}{2} \right) \right] = \frac{A_{\text{hfs}}}{4} \\ E_{\text{hfs}}^{F=0} &= \frac{A_{\text{hfs}}}{2} \left[0(1) - \frac{3}{4} - \frac{3}{4} \right] = -\frac{3}{4} A_{\text{hfs}}.\end{aligned}$$

We can calculate explicitly A_{hfs} to be

$$A_{\text{hfs}} = \frac{2}{3} \mu_0 g_s \mu_B g_I \mu_N \frac{1}{\pi a_0^3 n^3} = h \cdot 1.420 \text{ GHz}.$$

Note that

$$|\psi_0(0)|^2 = \frac{1}{\pi} \left(\frac{1}{na_0} \right)^3, \quad g_i \approx 5.6 \text{ for a proton}.$$

Thus, we can determine a specific wavelength

$$\lambda = \frac{hc}{\langle E_{\text{hfs}} \rangle} = 21 \text{ cm}$$

which is a famous line in radio astronomy and other related fields of study.

1.6 Anomalous Zeeman Effect

So far we have considered only the states of the hydrogen atomic (and hydrogenic atom) as they respond to an internal effective magnetic field. Now, let us consider the effect of external fields. We can write the atom's total magnetic moment as

$$\boldsymbol{\mu} = -\mu_B \mathbf{L} - g_s \mu_B \mathbf{S} - g_n \mu_N \mathbf{I}$$

where the last term is quite a bit smaller than the first two terms. Since this is the case, we will look at the first two terms, which just describe the electron

$$\boldsymbol{\mu} = -\mu_B \mathbf{L} - g_s \mu_B \mathbf{S}.$$

The eigenstates describing this system (where we neglect nuclear spin) are $|nLSJm_J\rangle$, and the Zeeman Hamiltonian is thus

$$\hat{H}_z = -\boldsymbol{\mu} \cdot \mathbf{B}_{\text{ext}}.$$

Observe by the projection theorem on $\boldsymbol{\mu}$, that

$$\hat{H}_z = -\frac{\langle \boldsymbol{\mu} \cdot \mathbf{J} \rangle}{J(J+1)} \mathbf{J} \cdot \mathbf{B}_{\text{ext}}$$

where

$$-\langle \boldsymbol{\mu} \cdot \mathbf{J} \rangle = \mu_B \langle \mathbf{L} \cdot \mathbf{J} \rangle + g_s \mu_B \langle \mathbf{S} \cdot \mathbf{J} \rangle.$$

Then, with $\mathbf{J} = \mathbf{L} + \mathbf{S}$, we have

$$\begin{aligned} \mathbf{L} \cdot \mathbf{J} &= \mathbf{L}^2 + \mathbf{L} \cdot \mathbf{S} = \mathbf{L}^2 + \frac{1}{2}(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2) = \frac{1}{2}(\mathbf{J}^2 + \mathbf{L}^2 - \mathbf{S}^2) \\ \mathbf{S} \cdot \mathbf{J} &= \frac{1}{2}(\mathbf{J}^2 + \mathbf{S}^2 - \mathbf{L}^2). \end{aligned}$$

With $g_s = 2$, we have

$$\begin{aligned} -\langle \boldsymbol{\mu} \cdot \mathbf{J} \rangle &= \mu_B \left[\frac{1}{2}J(J+1) + \frac{1}{2}L(L+1) - \frac{1}{2}S(S+1) + J(J+1) + S(S+1) - L(L+1) \right] \\ &= \mu_B \left[\frac{3}{2}J(J+1) + \frac{1}{2}S(S+1) - \frac{1}{2}L(L+1) \right]. \end{aligned}$$

So,

$$-\frac{\langle \boldsymbol{\mu} \cdot \mathbf{J} \rangle}{J(J+1)} = \mu_B \left[\frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)} \right] = \mu_B g_J, \quad g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}.$$

Then, the energy associated with the Zeeman Hamiltonian is

$$E_z = \mu_B g_J (\mathbf{J} \cdot \mathbf{B}_{\text{ext}}) = \mu_B g_J m_J B_{\text{ext}}$$

where we define the direction of field (for m_J) as "z." To see how this affects our system, let us consider an example of the $^2S_{1/2}$, $^2P_{1/2}$, and $^2P_{3/2}$ states as a function of the B -field. We determine each eigenvalue so we find the g_J factor,

$$\begin{aligned} ^2S_{1/2} : L=0, S=\frac{1}{2}, J=\frac{1}{2} &\implies g_J = \frac{3}{2} + \frac{1}{2} = g_s \\ ^2P_{1/2} : L=1, S=\frac{1}{2}, J=\frac{1}{2} &\implies g_J = \frac{3}{2} + \frac{(3/4-2)}{2(3/4)} = \frac{3}{2} - \frac{5}{6} = \frac{2}{3} \\ ^2P_{3/2} : L=1, S=\frac{1}{2}, J=\frac{3}{2} &\implies g_J = \frac{3}{2} + \frac{3/4-2}{2(15/4)} = \frac{3}{2} - \frac{1}{6} = \frac{4}{3}. \end{aligned}$$

And, the energy of each state is given by $E = \mu_B g_J m_J B$.

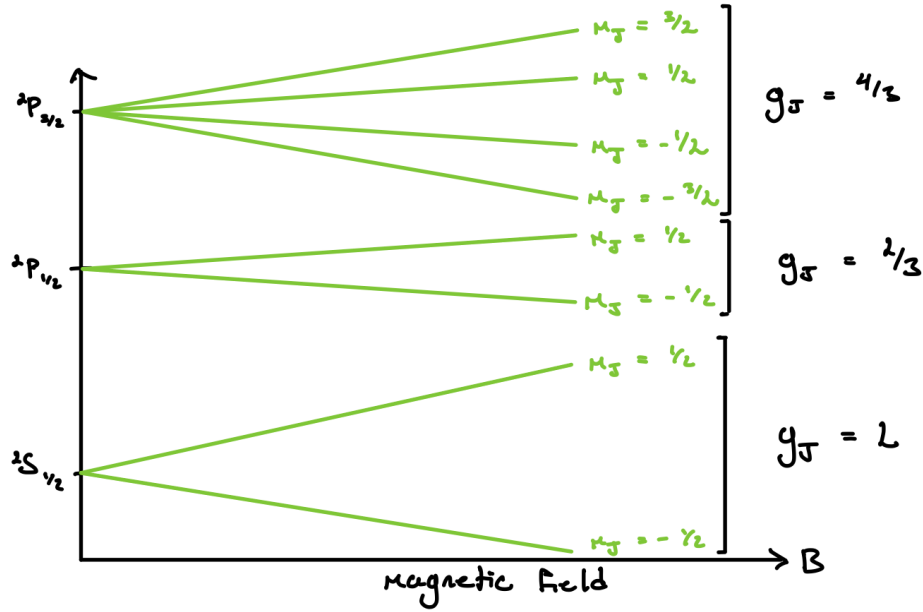


Figure 8: $^2S_{1/2}$, $^2P_{1/2}$ and $^2P_{3/2}$ energy states as a function of the magnetic field.

Observe that as B increases, E increases or decreases depending on the sign of m_J . Next, if we do consider the hyperfine interaction, we have

$$\boldsymbol{\mu} = -g_J \mu_B \mathbf{J} + g_I \mu_N \mathbf{I}.$$

If the Zeeman shift is smaller than the hyperfine splitting, we can use eigenstates $|\cdots F, m_F\rangle$ and apply the projection theorem to this magnetic moment,

$$\langle \boldsymbol{\mu} \cdot \mathbf{B} \rangle = \frac{\langle \boldsymbol{\mu} \cdot \mathbf{F} \rangle}{F(F+1)} \langle \mathbf{F} \cdot \mathbf{B} \rangle.$$

Since $\mu_N \ll \mu_B$, we can also approximate $\boldsymbol{\mu} \approx g_J \mu_B \mathbf{J}$, so

$$g_J \mu_B \frac{\langle \mathbf{J} \cdot \mathbf{F} \rangle}{F(F+1)} = g_F \mu_B m_F B$$

where

$$\begin{aligned} g_F &= g_J \frac{\langle \mathbf{J} \cdot \mathbf{F} \rangle}{F(F+1)} \\ &= g_J \frac{\langle \mathbf{J} \cdot \mathbf{J} \rangle + \langle \mathbf{J} \cdot \mathbf{I} \rangle}{F(F+1)} \\ &= g_J \left[\frac{\langle \mathbf{J}^2 + \mathbf{F}^2/2 - \mathbf{J}^2/2 - \mathbf{I}^2/2 \rangle}{F(F+1)} \right] \\ &= g_J \left[\frac{1}{2} + \frac{J(J+1) - I(I+1)}{2F(F+1)} \right] \\ g_F &= \frac{g_J}{2} \left[1 + \frac{J(J+1) - I(I+1)}{F(F+1)} \right]. \end{aligned}$$

We consider the ground state of alkali atoms as an example, specifically ^{87}Rb . The ground state has $L = 0$, and for a single outer electron we have $S = 1/2$ so that $J = 1/2$. The nuclear spin has a value $I = 3/2$, so that $F = 3/2 + 1/2 = 2$ or $F = 3/2 - 1/2 = 1$. We wish to find g_J and g_F . $g_J = 2$ as for the $^2S_{1/2}$ state above, and

$$g_F = \frac{g_J}{2} \left[1 + \frac{(1/2)(3/2) - (3/2)(5/2)}{F(F+1)} \right].$$

For our F values, we have

$$F = 1 : g_F = \left[1 + \frac{3/4 - 15/4}{2} \right] = -\frac{1}{2}$$

$$F = 2 : g_F = \left[1 + \frac{-12/4}{6} \right] = \frac{1}{2}.$$

The energy in a small magnetic field, of course, is given by $E_z = \mu_B g_F m_F B$.

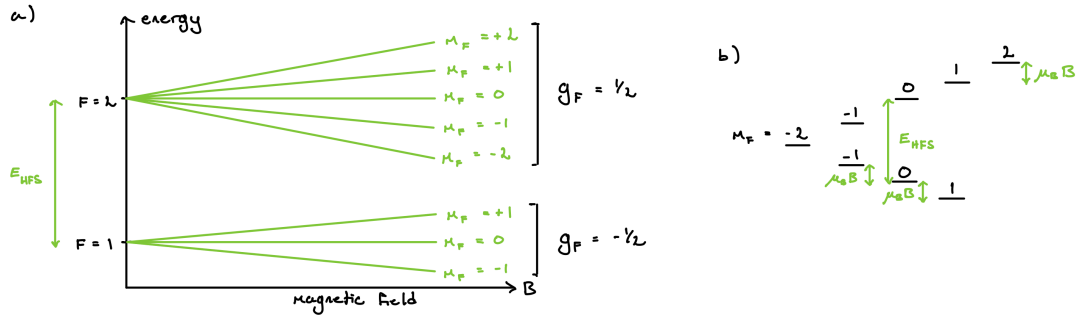


Figure 9: a) Energy function of the $F = 1, 2$ state as a function of the magnetic field, B . b) Alternative view of the same function.

1.7 Application: Magnetic Trapping

Many atomic physics experiments use magnetic trapping to confine atoms to a particular region of space. We consider the popular atom ^{87}Rb .

The Idea: atoms with $g_F m_F > 0$ are "magnetically trappable" within a magnetic-field minimum. Though Maxwell's equations prevent a global maximum of field, a global minimum is possible. We consider a pair of *anti-Helmholtz* coils

Observe in Fig. 10 that the magnetic field amplitude approaches zero at the centre, since the fields from the two coils cancel each other. We can explicitly calculate this result by determining the magnetic field

$$\mathbf{B} = -bx\mathbf{e}_x - by\mathbf{e}_y + 2bz\mathbf{e}_z$$

$$\nabla \cdot \mathbf{B} = -b - b + 2b = 0$$

which is zero as expected.

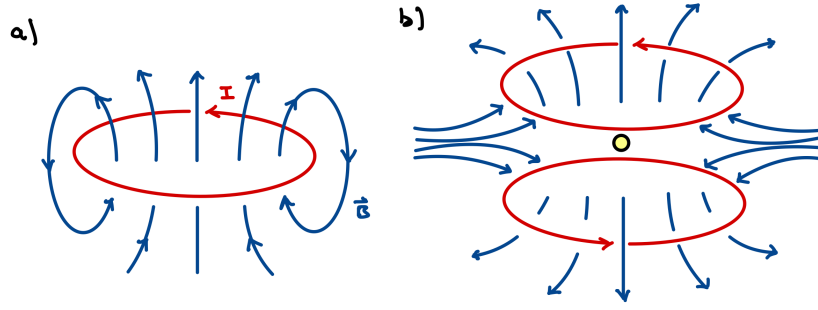


Figure 10: a) Single Helmholtz coil showing the current and magnetic field induced. b) Anti-Helmholtz coil setup showing the "magnetic trap" induced by the opposing currents.

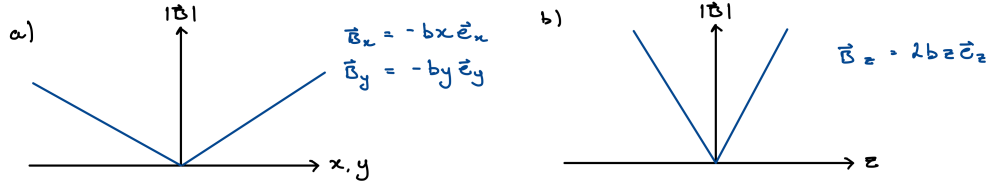


Figure 11: Magnitude against position of the magnetic field for the anti-Helmholtz coil setup. a) Plots for x and y direction, with the associated axis field. b) Plot for the z direction, with the associated axis field.

Next, we will assume that the atoms move slowly enough that their magnetic moments "follow" the local field direction, so the local field always defines the quantization axis.

$$E = m_F g_F \mu_B |\mathbf{B}| \implies E(x, y, z) = m_F g_F \mu_B b \sqrt{x^2 + y^2 + 4z^2}.$$

Observe that E has a minimum at $(x, y, z) = (0, 0, 0)$ for $m_F g_F > 1$, so the atom's energy is minimized at this point. However, is this strong enough to counteract the effect of gravity? We know that the force in the vertical direction must exceed gravity. Specifically, we determine

$$\mathbf{F} = \nabla E \implies F_z = \partial_z E = \frac{8z m_F g_F \mu_B b}{2\sqrt{x^2 + y^2 + 4z^2}}.$$

Along the central axis, $x = y = 0$, so

$$F_z = \frac{4z m_F g_F \mu_B b}{\sqrt{0 + 0 + 4z^2}} = 2b m_F g_F \mu_B.$$

This must at the very least be equal to $F_g = mg$. We can calculate explicitly the force experienced by an ^{87}Rb atom, with $L = 0$, $S = J = 1/2$, and $I = 3/2$, so that $F = 1, 2$, and

$$g_F = \frac{g_J}{2} \left[1 + \frac{J(J+1) - I(I+1)}{F(F+1)} \right].$$

Then,

$$\begin{aligned} F = 1 &\implies 1 + \frac{(1/2)(3/2) - (3/2)(5/2)}{2} = 1 + \frac{-12/4}{2} = -\frac{1}{2} \\ F = 2 &\implies 1 + \frac{(1/2)(3/2) - (3/2)(5/2)}{6} = 1 - \frac{12/4}{6} = \frac{1}{2}. \end{aligned}$$

The strongest force will be for the largest (positive) product of $m_F g_F$, so

$$F = 2, m_F = 2 \implies m_F g_F = 1.$$

We ask at what magnetic field gradient is $F_z > F_g$:

$$2b m_F g_F \mu_B > mg$$

$$\begin{aligned} b &\geq \frac{mg}{2m_F g_F \mu_B} = \frac{87(1.66 \times 10^{-27} \text{ kg})(9.81 \text{ m/s}^2)}{2(9.27 \times 10^{-24} \text{ J/T})} \\ &\geq 0.0764 \text{ T/m} \times 10000 \text{ G/T} \times \frac{1 \text{ m}}{100 \text{ cm}} \\ b &\geq 7.64 \text{ G/cm} \end{aligned}$$

which is very achievable.

1.8 Concluding Remarks: Atomic Hamiltonian

We will wrap up our discussion of atomic physics by thinking about the Hamiltonian we have put together. The total Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{so}} + \hat{H}_{\text{hfs}} + \hat{H}_z$$

where

$$\hat{H}_z = g_J \mu_B \mathbf{J} \cdot \mathbf{B} + g_I \mu_N \mathbf{I} \cdot \mathbf{B} \cong g_F \mu_B \mathbf{F} \cdot \mathbf{B}.$$

We have dealt with different quantum numbers in different situations and may sometimes need to determine when one set of quantum numbers may be the most appropriate compared to another. Consider the hydrogen atom's energy scale.

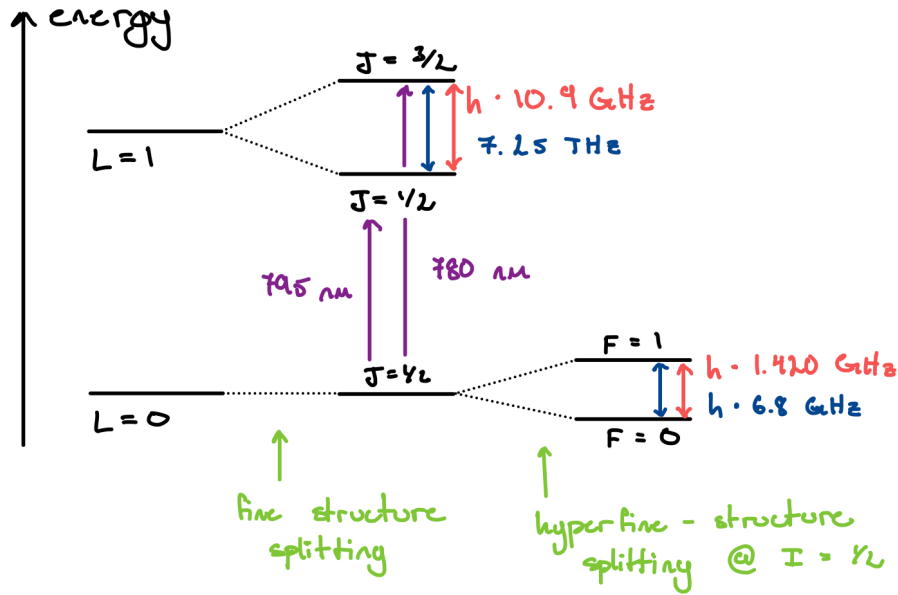


Figure 12: Hydrogen and rubidium energy structure with the two D-lines for rubidium. The rubidium splitting is coloured in blue, while the hydrogen splitting is coloured in pink.

The Zeeman energy scale for the hydrogen atom is

$$E_z = g_J m_J \mu_B |\mathbf{B}| = m_J \cdot 2(9.27 \times 10^{-24} \text{ J/T}) \cdot \frac{1 \text{ T}}{104 \text{ G}}$$

$$\frac{E_z}{h} = g_J \cdot 2.8 \text{ MHz/G} \cdot B[\text{G}].$$

So, for $g_J = 1/2$ (ground state), we have

$$\frac{E_z}{h} = 1.4 \text{ MHz/G} \cdot B[\text{G}].$$

This is approximately the hyperfine splitting for

$$B^* \approx \frac{E_z/h}{1.420 \text{ GHz}} \cong 1000 \text{ G}.$$

At this value, the Zeeman term is about equal to the hyperfine term, and any assumption about $|F\rangle, m_F$ being the "good quantum numbers" will not stand.

Now we will consider a heavier atom like ^{87}Rb . Here, the fine structure splitting is separated by

$$\Delta E_{\text{fs}} = \frac{hc}{780 \text{ nm}} - \frac{hc}{795 \text{ nm}} = h \cdot 7.25 \times 10^{12} \text{ Hz} = h \cdot 7.25 \text{ THz}.$$

The Zeeman energy scale of this is

$$\frac{E_z}{h} = 1.4 \text{ MHz/G} \cdot B[\text{G}]$$

with

$$B^* \approx \frac{6.8 \text{ GHz}}{1.4 \text{ MHz/G}} \approx 4900 \text{ G} \sim 0.5 \text{ T}.$$

Why exactly are these energy scales important? The assumption we made about the projection theorem and our ability to define eigenstates based on this relied on certain terms being small compared to others. Trouble may arise when, for example, a Zeeman term (due to a large magnetic field) larger than (or near) the hyperfine term means that defining F, m_F state is no longer valid.

In considering the case where the magnetic field may be large, the electronic part (first term) of the Zeeman Hamiltonian will dominate

$$\hat{H}_z = g_J \mu_B \mathbf{J} \cdot \mathbf{B} + g_I \mu_B \mathbf{I} \cdot \mathbf{B}.$$

Comparing this to the hyperfine term,

$$\hat{H}_{\text{Hfs}} = A_{\text{hfs}} \mathbf{I} \cdot \mathbf{J}.$$

Notably, eigenstates of $\mathbf{I} \cdot \mathbf{J}$ are not simultaneous eigenstates of $\mathbf{J} \cdot \mathbf{B} = J_z B$. In the extremes, we can say

- (1) if $\langle \hat{H}_{\text{hfs}} \rangle \gg \langle \hat{H}_z \rangle$, then F, m_F are "good quantum numbers" and we should use $|F, m_F\rangle$;
- (2) if $\langle \hat{H}_z \rangle \gg \langle \hat{H}_{\text{hfs}} \rangle$, then J, I, m_J , and m_I are "good quantum numbers" and we can use eigenstates $|m_I m_J\rangle$.

Suppose we have the between case, where $\langle \hat{H}_{\text{hfs}} \rangle$ is not necessarily much greater or less than $\langle \hat{H}_z \rangle$. For the ground state of hydrogen, we may have $L = 0$, $S = I = J = 1/2$, for $F = 0, 1$. Indeed, this gives us four basis states

$$|F, m_F\rangle \rightarrow \begin{cases} |1, 1\rangle \\ |1, 0\rangle \\ |1, -1\rangle \\ |0, 0\rangle. \end{cases}$$

We can decompose these states into their $|m_I m_J\rangle$ equivalents.

$$\begin{aligned} |F = 1, m_F = 1\rangle &= (-1)\sqrt{2+1} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 1 \\ \frac{1}{2} & \frac{1}{2} & -1 \end{pmatrix} |m_I = 1/2, m_J = 1/2\rangle \\ &= (-1)\sqrt{3} \left(-\frac{1}{\sqrt{3}} \right) |1/2, 1/2\rangle \\ &= |m_I = 1/2, m_J = 1/2\rangle. \\ |F = 1, m_F = 0\rangle &= (-1)^0 \sqrt{3} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{pmatrix} |m_I = 1/2, m_J = 1/2\rangle \\ &\quad + (-1)^0 \sqrt{3} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 1 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} |m_I = -1/2, m_J = 1/2\rangle \\ &= \frac{1}{\sqrt{2}} |m_I = 1/2, m_J = -1/2\rangle + \frac{1}{\sqrt{2}} |m_I = -1/2, m_J = 1/2\rangle \\ |F = 1, m_F = -1\rangle &= |m_I = -1/2, m_J = -1/2\rangle \\ |F = 0, m_F = 0\rangle &= \frac{1}{\sqrt{2}} |m_I = 1/2, m_J = -1/2\rangle - \frac{1}{\sqrt{2}} |m_I = -1/2, m_J = 1/2\rangle. \end{aligned}$$

For the $|F = 1, m_F = -1\rangle$ state, we have only $m_I = m_J = -1/2$ that satisfies the rule $m_I + m_J = m_F$. For the $|F = 0, m_F = 0\rangle$ state, we require m_I and m_J such that it is orthogonal to the other states.

Or, if we want to write them in another way,

$$\begin{aligned} |m_I = 1/2, m_J = 1/2\rangle &= |F = 1, m_F = 1\rangle \\ |m_I = 1/2, m_J = -1/2\rangle &= \frac{1}{\sqrt{2}} |F = 1, m_F = 0\rangle + \frac{1}{\sqrt{2}} |F = 0, m_F = 0\rangle \\ |m_I = -1/2, m_J = 1/2\rangle &= \frac{1}{\sqrt{2}} |F = 1, m_F = 0\rangle - \frac{1}{\sqrt{2}} |F = 0, m_F = 0\rangle \\ |m_I = -1/2, m_J = -1/2\rangle &= |F = 1, m_F = -1\rangle. \end{aligned}$$

Our Hamiltonian of interest is

$$\hat{H}_{\text{hfs}} + \hat{H}_z = A_{\text{hfs}} \mathbf{I} \cdot \mathbf{J} + g_J \mu_B B \hat{J}_z.$$

In the $|F, m_F\rangle$ basis, we can write this in matrix form using the states

$$\begin{pmatrix} |1, 1\rangle \\ |1, 0\rangle \\ |0, 0\rangle \\ |1, -1\rangle \end{pmatrix}$$

and

$$\begin{aligned}
\langle \mathbf{I} \cdot \mathbf{J} \rangle &= \frac{1}{2}[F(F+1) - J(J+1) - I(I+1)] \\
&= \frac{1}{2}\left[2 - \frac{3}{4} - \frac{3}{4}\right] = \frac{1}{4} \quad (\text{for } F = 1) \\
&= \frac{1}{2}\left[0 - \frac{3}{4} - \frac{3}{4}\right] = -\frac{3}{4} \quad (\text{for } F = 0).
\end{aligned}$$

So, the hyperfine-splitting Hamiltonian is

$$\hat{H}_{\text{hfs}} = \begin{pmatrix} A/4 & 0 & 0 & 0 \\ 0 & A/4 & 0 & 0 \\ 0 & 0 & -3A/4 & 0 \\ 0 & 0 & 0 & A/4 \end{pmatrix}, \quad A = A_{\text{hfs}}.$$

However, as a trade-off, $\hat{H}_z$ is more complicated in this basis. We have

$$\begin{aligned}
\langle 1, 1 | \hat{H}_z | 1, 1 \rangle &= \langle m_I = 1/2, m_J = 1/2 | \hat{H}_z | m_I = 1/2, m_J = 1/2 \rangle = g_J \left(\frac{1}{2} \right) \mu_B B = \mu_B B \\
\langle 1, -1 | \hat{H}_z | 1, -1 \rangle &= -\mu_B B \\
\langle 1, 0 | \hat{H}_z | 1, 0 \rangle &= \frac{1}{2} [\langle 1/2, -1/2 | \hat{H}_z | 1/2, -1/2 \rangle + \langle -1/2, 1/2 | \hat{H}_z | -1/2, 1/2 \rangle] = 0 \\
\langle 1, 0 | \hat{H}_z | 1, 0 \rangle &= g_J \mu_B B \times \frac{1}{2} (\langle -1/2, 1/2 | + \langle 1/2, -1/2 |) \hat{J}_z (| -1/2, 1/2 \rangle + | 1/2, -1/2 \rangle) \\
&= \frac{1}{2} g_J \mu_B B [\langle -1/2, 1/2 | \hat{J}_z | -1/2, 1/2 \rangle + \langle -1/2, 1/2 | \hat{J}_z | 1/2, -1/2 \rangle \\
&\quad + \langle 1/2, -1/2 | \hat{J}_z | -1/2, 1/2 \rangle + \langle 1/2, -1/2 | \hat{J}_z | 1/2, -1/2 \rangle] \\
&= \frac{1}{2} g_J \mu_B B [\langle -1/2, 1/2 | -1/2, 1/2 \rangle / 2 - \langle -1/2, 1/2 | 1/2, -1/2 \rangle / 2 \\
&\quad + \langle 1/2, -1/2 | -1/2, 1/2 \rangle / 2 - \langle 1/2, -1/2 | 1/2, -1/2 \rangle / 2] \\
&= \frac{1}{2} g_J \mu_B B \left[\frac{1}{2} - \frac{1}{2} \right] \\
&= 0 \\
\langle 0, 0 | \hat{H}_z | 0, 0 \rangle &= g_J \mu_B B \times \frac{1}{2} (\langle 1/2, -1/2 | - \langle -1/2, 1/2 |) \hat{J}_z (| 1/2, -1/2 \rangle - | -1/2, 1/2 \rangle) \\
&= 0 \\
\langle 1, 0 | \hat{H}_z | 0, 0 \rangle &= g_J \mu_B B \times \frac{1}{2} (\langle -1/2, 1/2 | + \langle 1/2, -1/2 |) \hat{J}_z (| 1/2, -1/2 \rangle - | -1/2, 1/2 \rangle) \\
&= \frac{1}{2} g_J \mu_B B \left(-\langle -1/2, 1/2 | \hat{J}_z | -1/2, 1/2 \rangle + \langle 1/2, -1/2 | \hat{J}_z | 1/2, -1/2 \rangle \right) \\
&= \frac{1}{2} g_J \mu_B B \left(-\frac{1}{2} \langle -1/2, 1/2 | -1/2, 1/2 \rangle - \frac{1}{2} \langle 1/2, -1/2 | 1/2, -1/2 \rangle \right) \\
&= \frac{1}{2} (2) \mu_B B (-1) \\
&= -\mu_B B. \\
\langle 0, 0 | \hat{H}_z | 1, 0 \rangle &= -\mu_B B \quad (\text{similar math as above}).
\end{aligned}$$

Thus, we arrive at the Zeeman Hamiltonian

$$\hat{H}_z = \begin{pmatrix} \mu_B B & 0 & 0 & 0 \\ 0 & 0 & -\mu_B B & 0 \\ 0 & -\mu_B B & 0 & 0 \\ 0 & 0 & 0 & -\mu_B B \end{pmatrix}.$$

Then, the combined Hamiltonian is

$$\hat{H}_{\text{hfs}} + \hat{H}_z = \begin{pmatrix} A/4 + \mu_B B & 0 & 0 & 0 \\ 0 & A/4 & -\mu_B B & 0 \\ 0 & -\mu_B B & -3A/4 & 0 \\ 0 & 0 & 0 & A/4 - \mu_B B \end{pmatrix}.$$

The "true eigenstates" of the combined Hamiltonian are found by diagonalizing the matrix. To do this, we need only to diagonalize the non-diagonal sub-matrix (which uses bases $|F=1, m_F=0\rangle$ and $|F=0, m_F=0\rangle$).

$$\hat{H}' = \begin{pmatrix} A/4 & -\mu_B B \\ -\mu_B B & -3A/4 \end{pmatrix} \rightarrow \begin{vmatrix} A/4 - \lambda & -\mu_B B \\ -\mu_B B & -3A/4 - \lambda \end{vmatrix} = 0.$$

This results in

$$\lambda = -\frac{A}{4} \pm \frac{A}{2} \left[1 + \left(\frac{2\mu_B B}{A} \right)^2 \right]^{1/2}.$$

We make a quick note here to say that the term $(2\mu_B B/A)^2$ is approximately the relative size of the Zeeman effect to the hyperfine splitting terms. Then, the eigenenergies are

$$E_{\pm} = -\frac{A_{\text{hfs}}}{4} \pm \frac{A_{\text{hfs}}}{2} \left[1 + \left(\frac{2\mu_B B}{A_{\text{hfs}}} \right)^2 \right]^{1/2}.$$

Now we must look at the high and low-field limits of our eigenenergies. In the low-field limit, we have $A_{\text{hfs}}/2 \gg 2\mu_B B$ so that

$$E_{\pm} \cong -\frac{A}{4} \pm \frac{A}{2} \left[1 + \left(\frac{2\mu_B B}{A} \right)^2 \right] = -\frac{A}{4} \pm \frac{A}{2} \pm \frac{2(\mu_B B)^2}{A}.$$

Thus,

$$E_+ = \frac{A}{4} + \frac{2(\mu_B B)^2}{A}, \quad E_- = -\frac{3A}{4} - \frac{2(\mu_B B)^2}{A}.$$

Our eigenstates are

$$|\psi_+\rangle : \hat{H} \begin{pmatrix} a \\ b \end{pmatrix} = E_+ \begin{pmatrix} a \\ b \end{pmatrix}$$

to give

$$\begin{aligned} \frac{A}{4}a - \mu_B Bb &= \left[\frac{A}{4} + \frac{2(\mu_B B)^2}{A} \right] a \\ -\mu_B Ba - \frac{3A}{4}b &= \left[\frac{A}{4} + \frac{2(\mu_B B)^2}{A} \right] b \\ b &= -2 \left(\frac{\mu_B B}{A} \right) a. \end{aligned}$$

Thus, the plus eigenstate is

$$|\psi_+\rangle = \frac{1}{N} \begin{pmatrix} 1 \\ -2\mu_B B/A \end{pmatrix} = \frac{1}{\sqrt{1 + 4(\mu_B B/A)^2}} \begin{pmatrix} 1 \\ -2\mu_B B/A \end{pmatrix}$$

where N is the normalization factor. We can rewrite this result as

$$|\psi_+\rangle = \frac{1}{\sqrt{1+x^2}} |F=1, m_F=0\rangle - \frac{x}{\sqrt{1+x^2}} |F=0, m_F=0\rangle, \quad x = \frac{2\mu_B B}{A}$$

and x is a small parameter. Similarly, we can, for $|\psi_-\rangle$, determine

$$\begin{aligned} \frac{A}{4}a - \mu_B Bb &= \left[-\frac{3A}{4} - \frac{2(\mu_B B)^2}{A} \right] a \\ -\mu_B Ba - \frac{3A}{4}b &= \left[-\frac{3A}{4} - \frac{2(\mu_B B)^2}{A} \right] b \\ a &= \frac{2\mu_B B}{A}b = xb. \end{aligned}$$

Thus,

$$|\psi_-\rangle = \frac{x}{\sqrt{1+x^2}} |F=1, m_F=0\rangle + \frac{1}{\sqrt{1+x^2}} |F=0, m_F=0\rangle.$$

Inspecting both $|\psi_+\rangle$ and $|\psi_-\rangle$, since x is a small parameter, for $2\mu_B B \ll A/2$, we have

$$|\psi_+\rangle \approx |F=1, m_F=0\rangle, \quad |\psi_-\rangle \approx |F=0, m_F=0\rangle.$$

The $|F, m_F\rangle$ basis is a "good" for low magnetic fields (see Fig. 13a). In fact, the true eigenstates are approximately the $|F, m_F\rangle$ eigenstates.

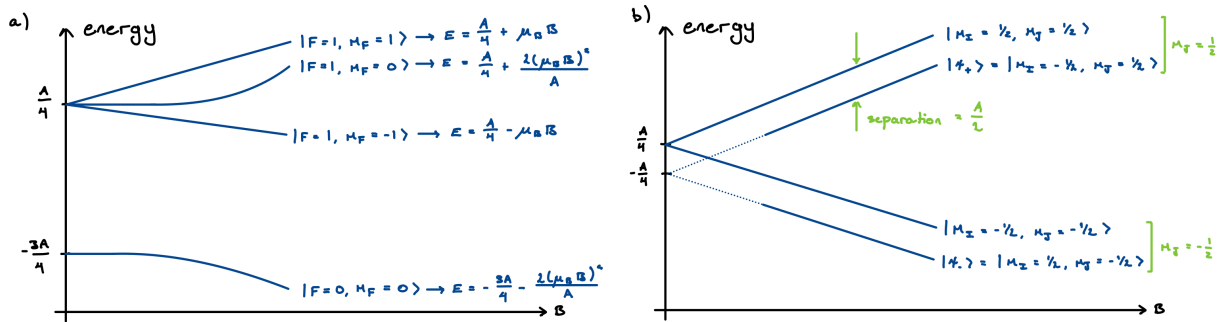


Figure 13: Sketch of the atomic Hamiltonian energy as a function of magnetic field. a) Low-field limit. b) High-field limit.

In the high-field limit, we have

$$\begin{aligned}
E_{\pm} &= -\frac{A}{4} \pm \frac{1}{2} [A^2 + (2\mu_B B)^2]^{1/2} \\
&= -\frac{A}{4} \pm \frac{1}{2} (2\mu_B B) \left[\frac{A^2}{(2\mu_B B)^2} + 1 \right]^{1/2} \\
&\approx -\frac{A}{4} \pm \mu_B B \left[1 + \frac{1}{2} \frac{A^2}{(4\mu_B B)^2} \right] \\
&\approx -\frac{A}{4} \pm \mu_B B + \frac{A^2}{8\mu_B B} \\
&= -\frac{A}{4} \pm \mu_B B
\end{aligned}$$

where we ignore the last term for large B . The eigenstates for these eigenenergies are

$$\begin{aligned}
|\psi_+\rangle : -\frac{A}{4}a - \mu_B Bb &= \left(-\frac{A}{4} + \mu_B B \right) a \implies a = -b \\
|\psi_+\rangle &= \frac{1}{\sqrt{2}} |F=1, m_F=0\rangle - \frac{1}{\sqrt{2}} |F=0, m_F=0\rangle \\
|\psi_-\rangle : -\frac{A}{4}a - \mu_B Bb &= \left(-\frac{A}{4} - \mu_B B \right) b \implies a = b \\
|\psi_-\rangle &= \frac{1}{\sqrt{2}} |F=1, m_F=0\rangle + \frac{1}{\sqrt{2}} |F=0, m_F=0\rangle.
\end{aligned}$$

Looking at these eigenstates, we can write

$$|\psi_+\rangle = |m_I = -1/2, m_J = 1/2\rangle, \quad |\psi_-\rangle = |m_I = 1/2, m_J = -1/2\rangle.$$

So, in the high-field limit, the $|m_I, m_J\rangle$ eigenstates are the approximate eigenstates, and I and J are the "good quantum numbers" (see Fig. 13b).

We put these results together in Table 4 and Fig. 14.

Low Field	Intermediate Field	High Field
$ F, m_F\rangle$ appropriate $E \approx m_F g_F \mu_B B$	Eigenstates not simply expressed Neither $\{F, m_F\}$ or $\{m_I, m_J\}$ hold	$ m_I, m_J\rangle$ appropriate $E = g_J m_J \mu_B B + m_I A$

Table 4: Summarized results of the high and low-field limits of the atomic Hamiltonian under a magnetic field.

We make a note on the added $m_I A$ term on the energy for the high-field regime. This shift term occurs when we diagonalize the m_I, m_J basis. Indeed, we can rewrite

$$\mathbf{I} \cdot \mathbf{J} = \frac{1}{2} [\mathbf{I}_+ \mathbf{J}_- + \mathbf{I}_- \mathbf{J}_+] + \mathbf{I}_z \mathbf{J}_z.$$

For our states, we have

$$\langle m_I = 1/2, m_J = 1/2 | \frac{1}{2} [\mathbf{I}_+ \mathbf{J}_- + \mathbf{I}_- \mathbf{J}_+] + \mathbf{I}_z \mathbf{J}_z | m_I = 1/2, m_J = 1/2 \rangle$$

so

$$\begin{aligned}\langle 1/2, 1/2 | \mathbf{I} \cdot \mathbf{J} | 1/2, 1/2 \rangle &= m_I m_J = \frac{1}{4} \\ \langle 1/2, -1/2 | \mathbf{I} \cdot \mathbf{J} | 1/2, -1/2 \rangle &= -\frac{1}{4} \\ \langle -1/2, 1/2 | \mathbf{I} \cdot \mathbf{J} | -1/2, 1/2 \rangle &= -\frac{1}{4} \\ \langle -1/2, -1/2 | \mathbf{I} \cdot \mathbf{J} | -1/2, -1/2 \rangle &= \frac{1}{4}.\end{aligned}$$

Thus, the hyperfine splitting is

$$\begin{aligned}E_{\text{hfs}} &= \frac{A}{4} \text{ for } |m_I = 1/2, m_J = 1/2\rangle \text{ and } |m_I = -1/2, m_J = -1/2\rangle \\ &= -\frac{A}{4} \text{ for } |m_I = 1/2, m_J = -1/2\rangle \text{ and } |m_I = -1/2, m_J = 1/2\rangle.\end{aligned}$$

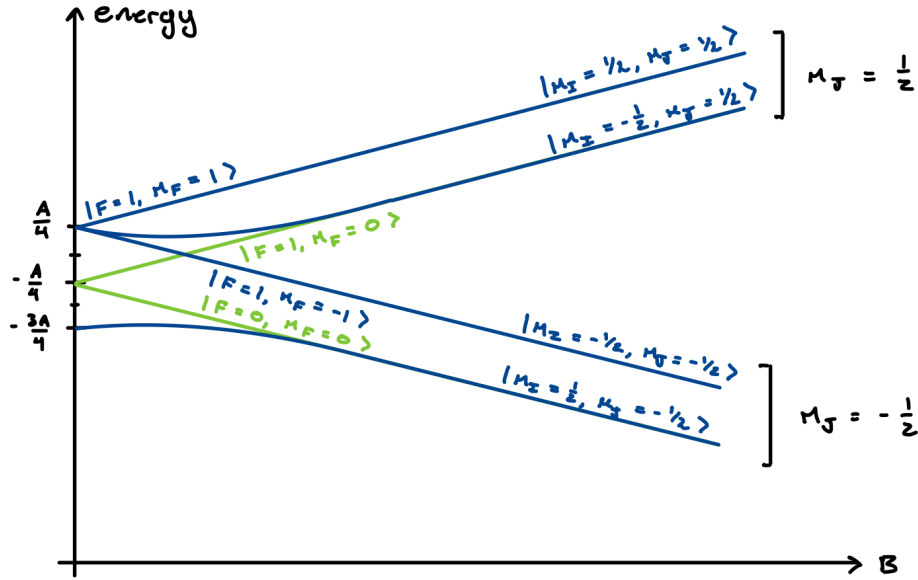


Figure 14: Complete sketch of atomic Hamiltonian energies as a function of the magnetic field.

2 Semiclassical Light-Matter Interactions

In this section, we study the intersection of classical light (using a field description) with quantum matter (atoms or other structures with discrete spectra). In general, this is a very good picture that explains most phenomena.

2.1 Introduction

We begin by revisiting the interaction Hamiltonian for a plane electromagnetic wave interacting with quantum matter. To describe the EM wave,¹⁰ we require the wavevector $\mathbf{k}$ with associated magnitude $k = |\mathbf{k}|$, and angular frequency $\omega = ck$. In free space, we have the relationships

$$k = \frac{2\pi}{\lambda}, \quad \omega = ck.$$

To capture the full effect of the EM wave, we will work with the vector potential $\mathbf{A}(\mathbf{r}, t)$, recalling that

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= -\frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t} - \nabla \phi \\ \mathbf{B}(\mathbf{r}, t) &= \nabla \times \mathbf{A} \end{aligned}$$

where ϕ is the *scalar potential*.

Assuming a suitable choice of gauge, we can make the scalar potential zero ($\phi = 0$) and

$$\mathbf{A}(\mathbf{r}, t) = A_0 \mathbf{e}_x e^{i(kz - \omega t)} + A_0^* \mathbf{e}_x e^{-i(kz - \omega t)}$$

where z is the assumed propagation direction and $\mathbf{A}$ is a real quantity. Thus, we have

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= i\omega A_0 \mathbf{e}_x e^{i(kz - \omega t)} - i\omega A_0^* \mathbf{e}_x e^{-i(kz - \omega t)} \\ \mathbf{B}(\mathbf{r}, t) &= ik A_0 \mathbf{e}_y e^{i(kz - \omega t)} - ik A_0^* \mathbf{e}_y e^{-i(kz - \omega t)}. \end{aligned}$$

Note that $\mathbf{e}_x$ is on the same plane as $\mathbf{A}(\mathbf{r}, t)$. We shall choose an origin such that A_0 is purely imaginary, so $A_0^* = -A_0$ and define the nominal electric and magnetic field strengths

$$\frac{E_0}{2} = i\omega A_0, \quad \frac{B_0}{2} = ik A_0.$$

This leaves us with

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= E_0 \cos(kz - \omega t) \mathbf{e}_x \\ \mathbf{B}(\mathbf{r}, t) &= B_0 \cos(kz - \omega t) \mathbf{e}_y. \end{aligned}$$

Next, we move on to the interaction Hamiltonian. A very general expression for a charged particle in a gauge potential is

$$\hat{H} = \frac{1}{2m} [\hat{p} - q\mathbf{A}(\mathbf{r}, t)]^2 + V(\mathbf{r}) - \frac{q}{m} \mathbf{s} \cdot \mathbf{B}(\mathbf{r}, t)$$

where $\mathbf{s}$ includes the spin degree of freedom. This Hamiltonian can be written as a combination of the pure atomic Hamiltonian plus the interaction part, so

$$\hat{H} = \hat{H}_{\text{at}} + \hat{H}_{\text{int}}.$$

¹⁰We will colloquially call this "light," recognizing that the description applies outside of the optical spectrum.

The atomic Hamiltonian is given by

$$\hat{H}_{\text{at}} = \frac{\hat{\mathbf{p}}^3}{2m} + V(\mathbf{r}) = -\frac{\hbar^2 \nabla^2}{2m} + V(\mathbf{r})$$

and the interaction Hamiltonian is given by

$$\hat{H}_{\text{int}} = -\frac{q}{2m}[\mathbf{A}(\mathbf{r}, t) \cdot \hat{\mathbf{p}} + \hat{\mathbf{p}} \cdot \mathbf{A}(\mathbf{r}, t)] + \frac{q^2}{2m}[\mathbf{A}(\mathbf{r}, t)]^2 - \frac{q}{m}\mathbf{s} \cdot \mathbf{B}(\mathbf{r}, t).$$

Using the conventions defined above, $\mathbf{A}(\mathbf{r}, t) \propto \mathbf{e}_x$. Then, we consider $\mathbf{p} = -i\hbar \nabla$ acting on some state:

$$\begin{aligned} \mathbf{A}(\mathbf{r}, t) \cdot \hat{\mathbf{p}} |\psi\rangle &= A_0 \sin(kz - \omega t) [\mathbf{e}_x \cdot (-i\hbar \nabla)] |\psi(\mathbf{r}, t)\rangle \\ &= -i\hbar A_0 \sin(kz - \omega t) \nabla_x |\psi(\mathbf{r}, t)\rangle \\ \hat{\mathbf{p}} \cdot \mathbf{A}(\mathbf{r}, t) |\psi\rangle &= -i\hbar (\nabla \cdot \mathbf{e}_x) A_0 \sin(kz - \omega t) |\psi(\mathbf{r}, t)\rangle \\ &= -i\hbar \nabla_x A_0 \sin(kz - \omega t) |\psi(\mathbf{r}, t)\rangle \\ &= -i\hbar A_0 \sin(kz - \omega t) \nabla_x |\psi(\mathbf{r}, t)\rangle \\ &= \mathbf{A}(\mathbf{r}, t) \cdot \hat{\mathbf{p}} |\psi\rangle. \end{aligned}$$

So,

$$\mathbf{A}(\mathbf{r}, t) \cdot \hat{\mathbf{p}} + \hat{\mathbf{p}} \cdot \mathbf{A}(\mathbf{r}, t) = 2\mathbf{A}(\mathbf{r}, t) \cdot \hat{\mathbf{p}}$$

which is a consequence of EM fields being transverse to propagation direction, since they must be in free space. Thus, our interaction Hamiltonian is

$$\hat{H}_{\text{int}} = -\frac{q}{m}[\mathbf{A}(\mathbf{r}, t) \cdot \hat{\mathbf{p}}] - \frac{q}{m}\mathbf{s} \cdot \mathbf{B} + \frac{q^2}{2m}[\mathbf{A}(\mathbf{r}, t)]^2.$$

We can generally neglect the last term for fields of reasonable strength, and, estimating the relative size of the other terms,

$$\begin{aligned} -\frac{q}{m}\mathbf{A}(\mathbf{r}, t) \cdot \hat{\mathbf{p}} &\sim \frac{q}{m}\langle p \rangle A_0 \approx \frac{q}{m} \left(\frac{h}{a_0} \right) A_0 \\ \frac{q}{m}\mathbf{s} \cdot \mathbf{B} &\sim \frac{q}{m} \hbar k A_0 \approx \frac{q}{m} \left(\frac{h}{\lambda} \right) A_0 \end{aligned}$$

where $p = h/\lambda dB \approx h/a_0$, and a_0 is the size of an atom. Here, the $\mathbf{A}$ term is generally larger by a factor of λ/a_0 , where λ is usually an optical (100's of nm) or larger scale, while $a_0 \sim 0.05$ nm. Thus, we rank terms in the Hamiltonian by size

$$\hat{H}_{\text{int}} = \hat{H}_I + \hat{H}_{II} = -\frac{q}{m}\mathbf{A}(\mathbf{r}, t) \cdot \hat{\mathbf{p}} - \frac{q}{m}\mathbf{s} \cdot \mathbf{B}(\mathbf{r}, t).$$

Consider first the dominant term,

$$\hat{H}_I = -\frac{q}{m} \left[A_0 e^{ikz} e^{-i\omega t} + A_0^* e^{-ikz} e^{i\omega t} \right] \hat{p}_x.$$

Since the spatial distance over which the interaction occurs is on the order of the atom's size, we can estimate

$$\begin{aligned} e^{ikz} &= 1 + ikz + \frac{(ikz)^2}{2} + \dots \\ &= 1 + i \frac{2\pi z}{\lambda} - \left(\frac{2\pi z}{\lambda} \right)^2 + \dots \\ &\approx 1 + i \left(\frac{2\pi a_0}{\lambda} \right) - \left(\frac{2\pi a_0}{\lambda} \right)^2 + \dots \\ &\approx 1 \end{aligned}$$

since $a_0 \ll \lambda$, as above. Thus, the simplified Hamiltonian is

$$\hat{H}_I \approx -\frac{q}{m}(A_0 e^{-i\omega t} + A_0^* e^{i\omega t})\hat{p}_x = -\frac{q}{m}\frac{E_0}{2i\omega}(e^{-i\omega t} - e^{i\omega t})\hat{p}_x = \frac{qE_0}{m\omega}\sin(\omega t)\hat{p}_x.$$

Note that we have used $A_0 = E_0/2i\omega$. The final step towards getting this expression in a "familiar form" is by considering the relationship between $\mathbf{r}$ and $\mathbf{p}$. Notably, the commutation relation $[\hat{r}, \hat{H}_0] = i\hbar\hat{p}/m$ will be useful.

The basis in which we will work in will be the one of eigenstates of $\hat{H}_0$, so we consider the operation of $\hat{p}$ for two such states, $|\psi_1\rangle$ and $|\psi_2\rangle$:

$$\begin{aligned}\langle\psi_2|\hat{p}|\psi_1\rangle &= \langle\psi_2|\frac{m}{2\hbar}[\hat{r}, \hat{H}_0]|\psi_1\rangle \\ &= -\frac{im}{\hbar}\langle\psi_2|\hat{r}\hat{H}_0|\psi_1\rangle + \frac{im}{\hbar}\langle\psi_2|\hat{H}_0\hat{r}|\psi_1\rangle \\ &= -\frac{im}{\hbar}[\langle\psi_2|\hat{r}E_1|\psi_1\rangle - \langle\psi_2|E_2\hat{r}|\psi_1\rangle] \\ &= \frac{im}{\hbar}(E_2 - E_1)\langle\psi_2|\hat{r}|\psi_1\rangle.\end{aligned}$$

Now, assuming that the interaction of interest is near resonance, then if $E_2 > E_1$ (absorption process), we have

$$\frac{(E_2 - E_1)}{\hbar} \approx \omega$$

and, if $E_1 > E_2$ (stimulated emission process), we have

$$\frac{(E_2 - E_1)}{\hbar} \approx -\omega.$$

In the Hamiltonian we are considering, the first term corresponds to absorption and the second to stimulated emission, as the $e^{\pm i\omega t}$ terms indicate, so for eigenstates

$$|\psi_1(t)\rangle \sim e^{-i\omega_1 t}|\psi_1\rangle, \quad |\psi_2(t)\rangle \sim e^{-i\omega_2 t}|\psi_2\rangle$$

we assign $\hat{p} \rightarrow im\omega\hat{r}$ for the absorption term and $\hat{p} \rightarrow -im\omega\hat{r}$ for the emission term. Thus,

$$\begin{aligned}\hat{H}_I &\cong -\frac{q}{2mi}\frac{E_0}{\omega}[im\omega\hat{x}e^{-i\omega t} - (-im\omega\hat{x})e^{i\omega t}] \\ &= -qE_0\hat{x}\frac{e^{-i\omega t} + e^{i\omega t}}{2} \\ &\cong -qE_0\hat{x}\cos(\omega t) \\ &= -q\mathbf{E}(t) \cdot \mathbf{r} \\ &\equiv \hat{E}_{E1}\end{aligned}$$

which is the electric (E) dipole (1) Hamiltonian. What about the other terms? We have

$$\hat{H}_{\text{int}} = \hat{H}_{E1} + (\hat{H}_I - \hat{H}_{E1}) + \hat{H}_{II}$$

where

$$\begin{aligned}\hat{H}_I - \hat{H}_{E1} &= -\frac{q}{m}\left\{A_0\left[ikz - \frac{(kz)^2}{2} + \dots\right]e^{-i\omega t} + A_0^*\left[-ikz - \frac{(kz)^2}{2} + \dots\right]e^{i\omega t}\right\}p_x \\ &= -\frac{q}{m}(ikA_0e^{-i\omega t} - ikA_0^*e^{i\omega t})zp_x + \frac{q}{m}\left(\frac{k^2}{2}A_0e^{-i\omega t} + \frac{k^2}{2}A_0^*e^{-i\omega t}\right)z^2p_x + \dots \\ &= -\frac{q}{m}B_0\cos(\omega t)zp_x + \dots\end{aligned}$$

We can write

$$zp_x = \frac{1}{2}(zp_x - xp_z) + \frac{1}{2}(zp_x + xp_z) = \frac{1}{2}l_y + \frac{1}{2}(zp_x + xp_z).$$

Thus,

$$\hat{H}_I - \hat{H}_{E1} = -\frac{q}{m}l_y B_0 \cos(\omega t) - \frac{q}{m}B_0 \cos(\omega t)(zp_x + xp_z).$$

Recall that the magnitude of $\hat{H}_{II}$ is

$$\frac{\langle \hat{H}_{II} \rangle}{\langle \hat{H}_I \rangle} \sim \frac{a_0}{\lambda}$$

which is the same order as the terms we just found

$$\langle \hat{H}_I - \hat{H}_{E1} \rangle \sim \frac{q}{2m}a_0 p k A_0 \sim \langle \hat{H}_I \rangle a_0 k \sim \langle \hat{H}_I \rangle \frac{a_0}{\lambda}.$$

So, we must consider the leading contribution from $\hat{H}_{II}$

$$\begin{aligned} \hat{H}_{II} &= -\frac{q}{m} \mathbf{s} \cdot \mathbf{B} \\ &= -\frac{q}{m} s_y B_0 \left[e^{i(kz - \omega t)} + e^{-i(kz - \omega t)} \right] \\ &= -\frac{q}{m} s_y B_0 \left\{ \left[1 + ikz + \frac{(kz)^2}{2} + \dots \right] e^{-i\omega t} + \left[1 - ikz + \frac{(kz)^2}{2} + \dots \right] e^{i\omega t} \right\} \\ &= -\frac{q}{m} s_y B_0 [\cos(\omega t) + ikz(\cdot) + \dots] \\ &\approx -\frac{q}{m} s_y B_0 \cos(\omega t). \end{aligned}$$

Thus, we are left with two contributions at this order

$$\hat{H}_I - \hat{H}_{E1} + \hat{H}_{II} \approx \hat{H}_{M1} + \hat{H}_{E2}$$

where

$$\hat{H}_{M1} = -\frac{q}{m}(l_y + 2s_y)B_0 \cos(\omega t)$$

is the "magnetic (M) dipole (1)" Hamiltonian, and

$$\hat{H}_{E2} = -\frac{q}{2mc}(zp_x + xp_z)E_0 \cos(\omega t)$$

is the electric (E) quadrupole (2) Hamiltonian.

2.2 Selection Rules

The selection rules for electric dipole (E1) are as discussed in §1 :

$$\begin{aligned} \Delta n &= \text{anything} \\ \Delta l &= \pm 1 \\ \Delta j &= 0, \pm 1, \quad \Delta m_j = 0, \pm 1 \quad (j = 0 \rightarrow j' = 0 \text{ not allowed}) \\ \Delta f &= 0, \pm 1, \quad \Delta m_f = 0, \pm 1 \quad (f = 0 \rightarrow f' = 0 \text{ not allowed}) \\ \Delta s &= 0 \\ \Delta i &= 0. \end{aligned}$$

For magnetic dipole transitions, the rules are different:

- (1) Neither $\hat{l}_y$ nor $\hat{s}_y$ changes the quantum number, l .

$$\Delta l = 0.$$

Or, in other words, there is no parity change.

- (2) $\hat{l}_y$ changes m_l by ± 1 if $\mathbf{B} \perp \mathbf{e}_z$ and $\Delta m_l = 0$ if $\mathbf{B} \parallel \mathbf{e}_z$.

$$\Delta m_l = 0, \pm 1.$$

- (3) $\hat{s}_y$ follows by the same argument as above, so $\Delta m_s = 0, \pm 1$.

- (4) When there is spin-orbit coupling, $\hat{l}_y$ and $\hat{s}_y$ do not commute with $\hat{\mathbf{J}}^2$, so $\hat{H}_\mu$ can connect states of different j but same l :

$$\Delta l = 0$$

$$\Delta j = 0, \pm 1, \quad \Delta m_j = 0, \pm 1.$$

In summary, we have the magnetic dipole transition rules:

$$\Delta j = 0, \pm 1 \quad (\text{but not } j = 0 \rightarrow j' = 0)$$

$$\Delta m_j = 0, \pm 1$$

$$\Delta f = 0, \pm 1 \quad (\text{but not } f = 0 \rightarrow f' = 0)$$

$$\Delta m_f = 0, \pm 1 \quad (\text{no parity change})$$

$$\Delta l = 0 \quad (\text{no change of configuration})$$

$$\Delta n = 0 \quad (\text{no change of configuration})$$

$$\Delta s = 0.$$

For electric quadrupole (E2) transitions, we have

$$zp_x + xp_z = \frac{m}{i\hbar} \left\{ \hat{z} [\hat{x}, \hat{H}_0] + [\hat{z}, \hat{H}_0] \hat{x} \right\} = \frac{m}{i\hbar} \left\{ \hat{z} \hat{x} \hat{H}_0 - \hat{x} \hat{z} \hat{H}_0 \right\}$$

$$\langle \psi_2 | \hat{H}_{\text{E2}} | \psi_1 \rangle = \frac{q}{2ic} \omega_{21} \langle \psi_2 | zx | \psi_1 \rangle E_0 \cos(\omega t).$$

Observe that the matrix element is proportional to zx . In terms of the spherical harmonics, zx is a linear superposition of $r^2 Y_{21}(\theta, \phi)$ and $r^2 Y_{2(-1)}(\theta, \phi)$. That is,

$$\begin{aligned} x &= r \sin \theta \cos \phi \\ z &= r \cos \theta \\ xz &= r^2 \sin \theta \cos \theta \cos \phi \\ &= r^2 \sin \theta \cos \theta \left(\frac{e^{i\phi} + e^{-i\phi}}{2} \right) \\ &= -\left(\frac{8\pi}{15} \right)^{1/2} r^2 [Y_{21}(\theta, \phi) - Y_{2(-1)}(\theta, \phi)] \end{aligned}$$

which means that matrix elements for $\hat{H}_{\text{E2}}$ have terms like

$$\int d\phi \int d\theta \sin \theta Y_{l_2 m_2}^*(\theta, \phi) Y_{2(\pm 1)}(\theta, \phi) Y_{l_1 m_1}(\theta, \phi)$$

and the only ways the integral is non-zero is where

$$\begin{aligned} -l^2 + 2 + l_1 &= 0, & \Delta l &= \pm 2 \\ -m_2 + 1m_1 &= 0, & \Delta m &= \pm 1 \end{aligned}$$

considering the different possible polarizations¹¹

$$\Delta l = 0, \pm 2, \quad \Delta m_l = 0, \pm 1, \pm 2.$$

Finally, note that we could keep expanding terms in the interaction Hamiltonian to find the electric octopole, magnetic quadrupole, etc, transitions. For the most part, however, we will consider E1 transitions, and, occasionally, M1 transitions.¹²

The relative importance of the terms decreases with increased "order" (see Fig. 15).

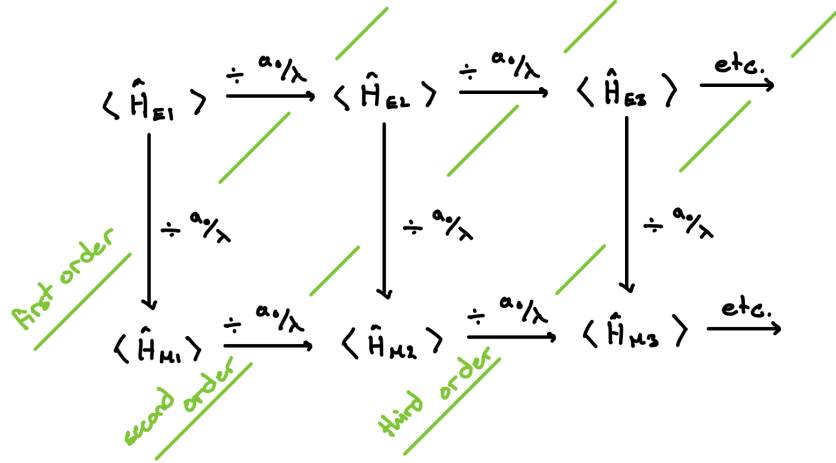


Figure 15: Relative importance of total atomic Hamiltonian as the order of terms increases.

2.3 The 2-Level Atom

A basic but powerful model of the light-matter interaction is to consider a two-level quantum atom interacting with monochromatic (single frequency) light, like that from a laser. This interaction is an excellent approximation for many atom-light interactions, and can be generalized to more complicated situations.

To begin, let us consider the time-dependent Schrödinger equation (TDSE):

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H}(t) |\Psi(t)\rangle, \quad \hat{H} = \hat{H}_{\text{at}} + \hat{H}_{\text{int}}.$$

We will work within the basis of two atomic eigenstates

$$\hat{H}_{\text{at}} |\Psi_1(t)\rangle = E_1 |\Psi_1(t)\rangle, \quad \hat{H}_{\text{at}} |\Psi_2(t)\rangle = E_2 |\Psi_2(t)\rangle$$

¹¹For more on selection rules, see Complement 2B from *Introduction to Quantum Optics* by Grynberg, Fabre, Aspect.

¹²For more info, see Complement A_{XIII} in *Quantum Mechanics, Volume II* by Cohen-Tannoudji et al.

The position-space representation is, for example,

$$\Psi_n(\mathbf{r}, t) = \psi_n(\mathbf{r})e^{-iE_n t/\hbar}$$

and an arbitrary state can be written as

$$\Psi(\mathbf{r}, t) = c_1(t)\psi_1(\mathbf{r})e^{-iE_1 t/\hbar} + c_2(t)\psi_2(\mathbf{r})e^{-iE_2 t/\hbar}$$

or, more abstractly, as

$$|\Psi(\mathbf{r}, t)\rangle = c_1(t)e^{-i\omega_1 t}|\psi_1\rangle + c_2(t)e^{-i\omega_2 t}|\psi_2\rangle$$

where

$$\omega_n = \frac{E_n}{\hbar}, \quad |c_1(t)|^2 + |c_2(t)|^2 = 1$$

and c_1 and c_2 are complex amplitudes. Let us take the E1 interaction Hamiltonian $\hat{H}_{\text{int}} \cong \hat{H}_{\text{E1}} = e\mathbf{r} \cdot \mathbf{E}_0 \cos(\omega t)$ and apply it to the TDSE:

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle &= (\hat{H}_{\text{at}} + \hat{H}_{\text{int}}) |\Psi(t)\rangle \\ i\hbar \frac{\partial}{\partial t} [c_1(t)e^{-i\omega_1 t}|\psi_1\rangle + c_2(t)e^{-i\omega_2 t}|\psi_2\rangle] &= E_1 c_1(t)e^{-i\omega_1 t}|\psi_1\rangle + e\mathbf{r} \cdot \mathbf{E}_0 \cos(\omega t) c_1(t)e^{-i\omega_1 t}|\psi_1\rangle \\ &\quad + E_2 c_2(t)e^{-i\omega_2 t}|\psi_2\rangle \\ &\quad + e\mathbf{r} \cdot \mathbf{E}_0 \cos(\omega t) c_2(t)e^{-i\omega_2 t}|\psi_2\rangle. \end{aligned}$$

Taking the inner product of both sides with $\langle\psi_1|$, we find

$$\begin{aligned} i\hbar \left[\frac{\partial c_1(t)}{\partial t} e^{-i\omega_1 t} + c_1(t)(-i\omega_1) e^{-i\omega_1 t} \right] \langle\psi_1|\psi_1\rangle &+ i\hbar \left[\frac{\partial c_2(t)}{\partial t} e^{-i\omega_2 t} + c_2(t)(-i\omega_2) e^{-i\omega_2 t} \right] \langle\psi_1|\psi_2\rangle \\ &= E_1 c_1(t)e^{-i\omega_1 t} \langle\psi_1|\psi_1\rangle + E_2 c_2(t)e^{-i\omega_2 t} \langle\psi_2|\psi_1\rangle + c_1(t)e^{-i\omega_1 t} e \cos(\omega t) \langle\psi_1|\mathbf{r} \cdot \mathbf{E}_0|\psi_1\rangle \\ &\quad + c_2(t)e^{-i\omega_2 t} e \cos(\omega t) \langle\psi_1|\mathbf{r} \cdot \mathbf{E}_0|\psi_2\rangle. \end{aligned}$$

Applying what we know about the matrix elements,

$$\begin{aligned} \langle\psi_1|\psi_1\rangle &= \langle\psi_2|\psi_2\rangle = 1 \\ \langle\psi_1|\psi_2\rangle &= \langle\psi_2|\psi_1\rangle = 0 \\ \langle\psi_1|\mathbf{r} \cdot \mathbf{E}_0|\psi_2\rangle &= 0 \\ \langle\psi_2|\mathbf{r} \cdot \mathbf{E}_0|\psi_1\rangle &\neq 0. \end{aligned}$$

We note that the $\mathbf{r}$ operator changes parity, so the matrix element $\langle\psi_1|\mathbf{r} \cdot \mathbf{E}_0|\psi_2\rangle$ is always zero, and the matrix element $\langle\psi_2|\mathbf{r} \cdot \mathbf{E}_0|\psi_1\rangle$ is zero if the selection rules are satisfied.

Thus, we have

$$i\hbar \left[\frac{\partial c_1(t)}{\partial t} e^{-i\omega_1 t} - i\omega_1 c_1(t) e^{-i\omega_1 t} \right] = E_1 c_1(t) e^{-i\omega_1 t} + c_2(t) e^{-i\omega_2 t} e E_0 \cos(\omega t) \langle\psi_1|\mathbf{r} \cdot \mathbf{e}_r|\psi_2\rangle$$

using $-i^2\hbar\omega_1 = E_1$. Then,

$$\begin{aligned} i\hbar \frac{\partial c_1(t)}{\partial t} e^{-i\omega_1 t} &= c_2(t) e^{-i\omega_2 t} e E_0 \cos(\omega t) \langle\psi_1|\mathbf{r} \cdot \mathbf{e}_r|\psi_2\rangle \\ i\hbar \dot{c}_1(t) &= c_2(t) e^{-i(\omega_2 - \omega_1)t} e E_0 \cos(\omega t) \langle\psi_1|\mathbf{r} \cdot \mathbf{e}_r|\psi_2\rangle. \end{aligned}$$

We follow the same process for an inner product with $\langle\psi_2|$, so

$$i\hbar\dot{c}_2(t) = c_1(t)e^{i(\omega_2-\omega_1)t}eE_0\cos(\omega t)\langle\psi_2|\mathbf{r}\cdot\mathbf{e}_r|\psi_1\rangle.$$

We define the *Rabi frequency*

$$\hbar\Omega_0 = \langle\psi_1|e\mathbf{r}\cdot\mathbf{E}_0|\psi_2\rangle$$

which leaves us with a set of two coupled equations

$$\begin{aligned} i\dot{c}_1(t) &= \Omega_0\cos(\omega t)c_2(t)e^{-i\omega_0 t} \\ i\dot{c}_2(t) &= \Omega_0^*\cos(\omega t)c_1(t)e^{i\omega_0 t} \end{aligned}$$

where

$$\omega_0 = \omega_2 - \omega_1 = \frac{E_2 - E_1}{\hbar}.$$

We would like to solve these equations to determine the time-dependence of the coefficients. To do so, we take the second time derivative of equation two

$$\ddot{c}_2(t) = -i\Omega_0^*\frac{\partial}{\partial t}\left[\frac{1}{2}c_1(t)(e^{i\omega t} + e^{-i\omega t})e^{i\omega_0 t}\right] = -i\frac{\Omega_0^*}{2}\frac{\partial}{\partial t}\left\{c_1(t)\left[e^{i(\omega+\omega_0)t} + e^{-i(\omega-\omega_0)t}\right]\right\}.$$

We make the *Rotating Wave Approximation* (RWA). This approximation works well when the frequency of the field ω is close to the frequency difference between the states, ω_0 , which we call "resonance." However, this does not well far from resonance.

Near resonance, $e^{-i(\omega-\omega_0)t}$ "rotates" slowly compared to $e^{i(\omega+\omega_0)t}$; the fast-rotating term averages to zero on the time scale of the dynamics of interest, $\tau \sim 1/(\omega - \omega_0)$. Thus,

$$\ddot{c}_2(t) = -i\frac{\Omega_0^*}{2}\left[\dot{c}_1(t)e^{-i(\omega-\omega_0)t} - i(\omega - \omega_0)c_1(t)e^{-i(\omega-\omega_0)t}\right].$$

After applying the RWA to the equation for $\dot{c}_1(t)$ and $\dot{c}_2(t)$, we find

$$\begin{aligned} \dot{c}_1(t) &= -i\frac{\Omega_0}{2}\left[c_2(t)e^{i(\omega-\omega_0)t}\right] \\ \dot{c}_2(t) &= -i\frac{\Omega_0^*}{2}\left[c_1(t)e^{-i(\omega-\omega_0)t}\right]. \end{aligned}$$

Rearranging, we find

$$c_1(t) = \frac{2i}{\Omega_0^*}e^{i(\omega-\omega_0)t}\dot{c}_2(t)$$

to which we can substitute into the differential equation (DE) for $\ddot{c}_2(t)$ to find

$$\begin{aligned} \ddot{c}_2(t) &= (-i)^2\frac{\Omega_0\Omega_0^*}{4}c_2(t)e^{i(\omega-\omega_0)t}e^{-i(\omega-\omega_0)t} - \frac{2i}{\Omega_0^*}\frac{\Omega_0^*}{2}(\omega - \omega_0)\dot{c}_2(t)e^{i(\omega-\omega_0)t}e^{-i(\omega-\omega_0)t} \\ &= -i(\omega - \omega_0)\dot{c}_2(t) - \frac{|\Omega_0|^2}{4}c_2(t). \end{aligned}$$

In other words,

$$\boxed{\ddot{c}_2(t) + i(\omega - \omega_0)\dot{c}_2(t) + \frac{|\Omega_0|^2}{4}c_2(t) = 0.}$$

We can define *detuning* as the frequency difference between ω and ω_0 , by

$$\delta = \omega - \omega_0$$

so that

$$\ddot{c}_2(t) + i\delta\dot{c}_2(t) + \frac{|\Omega_0|^2}{4}c_2(t) = 0.$$

To find the time-dependence of $|\Psi\rangle = c_1(t)|\psi_1\rangle + c_2(t)|\psi_2\rangle$, we can solve this DE by using some initial conditions: for example, $c_1(0) = 1$ and $c_2(0) = 0$; looking for stationary solutions of the form

$$\begin{aligned} c_2(t) &= \frac{A}{2i}(e^{i\omega_1 t} - e^{-i\omega_2 t}) \\ \dot{c}_2(t) &= \frac{A}{2i}(i\omega_1 e^{i\omega_1 t} + i\omega_2 e^{-i\omega_2 t}) = \frac{A}{2}(\omega_1 e^{i\omega_1 t} + \omega_2 e^{-i\omega_2 t}) \\ \ddot{c}_2(t) &= \frac{iA}{2}(\omega_1^2 e^{i\omega_1 t} - \omega_2^2 e^{-i\omega_2 t}). \end{aligned}$$

Substituting trial solutions, we find

$$i\frac{A}{2}(\omega_1^2 e^{i\omega_1 t} - \omega_2^2 e^{-i\omega_2 t}) + \frac{iA}{2}\delta(\omega_1 e^{i\omega_1 t} + \omega_2 e^{-i\omega_2 t}) + \frac{|\Omega_0|^2}{4}\frac{A}{2i}(e^{i\omega_1 t} - e^{-i\omega_2 t}) = 0.$$

Each of the sets of $e^{i\omega_1 t}$ and $e^{-i\omega_2 t}$ terms must satisfy the equation independently, so

$$\begin{aligned} e^{i\omega_1 t} : \frac{iA}{2}\omega_1^2 + \frac{iA}{2}\omega_1 + \frac{|\Omega_0|^2}{4}\frac{A}{2i} &= 0 \\ \omega_1^2 + \delta\omega_1 - \frac{|\Omega_0|^2}{4} &= 0 \\ e^{i\omega_2 t} : -i\omega_2^2 + i\delta\omega_2 - \frac{|\Omega_0|^2}{4i} &= 0 \\ \omega_2^2 - \delta\omega_2 - \frac{|\Omega_0|^2}{4} &= 0. \end{aligned}$$

Thus,

$$\begin{aligned} \omega_{1\pm} &= \frac{1}{2}\left(-\delta \pm \sqrt{\delta^2 + |\Omega_0|^2}\right) \\ \omega_{2\pm} &= \frac{1}{2}\left(\delta \pm \sqrt{\delta^2 + |\Omega_0|^2}\right). \end{aligned}$$

We define the generalized Rabi frequency

$$\Omega = \sqrt{\delta^2 + \Omega_0^2}$$

to yield four solutions:

$$\begin{aligned} c_2(t) &= \frac{A}{2i}(e^{i\omega_{1+}t} - e^{-i\omega_{2+}t}) \\ &= \frac{A}{2i}\left(e^{-i\delta t/2}e^{i\Omega t/2} - e^{-i\delta t/2}e^{i\Omega t/2}\right) \\ &= \frac{A}{2i}e^{-i\delta t/2}\left(e^{i\Omega t/2} - e^{-i\Omega t/2}\right) \\ &= \frac{A}{2}e^{-i\delta t/2}\sin\left(\frac{\Omega t}{2}\right) \end{aligned} \tag{i}$$

$$c_2(t) = \frac{A}{2i} (e^{i\omega_1 t} - e^{-i\omega_2 t}) = \frac{A}{2i} (e^{-i\delta t/2} e^{i\Omega t/2} - e^{-i\delta t/2} e^{i\Omega t/2}) = 0 \quad (\text{ii})$$

$$c_2(t) = \frac{A}{2i} (e^{i\omega_1 t} - e^{-i\omega_2 t}) = \frac{A}{2i} (e^{-i\omega t/2} e^{-i\Omega t/2} - e^{-i\delta t/2} e^{-i\Omega t/2}) = 0 \quad (\text{iii})$$

$$\begin{aligned} c_2(t) &= \frac{A}{2i} e^{-i\delta t/2} (e^{-i\Omega t/2} - e^{i\Omega t/2}) = 0 \\ &= -\frac{A}{2} e^{-i\delta t/2} \sin\left(\frac{\Omega t}{2}\right). \end{aligned} \quad (\text{iv})$$

We note that solution (iv) gives the same result as solution (i) after solving for A . Thus,

$$c_2(t) = A e^{-i\delta t/2} \sin\left(\frac{\Omega t}{2}\right)$$

and so

$$\begin{aligned} \dot{c}_2(t) &= -\frac{i\delta A}{2} e^{-i\delta t/2} \sin\left(\frac{\Omega t}{2}\right) + A \frac{\Omega}{2} e^{-i\delta t/2} \cos\left(\frac{\Omega t}{2}\right) \\ c_1(t) &= \frac{2i}{\Omega_0^*} e^{i(\omega - \omega_0)t} \dot{c}_2(t) = \frac{2i}{\Omega_0} e^{i\delta t} \dot{c}_2(t). \end{aligned}$$

This yields the full solution

$$c_1(t) = \frac{\delta A}{\Omega_0} e^{i\delta t/2} \sin\left(\frac{\Omega t}{2}\right) + iA \frac{\Omega}{\Omega_0} e^{i\delta t/2} \cos\left(\frac{\Omega t}{2}\right).$$

Applying our initial conditions $c_1(0) = 1$ and $c_2(0) = 0$, we find

$$c_1(0) = 1 = \frac{\delta A}{\Omega_0} e^0 \sin(0) + iA \frac{\Omega}{\Omega_0} e^0 \cos(0) = \frac{iA\Omega}{\Omega_0} \implies A = -i \frac{\Omega_0}{\Omega}.$$

Thus,

$$\boxed{\begin{aligned} c_1(t) &= \left[-i \frac{\delta}{\Omega} e^{i\delta t/2} \sin\left(\frac{\Omega t}{2}\right) + e^{i\delta t/2} \cos\left(\frac{\Omega t}{2}\right) \right] \\ c_2(t) &= -i \frac{\Omega_0}{\Omega} \sin\left(\frac{\Omega t}{2}\right). \end{aligned}}$$

Confirming that the solution is indeed correct by the normalization condition,

$$\begin{aligned} |c_1|^2 &= \frac{\delta^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right) + \cos^2\left(\frac{\Omega t}{2}\right) \\ |c_2|^2 &= \frac{\Omega_0^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right) \\ |c_1|^2 + |c_2|^2 &= \frac{(\Omega_0^2 + \delta^2)}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right) + \cos^2\left(\frac{\Omega t}{2}\right) \\ &= \sin^2\left(\frac{\Omega t}{2}\right) + \cos^2\left(\frac{\Omega t}{2}\right) \\ &= 1. \end{aligned}$$

2.4 2-Level Population Dynamics

Consider the population dynamics in a 2-level system, where the population in $|\psi_1\rangle$ is

$$|\langle\psi_1|\Psi(t)\rangle|^2 = |c_1(t)|^2 = \frac{\delta^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right) + \cos^2\left(\frac{\Omega t}{2}\right)$$

and the population in $|\psi_2\rangle$ is

$$|\langle\psi_2|\Psi(t)\rangle|^2 = |c_2(t)|^2 = \frac{\Omega_0^2}{\Omega} \sin^2\left(\frac{\Omega t}{2}\right).$$

There are some special cases of this 2-level system that we will analyze.

2.4.1 On-Resonance ($\delta = 0$, $\omega = \omega_0$)

Here, we have the generalized Rabi frequency

$$\Omega = \sqrt{\Omega_0^2 + 0} = \Omega_0$$

with time-dependent coefficients

$$|c_1(t)|^2 = \cos^2\left(\frac{\Omega_0 t}{2}\right) = \frac{1}{2}[1 + \cos(\Omega_0 t)]$$

$$|c_2(t)|^2 = \sin^2\left(\frac{\Omega_0 t}{2}\right) = \frac{1}{2}[1 - \cos(\Omega_0 t)].$$

Here, the population fully cycles with an angular frequency

$$\Omega_0 = \frac{1}{\hbar} \langle\psi_2| e\mathbf{r} \cdot \mathbf{E}_0 |\psi_1\rangle$$

which is the frequency of population oscillation (twice as fast as the amplitude). Notice that this frequency scales with the electric field amplitude, not the power (which would be $\propto E_0^2$). All of atomic physics is encapsulated by this frequency Ω_0 (Rabi frequency).

We can consider some "special" times. Observe from Fig. 16 that at $t = \pi/\Omega_0$ or $\Omega_0 t = \pi$, we have what is called a π -pulse, which is when the population is inverted from its initial state. At $t = \pi/2\Omega_0$ or $\Omega_0 t = \pi/2$, we have a $\pi/2$ -pulse, which is when the populations are equal.

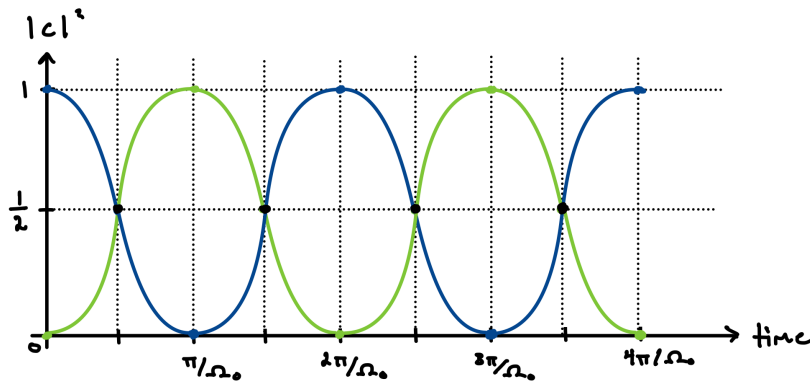


Figure 16: 2-level population on resonance over time.

We can calculate the amplitudes of the $\pi/2$ -pulse explicitly to find that

$$c_1(t_{\pi/2}) = \cos\left(\frac{\pi}{2}\right)e^{i\delta t/2} = \frac{1}{\sqrt{2}}$$

$$c_2(t_{\pi/2}) = -i\sin\left(\frac{\pi}{2}\right)e^{-i\delta t/2} = -\frac{i}{\sqrt{2}}.$$

These coefficients give a population state

$$|\Psi(t_{\pi/2})\rangle = \frac{1}{\sqrt{2}}(|\psi_1\rangle - i|\psi_2\rangle).$$

2.4.2 Off-Resonant Excitation ($\delta \neq 0$, $\Omega \neq \Omega_0$)

Here, we have the generalized Rabi frequency

$$\Omega = \sqrt{\Omega_0^2 + \delta^2} > \Omega_0.$$

The time-dependent coefficients are

$$|c_1(t)|^2 = \cos^2\left(\frac{\Omega t}{2}\right) - \frac{\delta^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right)$$

$$= \frac{1}{2}[1 + \cos(\Omega t)] - \frac{1}{2} \frac{\delta^2}{\Omega^2} [1 - \cos(\Omega t)]$$

$$|c_2(t)|^2 = \left(\frac{\Omega_0}{\Omega}\right)^2 \sin^2\left(\frac{\Omega t}{2}\right) = \frac{1}{2} \left(\frac{\Omega_0}{\Omega}\right)^2 [1 - \cos(\Omega t)].$$

The populations now oscillate at a higher frequency.

Here, the maximum possible amplitude of $|c_2|^2$ is Ω_0^2/Ω^2 . We cannot achieve full population transfer off-resonance. Observe also that the frequency of collisions is faster off-resonance.

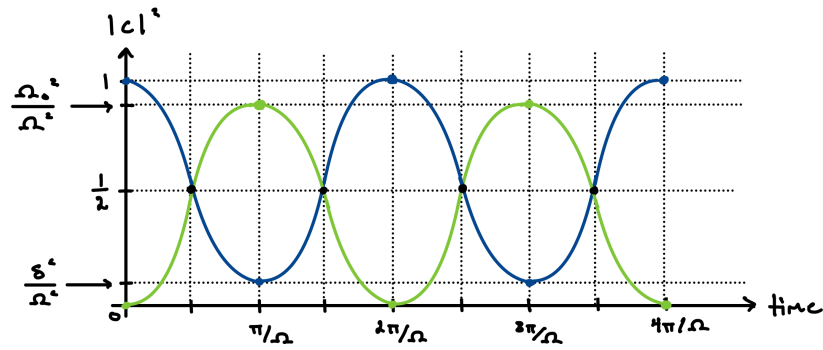


Figure 17: 2-level population off-resonance over time. Notice that the time-scale has changed from $n\pi/\Omega_0$ to $n\pi/\Omega$.

2.5 "Pictures" in Quantum Dynamics

To continue our study of these solutions, it is useful to review the different "pictures" of the time-dependent solutions in QM.

Schrödinger's Picture. In the Schrödinger picture, observables are stationary and the states have time-dependence instead. Thusly, we might have

$$|\psi\rangle \rightarrow |\Psi(t)\rangle_S, \quad \rho_s(t) = |\Psi(t)\rangle_S \langle \Psi(t)|_s.$$

The dynamics that follow are

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle_S &= \hat{H} |\Psi(t)\rangle_S \\ \frac{\partial}{\partial t} \hat{\rho}_s(t) &= \left(\frac{\partial}{\partial t} |\Psi(t)\rangle_S \right) \langle \Psi(t)|_s + |\Psi(t)\rangle_S \left(\frac{\partial}{\partial t} \langle \Psi(t)|_s \right) \\ &= \frac{1}{i\hbar} [\hat{H} |\Psi(t)\rangle_S] \langle \Psi(t)|_s + |\Psi(t)\rangle_S \left[-\frac{1}{i\hbar} \langle \Psi(t)|_s \hat{H} \right] \\ &= \frac{1}{i\hbar} [\hat{H}, \hat{\rho}_s] \\ i\hbar \frac{\partial \hat{\rho}_s(t)}{\partial t} &= [\hat{H}, \hat{\rho}_s]. \end{aligned}$$

Heisenberg's Picture. In the Heisenberg picture, the observables are the quantities that have time-dependence instead while the states are stationary. Thus, we might have

$$|\psi\rangle \rightarrow |\psi\rangle, \quad \hat{A}_H(t) = e^{i\hat{H}t/\hbar} \hat{A}_s e^{-i\hat{H}t/\hbar}, \quad |\psi\rangle_H = e^{i\hat{H}t/\hbar} |\Psi(t)\rangle_S \leftrightarrow |\Psi_s(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi\rangle_H$$

where $\hat{A}$ is an operator. The dynamics that follow are

$$\begin{aligned} \frac{\partial}{\partial t} \hat{A}_H(t) &= \frac{i\hat{H}}{\hbar} \hat{A}_H(t) - e^{i\hat{H}t/\hbar} \hat{A}_s \left(\frac{-i\hat{H}}{\hbar} \right) e^{-i\hat{H}t/\hbar} \\ &= \frac{i}{\hbar} (\hat{H} \hat{A}_H(t) - \hat{A}_H(t) \hat{H}) \\ i\hbar \frac{\partial}{\partial t} \hat{A}_H(t) &= [\hat{A}_H, \hat{H}]. \end{aligned}$$

However, regardless of picture, the measurable quantities, expectation values, remain unchanged, and are

$$\langle \hat{A} \rangle = \langle \Psi(t) | \hat{A}_s | \Psi(t) \rangle = \langle \psi | e^{-i\hat{H}t/\hbar} \hat{A}_H(t) e^{i\hat{H}t/\hbar} | \psi \rangle = \langle \psi | \hat{A}_H(t) | \psi \rangle.$$

Interaction Picture. We may also want to consider the interaction picture, where we split the Hamiltonian into non-interacting and interacting parts:

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}.$$

Here, we let the states evolve with $\hat{H}_0$, as in the Schrödinger picture, and the operators evolve with $\hat{H}_{\text{int}}$, as in the Heisenberg picture.

$$\begin{aligned} |\Psi(t)\rangle_I &= e^{i\hat{H}_0 t/\hbar} |\Psi(t)\rangle_S = e^{i\hat{H}_0 t/\hbar} e^{-i\hat{H}_0 t/\hbar} e^{-i\hat{H}_{\text{int}} t/\hbar} |\psi\rangle_H = e^{-i\hat{H}_{\text{int}} t/\hbar} |\psi\rangle_H \\ \hat{A}_I(t) &= e^{i\hat{H}_0 t/\hbar} \hat{A}_s e^{-i\hat{H}_0 t/\hbar} = e^{-i\hat{H}_{\text{int}} t/\hbar} \hat{A}_H(t) e^{i\hat{H}_{\text{int}} t/\hbar}. \end{aligned}$$

This leads, as one would expect, into the same expectation values

$$\begin{aligned}
\langle \hat{A}_I(t) \rangle &= \langle \Psi(t) |_I \hat{A}_I(t) | \Psi(t) \rangle_I \\
&= \langle \Psi(t) |_s e^{-i\hat{H}_0 t/\hbar} e^{-i\hat{H}_0 t/\hbar} \hat{A}_s e^{-i\hat{H}_0 t/\hbar} e^{i\hat{H}_0 t/\hbar} | \Psi(t) \rangle_S \\
&= \langle \Psi(t) |_s \hat{A}_s | \Psi(t) \rangle_S \\
&= \langle \psi |_H e^{i\hat{H}_{\text{int}} t/\hbar} e^{-i\hat{H}_{\text{int}} t/\hbar} \hat{A}_H(t) e^{i\hat{H}_{\text{int}} t/\hbar} e^{-i\hat{H}_{\text{int}} t/\hbar} | \psi \rangle_H \\
&= \langle \psi |_H \hat{A}_H(t) | \psi \rangle_H.
\end{aligned}$$

Both states and operators have dynamics, so

$$\begin{aligned}
\hat{\rho}_I &= | \Psi(t) \rangle_I \langle \Psi(t) |_I = e^{i\hat{H}_0 t/\hbar} \hat{\rho}_s(t) e^{-i\hat{H}_0 t/\hbar} = e^{-i\hat{H}_{\text{int}} t/\hbar} \hat{\rho}_H e^{i\hat{H}_{\text{int}} t/\hbar} \\
\frac{\partial}{\partial t} \hat{\rho}_I(t) &= -\frac{i\hat{H}_{\text{int}}}{\hbar} \hat{\rho}_I + e^{-i\hat{H}_{\text{int}} t/\hbar} \hat{\rho}_H \left(\frac{i\hat{H}_{\text{int}}}{\hbar} \right) e^{i\hat{H}_{\text{int}} t/\hbar} \\
&= -\frac{i}{\hbar} (\hat{H}_{\text{int}} \hat{\rho}_I - \hat{\rho}_I \hat{H}_{\text{int}}) \\
i\hbar \frac{\partial}{\partial t} \hat{\rho}_I(t) &= [\hat{H}_{\text{int}}, \hat{\rho}_I] \\
\frac{\partial}{\partial t} \hat{A}_I(t) &= \frac{i\hat{H}_0}{\hbar} \hat{A}_I + e^{i\hat{H}_0 t/\hbar} \hat{A}_s \left(\frac{-i\hat{H}_0}{\hbar} \right) e^{-i\hat{H}_0 t/\hbar} \\
&= -\frac{i}{\hbar} (\hat{H}_0 \hat{A}_I - \hat{A}_I \hat{H}_0) \\
i\hbar \frac{\partial}{\partial t} \hat{A}_I(t) &= [\hat{H}_0, \hat{A}_I].
\end{aligned}$$

In general, we can transform the system to a "rotating" frame of reference that is described by a unitary transformation. In this case, the interaction picture is a unitary transformation from the Schrödinger picture by the operator

$$\hat{U}_I = e^{i\hat{H}_0 t/\hbar} \implies | \Psi(t) \rangle_I = \hat{U}_I | \Psi(t) \rangle_S, \quad \hat{A}_I = \hat{U}_I \hat{A}_s \hat{U}_I^\dagger.$$

We can use this idea to transform the Hamiltonian and states under an arbitrary unitary transformation:

$$\begin{aligned}
\hat{H} &\rightarrow \hat{\tilde{H}} = \hat{U} \hat{H} \hat{U}^\dagger + i\hbar \frac{\partial \hat{U}}{\partial t} \hat{U}^\dagger \\
| \Psi \rangle &\rightarrow | \tilde{\Psi} \rangle = \hat{U} | \Psi \rangle
\end{aligned}$$

with

$$\hat{\tilde{H}} | \tilde{\psi} \rangle = i\hbar \frac{\partial}{\partial t} | \tilde{\psi} \rangle.$$

We will show this explicitly by inserting $\hat{1} = \hat{U}^\dagger \hat{U}$ onto both sides of the equation

$$\begin{aligned}
\hat{H} | \Psi(t) \rangle &= i\hbar \frac{\partial}{\partial t} | \Psi(t) \rangle \\
\hat{H} \hat{U}^\dagger \hat{U} | \Psi(t) \rangle &= i\hbar \frac{\partial}{\partial t} (\hat{U}^\dagger \hat{U} | \Psi(t) \rangle) \\
&= i\hbar \frac{\partial \hat{U}^\dagger}{\partial t} \hat{U} | \Psi(t) \rangle + i\hbar \hat{U}^\dagger \frac{\partial \hat{U}}{\partial t} | \Psi(t) \rangle + i\hbar \hat{U}^\dagger \hat{U} \frac{\partial}{\partial t} | \Psi(t) \rangle.
\end{aligned}$$

Applying $\hat{U}$ from the left, we see

$$\begin{aligned}\hat{U}\hat{H}\hat{U}^\dagger(\hat{U}|\Psi(t)\rangle) &= i\hbar\hat{U}\frac{\partial\hat{U}^\dagger}{\partial t}(\hat{U}|\Psi(t)\rangle) + i\hbar\frac{\partial\hat{U}}{\partial t}|\Psi(t)\rangle + i\hbar\hat{U}\frac{\partial}{\partial t}|\Psi(t)\rangle \\ &= i\hbar\hat{U}\frac{\partial\hat{U}^\dagger}{\partial t} + i\hbar\frac{\partial}{\partial t}[\hat{U}|\Psi(t)\rangle].\end{aligned}$$

So,

$$\left(\hat{U}\hat{H}\hat{U}^\dagger - i\hbar\hat{U}\frac{\partial\hat{U}^\dagger}{\partial t}\right)(\hat{U}|\Psi(t)\rangle) = i\hbar\frac{\partial}{\partial t}(\hat{U}|\Psi(t)\rangle).$$

Recall $\hat{U}\hat{U}^\dagger = 1$, so

$$\begin{aligned}\frac{\partial}{\partial t}(\hat{U}\hat{U}^\dagger) &= 0 \\ \frac{\partial\hat{U}}{\partial t}\hat{U}^\dagger + \hat{U}\frac{\partial\hat{U}^\dagger}{\partial t} &= \\ \hat{U}\frac{\partial\hat{U}^\dagger}{\partial t} &= -\frac{\partial\hat{U}}{\partial t}\hat{U}^\dagger\end{aligned}$$

and $\hat{U}|\Psi(t)\rangle = |\tilde{\Psi}(t)\rangle$, thus

$$\left(\hat{U}\hat{H}\hat{U}^\dagger + i\hbar\frac{\partial\hat{U}}{\partial t}\hat{U}^\dagger\right)|\tilde{\Psi}(t)\rangle\tilde{H}|\tilde{\Psi}(t)\rangle = i\hbar\frac{\partial}{\partial t}|\tilde{\Psi}(t)\rangle.$$

Similar to the interaction picture, we can move to an arbitrary "rotating frame." For our purposes, we wish to evaluate the effects of the interaction and the response at the frequency of the interaction.¹³ Additionally, we can use a unitary transformation to shift the "zero" of our energy levels through a "boost."

Let us consider our 2-level system, for which we determined the time-dependent states in the Schrödinger picture

$$|\Psi(t)\rangle_S = c_1(t)e^{-i\omega_1 t}|\psi_1\rangle + c_2(t)e^{-i\omega_2 t}|\psi_2\rangle.$$

First, let us set the energy-zero at the lower level:

$$\omega_1 = 0 \ (E_1 = 0) \implies \omega_2 = \omega_0 \ (\omega_0 = \omega_2 - \omega_1).$$

So,

$$|\Psi(t)\rangle_S = c_1(t)|\psi_1\rangle + c_2(t)e^{-i\omega_0 t}|\psi_2\rangle.$$

We now rewrite the Hamiltonian, as it applies to this 2-level system, in Dirac notation

$$\hat{H} = \hat{H}_{\text{at}} + \hat{H}_{\text{int}}$$

where

$$\begin{aligned}\hat{H}_{\text{at}} &= \hbar\omega_0|\psi_2\rangle\langle\psi_2| \\ \hat{H}_{\text{int}} &= \mathbf{e}\mathbf{r} \cdot \mathbf{E}_0 \cos(\omega t)(|\psi_1\rangle\langle\psi_2| + |\psi_2\rangle\langle\psi_1|) \\ &= \hbar\Omega_0^* \cos(\omega t)|\psi_1\rangle\langle\psi_2| + \hbar\Omega_0 \cos(\omega t)|\psi_2\rangle\langle\psi_1|.\end{aligned}$$

¹³We can think of this like lock-in amplification.

Note that the second term in $\hat{H}_{\text{int}}$ is not yet in our RWA. We could have alternatively written this in matrix form as

$$\hat{H} = \hat{H}_{\text{at}} + \hat{H}_{\text{int}} = \begin{pmatrix} 0 & 0 \\ 0 & \hbar\omega_0 \end{pmatrix} + \begin{pmatrix} 0 & \hbar\Omega_0 \cos(\omega t) \\ \hbar\Omega_0^* \cos(\omega t) & 0 \end{pmatrix} = \begin{pmatrix} 0 & \hbar\Omega_0 \cos(\omega t) \\ \hbar\Omega_0^* \cos(\omega t) & \hbar\omega_0 \end{pmatrix}.$$

We now apply a unitary operator that will do two things for us: (a) move our system to a frame "rotating at ω "; (b) shift the energy level of the system by $\delta/2$. This unitary operator must be of the form

$$\hat{U}_\omega = \exp \left[i\omega t |\psi_2\rangle \langle \psi_2| - \frac{i\delta t}{2} (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|) \right].$$

The first term in the exponential applies to one state, which create the frame "rotating at ω ", and the second term applies equally to both states, our "boost."

Evaluating the Hamiltonian in the rotating frame. we have

$$\hat{\tilde{H}} = \hat{\tilde{H}}_R = \hat{U}_\omega \hat{H} \hat{U}_\omega^\dagger + i\hbar(\partial_t \hat{U}_\omega) \hat{U}_\omega^\dagger.$$

Recalling the Taylor expansion for $e^{i\hat{\theta}}$, we find that

$$\begin{aligned} \hat{U}_\omega \hat{H}_{\text{at}} \hat{U}_\omega &= \exp \left[i\omega t |\psi_2\rangle \langle \psi_2| - \frac{i\delta t}{2} (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|) \right] \hbar\omega_0 |\psi_2\rangle \langle \psi_2| \\ &\quad \times \exp [-i\omega t |\psi_2\rangle \langle \psi_2|] \exp \left[\frac{i\delta t}{2} (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|) \right] \\ &= \exp \left(i\omega t - \frac{i\delta t}{2} \right) \exp \left(-i\omega t + \frac{i\delta t}{2} \right) \hbar\omega_0 |\psi_2\rangle \langle \psi_2| \\ &= \hat{H}_{\text{at}}. \end{aligned}$$

and

$$\begin{aligned} \hat{U}_\omega \hat{H}_{\text{int}} \hat{U}_\omega &= \exp \left[i\omega t |\psi_2\rangle \langle \psi_2| - \frac{i\delta t}{2} (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|) \right] \hbar\Omega_0^* \cos(\omega t) |\psi_1\rangle \langle \psi_2| \\ &\quad \times \exp (-i\omega t |\psi_2\rangle \langle \psi_2|) \exp \left[\frac{i\delta t}{2} (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|) \right] \\ &\quad + \exp \left[i\omega t |\psi_2\rangle \langle \psi_2| - \frac{i\delta t}{2} (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|) \right] \hbar\Omega_0 \cos(\omega t) \\ &\quad \times |\psi_2\rangle \langle \psi_1| \exp [-i\omega t |\psi_2\rangle \langle \psi_2|] \exp \left[\frac{i\delta t}{2} (|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|) \right] \\ &= \exp \left(0 - \frac{i\delta t}{2} \right) \hbar\Omega_0^* \cos(\omega t) |\psi_1\rangle \langle \psi_2| \exp(-i\omega t) \exp \left(\frac{i\delta t}{2} \right) \\ &\quad + \exp \left(i\omega t - \frac{i\delta t}{2} \right) \hbar\Omega_0 \cos(\omega t) |\psi_2\rangle \langle \psi_1| \exp \left(\frac{i\delta t}{2} \right) \\ &= \exp(-i\omega t) \hbar\Omega_0 \cos(\omega t) |\psi_1\rangle \langle \psi_2| + \exp(i\omega t) \hbar\Omega_0 \cos(\omega t) |\psi_2\rangle \langle \psi_1|. \end{aligned}$$

Applying the RWA (so that terms like $e^{2i\omega t}$, $e^{-2i\omega t} \rightarrow 0$), we find that

$$\begin{aligned} \hat{U}_\omega \hat{H}_{\text{int}} \hat{U}_\omega &= \frac{\hbar\Omega_0}{2} e^{-i\omega t} (e^{i\omega t} + e^{-i\omega t}) |\psi_1\rangle \langle \psi_2| + \frac{\hbar\Omega_0^*}{2} e^{i\omega t} (e^{i\omega t} + e^{-i\omega t}) |\psi_2\rangle \langle \psi_1| \\ &\approx \frac{\hbar\Omega_0}{2} |\psi_1\rangle \langle \psi_2| + \frac{\hbar\Omega_0^*}{2} |\psi_2\rangle \langle \psi_1| \approx \hat{U}_\omega \hat{H}_{\text{int}} \hat{U}_\omega^\dagger. \end{aligned}$$

We will also evaluate

$$\partial_t \hat{U}_\omega = \left[i\omega |\psi_2\rangle \langle \psi_2| - i\frac{\delta}{2}(|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|) \right].$$

So,

$$i\hbar(\partial_t \hat{U}_\omega) \hat{U}_\omega^\dagger = i\hbar[i\omega |\psi_2\rangle \langle \psi_2| - i\frac{\delta}{2}(|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|)].$$

Putting this all together, the rotating frame Hamiltonian is

$$\begin{aligned} \hat{H}_\omega &= \hbar\omega_0 |\psi_2\rangle \langle \psi_2| + \frac{\hbar\Omega_0}{2} |\psi_2\rangle \langle \psi_2| + \frac{\hbar\Omega - 0^*}{2} |\psi_2\rangle \langle \psi_1| - \hbar\omega |\psi_2\rangle \langle \psi_2| + \frac{\delta}{2}(|\psi_1\rangle \langle \psi_1| + |\psi_2\rangle \langle \psi_2|) \\ &= \hbar\left(\omega_0 - \omega + \frac{\delta}{2}\right) |\psi_2\rangle \langle \psi_2| + \frac{\hbar\delta}{2} |\psi_1\rangle \langle \psi_1| + \frac{\hbar\Omega_0}{2} |\psi_1\rangle \langle \psi_2| + \frac{\hbar\Omega_0^*}{2} |\psi_2\rangle \langle \psi_1| \\ &= \frac{\hbar\delta}{2} |\psi_1\rangle \langle \psi_1| - \frac{\hbar\delta}{2} |\psi_2\rangle \langle \psi_2| + \frac{\hbar\Omega_0}{2} |\psi_1\rangle \langle \psi_2| + \frac{\hbar\Omega_0^*}{2} |\psi_2\rangle \langle \psi_1|. \end{aligned}$$

The states are $\hat{U} |\Psi(t)\rangle_S$,

$$\begin{aligned} \hat{U}_\omega |\Psi_1(t)\rangle &= \hat{U}_\omega c_1(t) |\psi_1\rangle \\ &= e^{-i\delta t/2} c_1(t) |\psi_1\rangle \\ &= \tilde{c}_1(t) |\psi_1\rangle \\ \hat{U}_\omega |\Psi_2(t)\rangle &= \hat{U}_\omega c_2(t) e^{-i\omega_0 t} |\psi_2\rangle \\ &= e^{i\omega t - i\delta t/2} c_2(t) e^{-i\omega_0 t} |\psi_2\rangle \\ &= e^{i(\omega - \omega_0)t - i\delta t/2} c_2(t) |\psi_2\rangle \\ &= e^{i\delta t/2} c_2(t) |\psi_2\rangle \\ &= \tilde{c}_2(t) |\psi_2\rangle. \end{aligned}$$

So,

$$|\tilde{\Psi}\rangle = \tilde{c}_1(t) |\psi_1\rangle + \tilde{c}_2(t) |\psi_2\rangle.$$

In matrix form, in the basis $(\tilde{c}_1, \tilde{c}_2)$, we have

$$\hat{H}_\omega = \frac{\hbar}{2} \begin{pmatrix} \delta & \Omega_0^* \\ \Omega_0 & -\delta \end{pmatrix}.$$

So, by moving to this rotating frame, we eliminate the explicit time dependence and can see the simple dynamics that are "above and beyond" the rotation driven by the external field. Additionally, with our choice of boost, we get a symmetric Hamiltonian.

Note that when $\delta = 0$, the states are stationary in this rotating frame. When $\delta \neq 0$, there is some phase evolution.

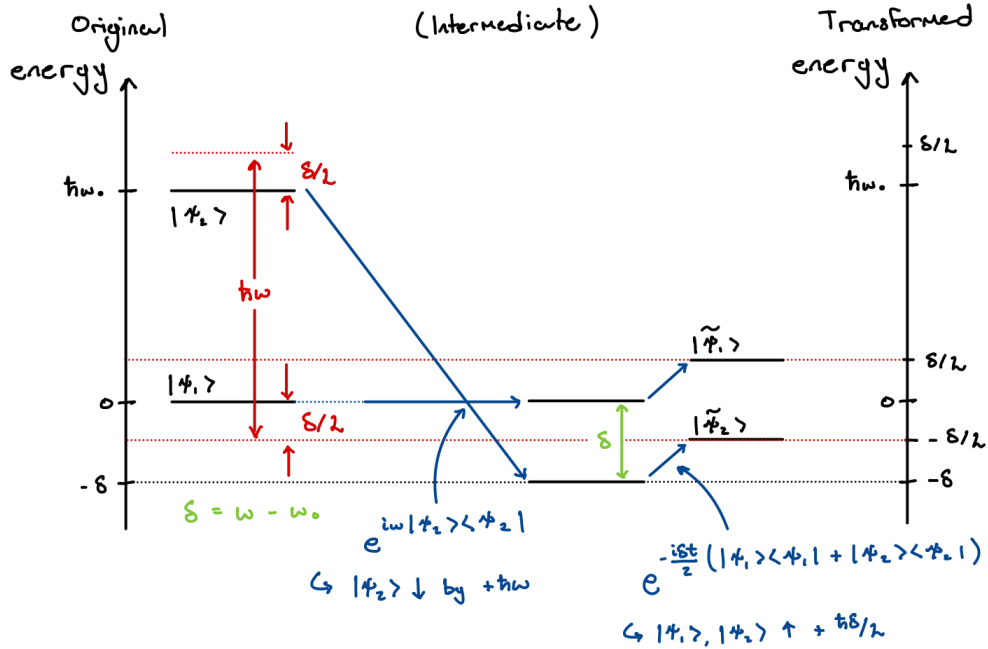


Figure 18: Picture of the transformation for the interaction picture. The first shift affects only $|\psi_2\rangle$ downward by $\hbar\omega$, while the second shift affects both states upward by $\hbar\delta/2$.

Using the results of the time-dependence of $c_1(t)$ and $c_2(t)$, we find that

$$\begin{aligned} c_1(t) &= \left[-i \frac{\delta}{\Omega} e^{i\delta t/2} \sin\left(\frac{\Omega t}{2}\right) + e^{i\delta t/2} \cos\left(\frac{\Omega t}{2}\right) \right] \\ c_2(t) &= -i \frac{\Omega_0}{\Omega} \sin\left(\frac{\Omega t}{2}\right) e^{-i\delta t/2}. \end{aligned}$$

To find the time-dependence in the rotating frame, we assume $\Omega = \Omega_0^*$, to get

$$\begin{aligned}\tilde{c}_1(t) &= -i\frac{\delta}{\Omega}\sin\left(\frac{\Omega t}{2}\right) + \cos\left(\frac{\Omega t}{2}\right) \\ \tilde{c}_2(t) &= -i\frac{\Omega_0}{\Omega}\sin\left(\frac{\Omega t}{2}\right).\end{aligned}$$

These are expressions for the *quantum amplitudes*; the populations are also important to consider. We will use the *density matrix* to quantify the population dynamics.

We stop here briefly to discuss the meaning of the density matrix. In a limited Hilbert space, like the one we are working with, the interpretation is straightforward: the operator $|\psi\rangle\langle\psi|$ projects states onto the direction $|\psi\rangle$ with the amplitude depending on the overlap with $|\psi\rangle$. For example, we may consider a state

$$|\psi\rangle = a|\psi_1\rangle + b|\psi_2\rangle$$

so that

$$|\psi\rangle\langle\psi| = aa^*|\psi_1\rangle\langle\psi_1| + ab^*|\psi_2\rangle\langle\psi_1| + ba^*|\psi_1\rangle\langle\psi_2| + bb^*|\psi_2\rangle\langle\psi_2|.$$

In matrix form, in the basis (a, b) , we the density matrix is

$$\hat{\rho} = |\psi\rangle\langle\psi| \rightarrow \begin{pmatrix} a \\ b \end{pmatrix} \begin{pmatrix} a^* & b^* \end{pmatrix} = \begin{pmatrix} aa^* & ab^* \\ ba^* & bb^* \end{pmatrix}$$

which are also the coefficients of $|\psi\rangle\langle\psi|$. In this matrix, the diagonal elements give the "populations" $|a|^2$ and $|b|^2$; the off-diagonal elements give the "coherences," which are a measure of "how much superposition" is in each state.

2.6 The Optical Bloch Sphere

The density matrix is usually written as

$$\hat{\rho} = |\psi\rangle\langle\psi|.$$

For the 2×2 matrix, representing a 2-level system like the one we have been discussing, we have

$$\hat{\rho} = |c_1|^2 |\psi_1\rangle\langle\psi_1| + c_1 c_2^* |\psi_1\rangle\langle\psi_2| + c_2 c_1^* |\psi_2\rangle\langle\psi_1| + |c_2|^2 |\psi_2\rangle\langle\psi_2| = \begin{pmatrix} |c_1|^2 & c_1 c_2^* \\ c_2 c_1^* & |c_2|^2 \end{pmatrix} \equiv \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix}$$

assuming the Schrödinger picture where $c_1 = c_1(t)$ and $c_2 = c_2(t)e^{-i\omega_0 t}$. The dynamics of the density matrix are

$$i\hbar \frac{\partial}{\partial t} \hat{\rho}(t) = [\hat{H}, \hat{\rho}]$$

and the analogue applies in our rotating frame

$$i\hbar \frac{\partial}{\partial t} \hat{\tilde{\rho}}(t) = [\hat{\tilde{H}}, \hat{\tilde{\rho}}], \quad \hat{\tilde{\rho}}(t) = \begin{pmatrix} |\tilde{c}_1|^2 & \tilde{c}_1 \tilde{c}_2^* \\ \tilde{c}_2 \tilde{c}_1^* & |\tilde{c}_2|^2 \end{pmatrix}.$$

Using this relationship, we can find the solutions for the $\tilde{c}$ -coefficients, in the form of the density matrix components

$$\begin{aligned} [\hat{\tilde{H}}, \hat{\tilde{\rho}}] &= \frac{\hbar}{2} \begin{pmatrix} \delta & \Omega_0 \\ \Omega_0 & -\delta \end{pmatrix} \begin{pmatrix} \tilde{\rho}_{11} & \tilde{\rho}_{12} \\ \tilde{\rho}_{21} & \tilde{\rho}_{22} \end{pmatrix} - \frac{\hbar}{2} \begin{pmatrix} \tilde{\rho}_{11} & \tilde{\rho}_{12} \\ \tilde{\rho}_{21} & \tilde{\rho}_{22} \end{pmatrix} \begin{pmatrix} \delta & \Omega_0 \\ \Omega_0 & -\delta \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} \delta \tilde{\rho}_{11} + \Omega_0 \tilde{\rho}_{21} & \delta \tilde{\rho}_{12} + \Omega_0 \tilde{\rho}_{22} \\ \Omega_0 \tilde{\rho}_{11} - \delta \tilde{\rho}_{21} & \Omega_0 \tilde{\rho}_{12} - \delta \tilde{\rho}_{22} \end{pmatrix} - \frac{\hbar}{2} \begin{pmatrix} \delta \tilde{\rho}_{11} + \Omega_0 \tilde{\rho}_{12} & \Omega_0 \tilde{\rho}_{11} - \delta \tilde{\rho}_{12} \\ \delta \tilde{\rho}_{21} + \Omega_0 \tilde{\rho}_{22} & \Omega_0 \tilde{\rho}_{21} - \delta \tilde{\rho}_{22} \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} \Omega_0(\tilde{\rho}_{21} - \tilde{\rho}_{12}) & \delta(\tilde{\rho}_{12} + \tilde{\rho}_{12}) + \Omega_0(\tilde{\rho}_{22} - \tilde{\rho}_{11}) \\ -\delta(\tilde{\rho}_{21} + \tilde{\rho}_{21}) + \Omega_0(\tilde{\rho}_{11} - \tilde{\rho}_{22}) & \Omega_0(\tilde{\rho}_{12} - \tilde{\rho}_{21}) \end{pmatrix}. \end{aligned}$$

So, we get 4 equations

$$\begin{aligned} \dot{\rho}_{11} &= \frac{1}{i\hbar} \frac{\hbar}{2} \Omega_0 (\rho_{21} - \rho_{12}) = \frac{i}{2} \Omega_0 (\rho_{12} - \rho_{21}) \\ \dot{\rho}_{22} &= \frac{1}{i\hbar} \frac{\hbar}{2} \Omega_0 (\rho_{12} - \rho_{21}) = -\frac{i}{2} \Omega_0 (\rho_{12} - \rho_{21}) \\ \dot{\rho}_{12} &= \frac{1}{i\hbar} \frac{\hbar}{2} [2\delta \rho_{12} - \Omega_0 (\rho_{11} - \rho_{22})] \\ &= -\frac{i}{2} [2\delta \rho_{12} - \Omega_0 (\rho_{11} - \rho_{22})] \\ \dot{\rho}_{21} &= \frac{\hbar}{i\hbar} [-2\delta \rho_{21} + \Omega_0 (\rho_{11} - \rho_{22})] \\ &= \frac{i}{2} [2\delta \rho_{21} - \Omega_0 (\rho_{11} - \rho_{22})]. \end{aligned}$$

We redefine this in terms of three variables (the fourth being constrained via $\rho_{11} + \rho_{22} = 1$ for normalization),

$$\begin{aligned} w &\equiv \rho_{11} - \rho_{22} && \text{(population difference)} \\ u &\equiv \rho_{12} + \rho_{21} && \text{(in-phase difference)} \\ v &= -i(\rho_{12} - \rho_{21}). && \text{(out-of-phase difference)} \end{aligned}$$

The dynamics of these variables are

$$\begin{aligned} \dot{w} &= \dot{\rho}_{11} - \dot{\rho}_{22} = \frac{i}{2}\Omega_0(\rho_{12} - \rho_{21} + \rho_{12} - \rho_{21}) \\ &= i\Omega_0(\rho_{12} - \rho_{21}) \\ &= -\Omega_0 v \\ \dot{u} &= \dot{\rho}_{12} + \dot{\rho}_{21} = -i\delta\rho_{12} + i\frac{\Omega_0}{2}w + i\delta\rho_{21} - i\Omega_0 w \\ &= -i\delta(\rho_{12} - \rho_{21}) \\ &= \delta v \\ \dot{v} &= -i(\dot{\rho}_{12} - \dot{\rho}_{21}) = -i\left(-i\delta\rho_{12} + i\frac{\Omega_0}{2}w - i\delta\rho_{21} + i\frac{\Omega_0}{2}w\right) \\ &= \Omega_0 w - \delta(\rho_{12} - \rho_{21}) \\ &= \Omega_0 w - \delta u. \end{aligned}$$

These dynamics can be expressed compactly (and pictorially) by the *Bloch vector*, where we define

$$\begin{aligned} \mathbf{R} &= (u, v, w) \\ \frac{\partial \mathbf{R}}{\partial t} &= \frac{\partial}{\partial t} \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} \delta v \\ \Omega_0 w - \delta u \\ -\Omega_0 v \end{pmatrix} = \mathbf{R} \times \boldsymbol{\Omega} \end{aligned}$$

where

$$\boldsymbol{\Omega} = (\Omega_0, 0, \delta).$$

This vector, $\boldsymbol{\Omega}$, has a length equal to the generalized Rabi frequency

$$|\boldsymbol{\Omega}| = \sqrt{\Omega_0^2 + \delta^2} \equiv \Omega.$$

The state vector $|\mathbf{R}|^2 = 1$ does not depend on the time-dependent components, and thus does not change length. We may wish to see this explicitly,

$$\begin{aligned} |\mathbf{R}|^2 &= |u|^2 + |v|^2 + |w|^2 \\ &= (\rho_{21}^* + \rho_{12}^*)(\rho_{21} + \rho_{12}) + i(\rho_{12}^* - \rho_{21}^*)(-i)(\rho_{12} - \rho_{21}) + (\rho_{11}^2 - 2\rho_{11}\rho_{22} + \rho_{22}^2) \end{aligned}$$

with $\rho_{12}^* = \rho_{21}$, we have

$$\begin{aligned}
|\mathbf{R}|^2 &= (\rho_{12}\rho_{21} + \rho_{21}\rho_{21} + \rho_{12}\rho_{12} + \rho_{21}\rho_{12}) \\
&\quad + (\rho_{21}\rho_{12} - \rho_{21}\rho_{21} - \rho_{12}\rho_{12} + \rho_{12}\rho_{21}) \\
&\quad + (\rho_{11}^2 - 2\rho_{11}\rho_{22} + \rho_{22}^2) \\
&= \rho_{11}^2 - 2\rho_{11}\rho_{22} + \rho_{22}^2 + 4\rho_{12}\rho_{21} \\
&= |c_1|^4 + |c_2|^2 - 2|c_1|^2|c_2|^2 + 4(c_1c_2^*)(c_2c_1^*) \\
&= |c_1|^4 + |c_2|^4 - 2|c_1|^2|c_2|^2 + 4|c_1|^2|c_2|^2 \\
&= (|c_1|^2 + |c_2|^2)^2 \\
&= 1.
\end{aligned}$$

The dynamics are those of a torque (pointed along $-\mathbf{\Omega}$) acting on the vector $\mathbf{R}$

$$\frac{d}{dt}\mathbf{V} = \boldsymbol{\tau} \times \mathbf{V} \rightarrow \frac{d}{dt}\mathbf{R} = -\mathbf{\Omega} \times \mathbf{R}.$$

If we solve to substitute for the dynamics, we find

$$\begin{aligned}
\tilde{c}_1(t) &= -\frac{i\delta}{\Omega} \sin\left(\frac{\Omega t}{2}\right) + \cos\left(\frac{\Omega t}{2}\right) \\
\tilde{c}_2(t) &= -\frac{i\Omega_0}{\Omega} \sin\left(\frac{\Omega t}{2}\right).
\end{aligned}$$

so that for $u(t)$,

$$\begin{aligned}
u(t) &= \tilde{\rho}_{12}(t) + \tilde{\rho}_{21}(t) \\
&= \tilde{c}_1\tilde{c}_2^* + \tilde{c}_2\tilde{c}_1^* \\
&= \left[-\frac{i\delta}{\Omega} \sin\left(\frac{\Omega t}{2}\right) + \cos\left(\frac{\Omega t}{2}\right)\right] \left(\frac{i\Omega_0}{\Omega}\right) \sin\left(\frac{\Omega t}{2}\right) \\
&\quad - \frac{i\Omega_0}{\Omega} \sin\left(\frac{\Omega t}{2}\right) \left[\frac{i\delta}{\Omega} \sin\left(\frac{\Omega t}{2}\right) + \cos\left(\frac{\Omega t}{2}\right)\right] \\
&= \frac{2\delta\Omega_0}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right) \\
&= \frac{\delta\Omega_0}{\Omega^2} [1 - \cos(\Omega t)]
\end{aligned}$$

for $v(t)$,

$$\begin{aligned}
v(t) &= -i(\tilde{\rho}_{12} - \tilde{\rho}_{21}) \\
&= -i\tilde{c}_1\tilde{c}_2^* + i\tilde{c}_2\tilde{c}_1^* \\
&= \left[-\frac{\delta}{\Omega} \sin\left(\frac{\Omega t}{2}\right) - i\cos\left(\frac{\Omega t}{2}\right)\right] \left(\frac{i\Omega_0}{\Omega}\right) \sin\left(\frac{\Omega t}{2}\right) \\
&\quad + \frac{\Omega_0}{\Omega} \sin\left(\frac{\Omega t}{2}\right) \left[\frac{i\delta}{\Omega} \sin\left(\frac{\Omega t}{2}\right) + \cos\left(\frac{\Omega t}{2}\right)\right] \\
&= \frac{2\Omega_0}{\Omega} \cos\left(\frac{\Omega t}{2}\right) \sin\left(\frac{\Omega t}{2}\right) \\
&= \frac{\Omega_0}{\Omega} \sin(\Omega t)
\end{aligned}$$

and for $w(t)$,

$$\begin{aligned}
w(t) &= \tilde{\rho}_{11} - \tilde{\rho}_{22} \\
&= \tilde{c}_1 \tilde{c}_1^* - \tilde{c}_2 \tilde{c}_2^* \\
&= \frac{\delta^2 + \Omega_0^2}{\Omega^2} \cos^2\left(\frac{\Omega t}{2}\right) + \frac{\delta^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right) - \frac{\Omega_0^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right) \\
&= \frac{\delta^2}{\Omega^2} + \frac{\Omega_0^2}{\Omega^2} \left[\cos^2\left(\frac{\Omega t}{2}\right) - \sin^2\left(\frac{\Omega t}{2}\right) \right] \\
&= \frac{\delta^2}{\Omega^2} + \frac{\Omega_0^2}{\Omega^2} \left\{ \frac{1}{2}[1 + \cos(\Omega t)] - \frac{1}{2}[1 - \cos(\Omega t)] \right\} \\
&= \frac{\delta^2}{\Omega^2} + \frac{\Omega_0^2}{\Omega^2} \cos(\Omega t).
\end{aligned}$$

So, in summary, we have

$$\begin{aligned}
u(t) &= \frac{\delta \Omega_0}{\Omega^2} [1 - \cos(\Omega t)] \\
v(t) &= \frac{\Omega_0}{\Omega} \sin(\Omega t) \\
w(t) &= \frac{\delta^2}{\Omega^2} + \frac{\Omega_0^2}{\Omega^2} \cos(\Omega t).
\end{aligned}$$

Observe that these dynamics are all harmonic, with a frequency equal to the generalized Rabi frequency.

To visualize the dynamics of a 2-level system, we will often use the "*Bloch sphere*": a representation of the parameter space spanned by (u, v, w) . There are a couple notes to make here:

- the state is represented by a vector, $\mathbf{R}$, from the origin to the surface of the sphere;
- the three dimensions represent the variables (u, v, w) ;
- the Bloch vector direction helps us see how the effective "torque" causes the state vector to rotate.

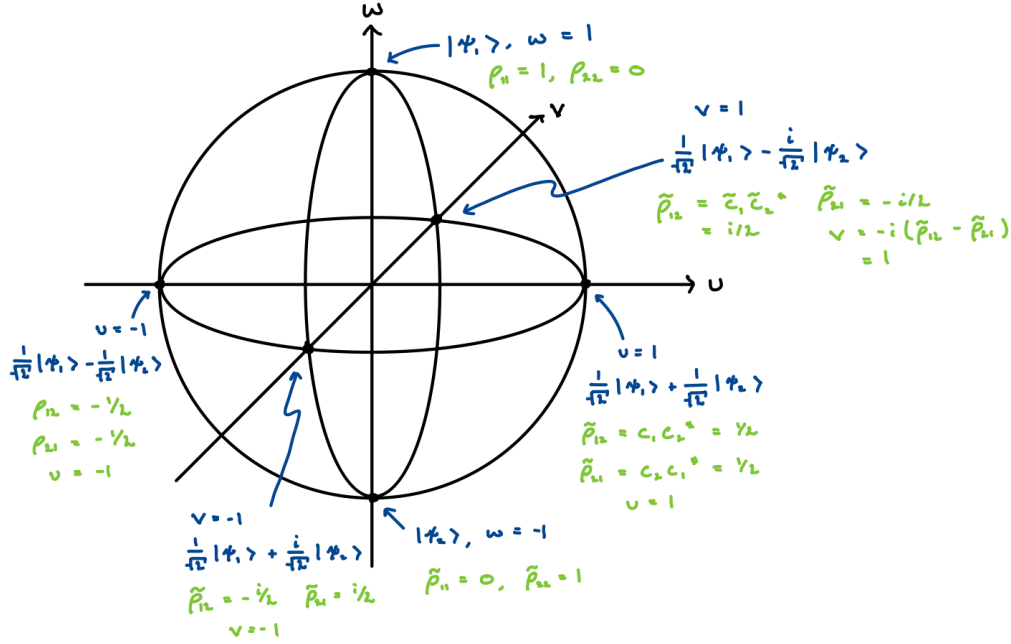


Figure 19: Bloch sphere with extrema related to states $|\psi_1\rangle$ and $|\psi_2\rangle$.

Let us look at some examples. Consider a π -pulse with detuning $\delta = 0$. Then, we have

$$\mathbf{\Omega} = (\Omega_0, 0, 0)$$

with an initial state $|\tilde{\psi}_1\rangle$. We have $\mathbf{R}_1 = (0, 0, 1)$ which suggests (from $u = v = 0$) that there is no coherence. We can determine $\mathbf{R}_2$ by considering conditions

$$\begin{aligned} w(0) &= 1, & w(\pi/\Omega) &= \cos(\pi) = -1 \\ u(0) &= 0, & u(\pi/\Omega) &= 0 \\ v(0) &= 0, & v(\pi/\Omega) &= \sin(\pi) = 0. \end{aligned}$$

Thus, $\mathbf{R}_2 = (0, 0, -1)$.

Looking at a $\pi/2$ -pulse with detuning $\delta = 0$, we have again

$$\mathbf{\Omega} = (\Omega_0, 0, 0)$$

with the same initial state. We have $\mathbf{R}_1 = (0, 0, 1)$ as well, and $\Omega t = \pi/2$. Using our conditions, we find

$$\begin{aligned} w(\pi/2\Omega) &= \cos(\pi/2) = 0 \\ u(\pi/2\Omega) &= 0 \\ v(\pi/2\Omega) &= \sin(\pi/2) = 1 \end{aligned}$$

and, we can determine the superimposed state

$$\rho_{12} - \rho_{21} = i \implies c_1 c_2^* - c_2 c_1^* = \frac{1}{\sqrt{2}} \frac{i}{\sqrt{2}} - \frac{(-i)}{\sqrt{2}} \frac{1}{\sqrt{2}} = i$$

so that

$$|\tilde{\psi}\rangle = \frac{1}{\sqrt{2}} |\psi_1\rangle - \frac{i}{\sqrt{2}} |\psi_2\rangle.$$

The last example we consider is another π -pulse but with detuning $\delta = \Omega_0/2$. Here, we have the same initial state $\mathbf{R}_1 = (0, 0, 1)$, with

$$\Omega = \sqrt{\Omega_0^2 + \frac{1}{4}\Omega_0^2} = \frac{\sqrt{5}}{2}\Omega_0.$$

Using the conditions,

$$\begin{aligned} w(\pi/\Omega) &= \left(1 - \frac{4}{5}\right) + \frac{4}{5} \cos(\pi) \\ &= \frac{1}{5} - \frac{4}{5} \\ &= -\frac{3}{5} \\ u(\pi/\Omega) &= \frac{\Omega_0^2}{2(5/4)\Omega_0^2} [1 - \cos(\pi)] \\ &= \frac{4}{5} \\ v(\pi/\Omega) &= \frac{2}{\sqrt{5}} \sin(\pi) = 0. \end{aligned}$$

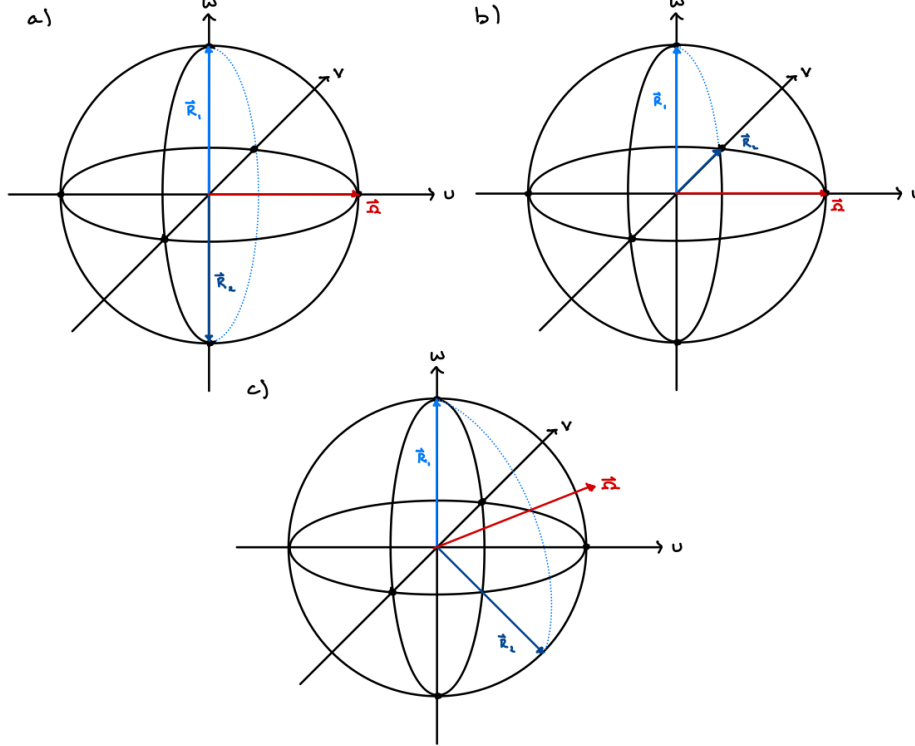


Figure 20: Bloch sphere representation of the pulses applied to a system. a) π -pulse with $\delta = 0$. b) $\pi/2$ -pulse with $\delta = 0$. c) π -pulse with $\delta = \Omega_0/2$.

The physical interpretation of the coherences u and v are found by considering the dipole moment of the 2-level atom under this interaction

$$\mathbf{d} = e\mathbf{r} \rightarrow \langle \mathbf{d} \rangle e\langle \mathbf{r} \rangle.$$

We will take the case for which the electric field, E_0 , is linearly polarized along z :

$$\mathbf{d} = e\langle \mathbf{z} \rangle, \quad |\mathbf{d}| = e|\langle \mathbf{z} \rangle| \equiv d.$$

In our 2-level basis, the operator for this dipole is

$$\hat{d} = e\mathbf{z}$$

and the matrix elements in this 2-level basis are always zero for the diagonal terms. That is,

$$\langle \psi_1 | \mathbf{z} | \psi_1 \rangle = 0, \quad \langle \psi_2 | \mathbf{z} | \psi_2 \rangle$$

because of parity. Thus, the operator is represented by

$$\hat{d} = ez |\psi_1\rangle \langle \psi_2| + ez |\psi_2\rangle \langle \psi_1|$$

in the rotating frame. Our solutions are in the rotating frame; let us rotate them:

$$|\tilde{\psi}_1\rangle = \tilde{c}_1(t) |\tilde{\psi}_1\rangle + \tilde{c}_2(t) |\tilde{\psi}_2\rangle$$

with

$$\begin{aligned} \tilde{c}_1(t) &= e^{-i\delta t/2} c_1(t), & |\tilde{\psi}_1\rangle &= e^{-i\delta t/2} |\psi_1\rangle \\ \tilde{c}_2(t) &= e^{i\delta t/2} c_2(t), & |\tilde{\psi}_2\rangle &= e^{i\omega t} e^{-i\delta t/2} |\psi_2\rangle. \end{aligned}$$

So,

$$\begin{aligned} |\psi_1\rangle &= e^{i\delta t/2} |\tilde{\psi}_1\rangle \\ |\psi_2\rangle &= e^{-i\omega t} e^{i\delta t/2} |\tilde{\psi}_2\rangle. \end{aligned}$$

We have the outer products

$$\begin{aligned} |\psi_1\rangle \langle \psi_1| &= e^{i\delta t/2} |\tilde{\psi}_1\rangle \langle \tilde{\psi}_1| e^{-i\delta t/2} \\ &= |\tilde{\psi}_1\rangle \langle \tilde{\psi}_1| \\ &= \tilde{\rho}_{11} \\ |\psi_1\rangle \langle \psi_2| &= e^{i\delta t/2} |\tilde{\psi}_1\rangle \langle \tilde{\psi}_2| e^{-i\delta t/2} e^{i\omega t} \\ &= e^{i\omega t} |\tilde{\psi}_1\rangle \langle \tilde{\psi}_2| \\ &= e^{i\omega t} \tilde{\rho}_{12} \\ |\psi_2\rangle \langle \psi_1| &= e^{-i\omega t} |\tilde{\psi}_2\rangle \langle \tilde{\psi}_1| \\ &= e^{-i\omega t} \tilde{\rho}_{21}. \end{aligned}$$

So,

$$\begin{aligned}\hat{d} &= ez[e^{i\omega t}\tilde{\rho}_{12} + e^{-i\omega t}\tilde{\rho}_{21}] \\ &= ez[\cos(\omega t)(\tilde{\rho}_{12} + \tilde{\rho}_{21}) + i\sin(\omega t)(\tilde{\rho}_{12} - \tilde{\rho}_{21})] \\ &= ez[u\cos(\omega t) - v\sin(\omega t)].\end{aligned}$$

The expectation value $\langle \hat{d} \rangle$ is the dipole moment, measurable, in the non-rotating frame. Here, we see that it oscillates with the drive frequency, as expected. The component u represents the "in-phase" response, while the component $-v$ represents the "out-of-phase" response.

The existence of superposition (a non-zero coherence) must be present for there to be a dipole moment. That is, there is no dipole moment if the state is in one of the eigenstates.

The resonant response ($\delta = 0$) means $u = 0$ implies a completely out-of-phase response.

2.7 Steady-State Configuration in the Rotating Frame

Suppose we are interested now in a steady-state configuration within the rotating frame. To analyze this, we will start with the time-independent Schrödinger equation, where

$$\tilde{H}|\tilde{\psi}\rangle = \frac{\hbar}{2} \begin{pmatrix} \delta & \Omega_0 \\ \Omega_0 & -\delta \end{pmatrix} |\tilde{\psi}\rangle = E|\tilde{\psi}\rangle.$$

To find solutions, we diagonalize the matrix with $\hbar\lambda/2 = E$, so

$$\frac{\hbar}{2} \begin{vmatrix} \delta - \lambda & \Omega_0 \\ \Omega_0 & -\delta - \lambda \end{vmatrix} = 0$$

which should give the solutions

$$\lambda = \pm \sqrt{\Omega_0^2 + \delta^2} \implies E_{\pm} = \pm \frac{\hbar}{2} \sqrt{\Omega_0^2 + \delta^2} = \pm \frac{\hbar\Omega}{2}.$$

The new energies are split by $\hbar\Omega = E_+ - E_-$. We might ask: what are these states? For the E_+ state, we have

$$\begin{aligned}E_+ \rightarrow |\tilde{\psi}_+\rangle : \frac{\hbar}{2} \begin{pmatrix} \delta & \Omega_0 \\ \Omega_0 & -\delta \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \frac{\hbar}{2} \Omega \begin{pmatrix} a \\ b \end{pmatrix} \\ \delta a + \Omega_0 b &= \Omega a \\ b &= \frac{1}{\Omega_0}(\Omega - \delta)a\end{aligned}$$

where $|\psi\rangle = a|\psi_1\rangle + b|\psi_2\rangle$. Thus, we find

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \Omega_0 \\ \Omega - \delta \end{pmatrix}.$$

Normalizing, we find that

$$\sqrt{a^2 + b^2} = (\Omega_0^2 + \Omega^2 - 2\Omega\delta + \delta^2)^{1/2} = (2\Omega^2 - 2\Omega\delta)^{1/2} = [2\Omega(\Omega - \delta)]^{1/2}.$$

Thus, we should find that

$$\left| \tilde{\psi}_+ \right\rangle = \frac{\Omega_0}{\sqrt{2\Omega(\Omega - \delta)}} |\psi_1\rangle + \frac{\Omega - \delta}{\sqrt{2\Omega(\Omega - \delta)}} |\psi_2\rangle.$$

Or, more compactly, we could have decomposed the coefficients into a triangle (Fig. 21) with

$$\tan \phi = \frac{\Omega - \delta}{\Omega_0}$$

so we could have written

$$\left| \tilde{\psi}_+ \right\rangle = \cos \phi |\psi_1\rangle + \sin \phi |\psi_2\rangle.$$

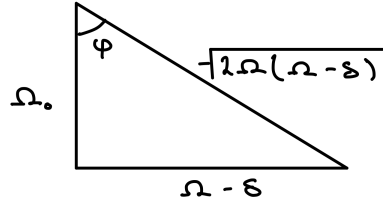


Figure 21: Coefficients of $|\psi\rangle = a|\psi_1\rangle + b|\psi_2\rangle$ decomposed into a triangle with angle ϕ .

Similarly, for E_- , we have

$$\begin{aligned} E_- \rightarrow \left| \tilde{\psi}_- \right\rangle : \begin{pmatrix} \delta & \Omega_0 \\ \Omega_0 & -\delta \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= -\Omega \begin{pmatrix} a \\ b \end{pmatrix} \\ \delta a + \Omega_0 b &= -\Omega a \\ b &= \frac{1}{\Omega_0}(\delta + \Omega) \end{aligned}$$

and

$$\begin{aligned} \Omega_0 a - \delta b &= -\Omega b \\ a &= \frac{1}{\Omega_0}(-\Omega + \delta) \\ &= -\frac{\Omega - \delta}{\Omega_0} b. \end{aligned}$$

Thus, we find

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -(\Omega - \delta) \\ \Omega_0 \end{pmatrix}.$$

Thus, we have the state

$$\left| \tilde{\psi}_+ \right\rangle = -\sin \phi |\psi_1\rangle + \cos \phi |\psi_2\rangle$$

or,

$$\left| \tilde{\psi}_+ \right\rangle = -\frac{\Omega - \delta}{\sqrt{2\Omega(\Omega - \delta)}} |\psi_1\rangle + \frac{\Omega_0}{\sqrt{2\Omega(\Omega - \delta)}} |\psi_2\rangle.$$

Thus, for our energy eigenstates, we have

$$\begin{aligned} E_+ &= \frac{\hbar\Omega}{2}, & |\tilde{\psi}_+\rangle &= \cos\frac{\theta}{2}|\psi_1\rangle + \sin\frac{\theta}{2}|\psi_2\rangle \\ E_- &= -\frac{\hbar\Omega}{2}, & |\tilde{\psi}_-\rangle &= -\sin\frac{\theta}{2}|\psi_1\rangle + \cos\frac{\theta}{2}|\psi_2\rangle, \end{aligned} \quad \text{with } \tan\theta = \frac{\Omega_0}{\delta}.$$

It is of our interest to analyze special cases. Suppose $\delta = 0$, then

$$\tan\theta \rightarrow \infty \implies \theta = \frac{\pi}{2}.$$

We have the energy eigenstates $E_{\pm} = \pm\hbar\Omega_0/2$ and the states

$$\begin{aligned} |\tilde{\psi}_+\rangle &= \cos\left(\frac{\pi}{4}\right)|\psi_1\rangle + \sin\left(\frac{\pi}{4}\right)|\psi_2\rangle = \frac{1}{\sqrt{2}}(|\psi_1\rangle + |\psi_2\rangle). \\ |\tilde{\psi}_-\rangle &= \frac{1}{\sqrt{2}}(-|\psi_1\rangle + |\psi_2\rangle). \end{aligned}$$

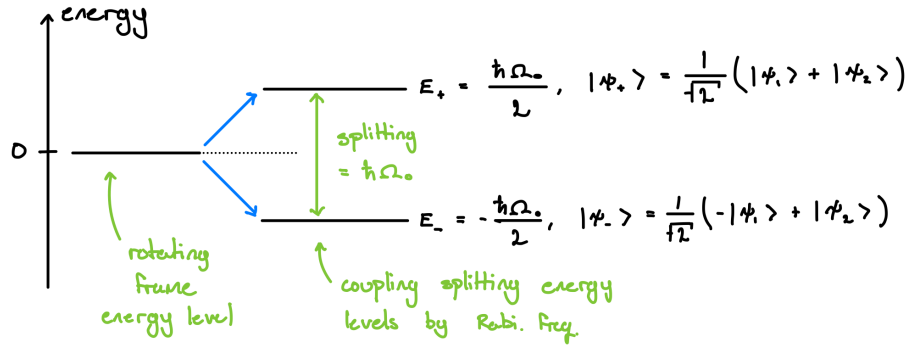


Figure 22: Ground state energy splitting under rotation for the case $\delta = 0$.

For $\delta \gg \Omega_0$ (large detuning), we have the energy eigenstates

$$E_{\pm} = \pm\frac{\hbar}{2}\sqrt{\Omega_0^2 + \delta^2} = \pm\frac{\hbar}{2}\delta\sqrt{1 + \frac{\Omega_0^2}{\delta^2}} \approx \pm\frac{\hbar}{2}\delta\left(1 + \frac{1}{2}\frac{\Omega_0^2}{\delta^2}\right) = \pm\frac{\hbar\delta}{2} \pm \frac{\hbar\Omega_0^2}{4\delta}.$$

Supposing $\delta > 0$, we have $\tan\theta = \Omega_0/\delta \rightarrow 0^+$ so that $\sin\theta/2 \approx \theta/2$ and $\cos\theta/2 \approx 1$. Thus,

$$\begin{aligned} |\tilde{\psi}_+\rangle &\sim \frac{1}{N}\left(|\psi_1\rangle + \frac{\Omega_0}{2\delta}|\psi_2\rangle\right) \\ &= \frac{1}{(1 + \Omega_0^2/4\delta^2)^{1/2}}\left(|\psi_1\rangle + \frac{\Omega_0}{2\delta}|\psi_2\rangle\right) \\ |\tilde{\psi}_-\rangle &= \frac{1}{(1 + \Omega_0^2/4\delta^2)^{1/2}}\left(-\frac{\Omega_0}{2\delta}|\psi_1\rangle + |\psi_2\rangle\right). \end{aligned}$$

Here, $|\tilde{\psi}_+\rangle$ is mostly $|\psi_1\rangle$, experiencing an energy shift of $+\hbar\Omega_0^2/4\delta = +\Delta$, while $|\tilde{\psi}_-\rangle$ is mostly $|\psi_2\rangle$, experiencing an energy shift of $-\hbar\Omega_0^2/4\delta = -\Delta$.

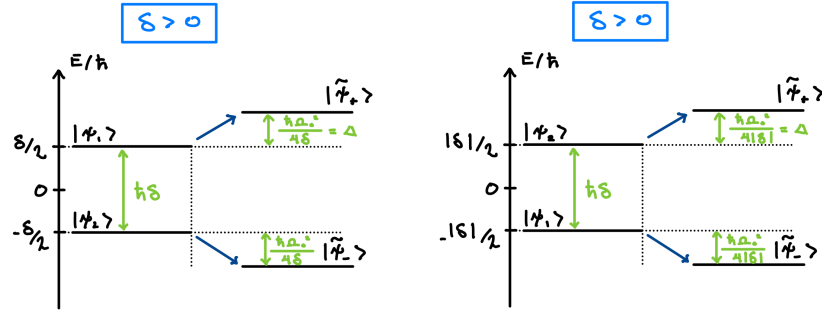


Figure 23: Red and blue-detuning for the ground state energy. The blue detuning corresponds to $\delta > 0$, where the ground state energy shifts up, while the red detuning corresponds to $\delta < 0$, where the ground state energy shifts down

We can also move back to the non-rotating frame (Fig. 24).

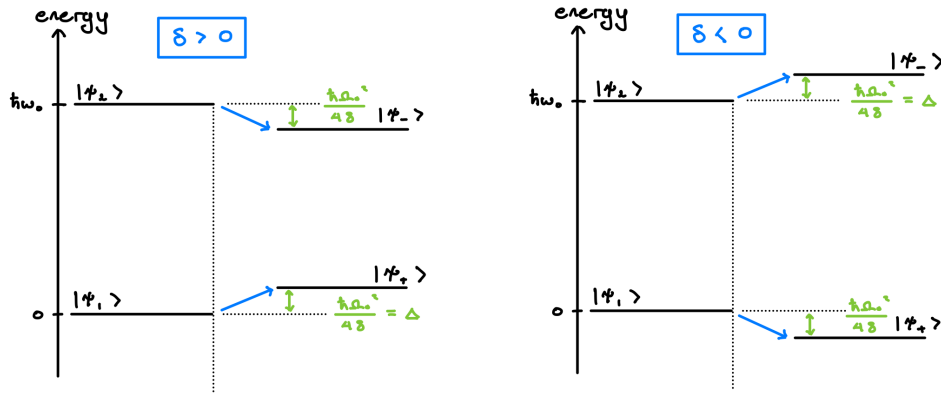


Figure 24: Non-rotating frame for the red and blue-detuning for the ground state energy. Observe how we have a shift $|\psi_2\rangle$ up by $\hbar\omega_0 = \hbar\delta/2$; and a shift $|\psi_1\rangle$ down by $\hbar\delta/2$.

These energy shifts are exactly the ac-Stark shifts due to the interaction of the oscillating field with our system. The same result can be found by doing second order time-dependent perturbation theory.

Further, the off-resonant radiation mixes a small portion of the "other" state into the new eigenstate, which shifts the energy levels by

$$\Delta = \frac{\hbar\Omega_0^2}{4\delta} \propto E_0^2 \propto \text{Intensity} \propto \text{Power}.$$

This shift can be quite useful in, for example, optical trapping.

2.8 Application: Optical Trapping and Optical Lattices

Suppose we wish to use this ac-Stark shift to tailor the potential energy landscape as seen by the atoms. Here, we have a light-shift of

$$\hbar\Delta = \frac{\hbar\Omega_0^2}{4\delta}.$$

For ^{87}Rb specifically, we shall consider the transition matrix element.

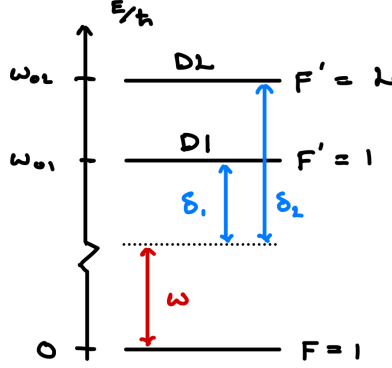


Figure 25: D1 and D2 line relative to the ground state $F = 1$ line.

Here, we have

$$\hbar\Omega_0 = \langle \psi_2 | \mathbf{e} \mathbf{r} \cdot \mathbf{E}_0 | \psi_1 \rangle = \mathbf{e}_r \cdot \mathbf{E}_0 \langle \psi_2 | e r | \psi_1 \rangle.$$

We can easily look up the energies of the D-lines, which are

$$\begin{aligned} \langle F' = 2 | e r | F = 1 \rangle &= 3.584 \times 10^{-29} \text{ Cm} \\ &= 4.226 e a_0 \\ \langle F' = 1 | e r | F = 1 \rangle &= 2.537 \times 10^{-29} \text{ Cm} \\ &= 2.992 e a_0. \end{aligned}$$

So, the total light shift is

$$\hbar\Delta = \frac{\hbar|\Omega_0|^2}{4\delta} = \frac{\hbar|\Omega_0|_{D1}^2}{4\delta_{D1}} + \frac{\hbar|\Omega_0|_{D2}^2}{4\delta_{D2}} = \frac{|E_0|^2}{4\hbar} \left[\frac{|\langle F' = 1 | e r | F = 1 \rangle|^2}{(\omega - \omega_{D1})} + \frac{|\langle F' = 2 | e r | F = 1 \rangle|^2}{(\omega - \omega_{D2})} \right].$$

We will use the "far-detuned light" at 815 nm, from a cw-Ti:sapph that outputs 1 Watt of power. The D-lines are D2 : 780.246 nm and D1 : 794.979 nm. So,

$$\begin{aligned} \omega_{D1} &= \frac{2\pi c}{\lambda_{D1}} = 2.3695 \times 10^{15} \text{ rad/s} \\ \omega_{D2} &= \frac{2\pi c}{\lambda_{D2}} = 2.4142 \times 10^{15} \text{ rad/s} \end{aligned}$$

with

$$\text{Intensity} = \frac{\text{Power}}{\text{Area}} = \frac{c\epsilon_0}{2} E_0^2 = \frac{P}{A} \implies E_0^2 = \frac{2P}{c\epsilon_0 A}.$$

We have two cases. The first is the case where we focus the light in a single beam, to a "waist" of $50 \mu\text{m}$.

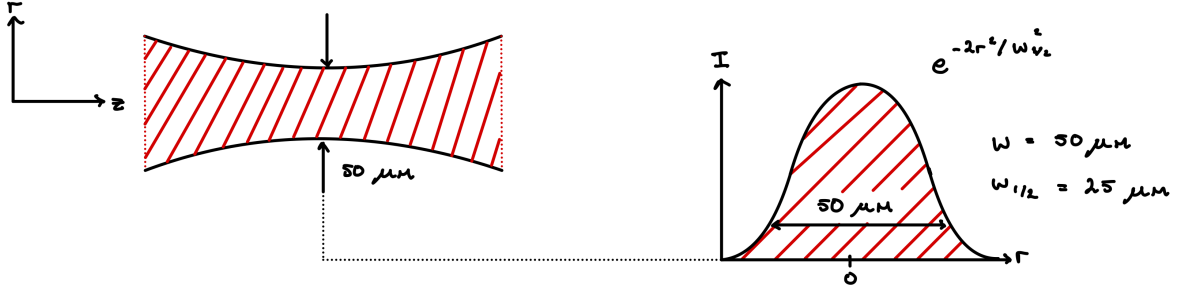


Figure 26: A single beam of light focused to have a waist of $50 \mu\text{m}$, along with its Gaussian spread.

Here, we have $A \approx \pi r^2 = \pi(25 \mu\text{m})^2$. So, we have an intensity of

$$E_0^2 = \frac{2(1 \text{ W})}{c\epsilon_0\pi(25 \times 10^{-6} \text{ m})^2} = 3.839 \times 10^{11} \text{ V}^2/\text{m}^2.$$

We also have a detuning frequency of

$$\begin{aligned}\delta_{D1} &= \frac{2\pi c}{815 \text{ nm}} - \frac{2\pi c}{794.979 \text{ nm}} = -5.821 \times 10^{13} \text{ rad/s} \\ \delta_{D2} &= \frac{2\pi c}{815 \text{ nm}} - \frac{2\pi c}{780.246 \text{ nm}} = -1.0295 \times 10^{14} \text{ rad/s}.\end{aligned}$$

Thus,

$$\hbar\Delta = \frac{E_0^2}{4\hbar} \left[\frac{(2.537 \times 10^{-29} \text{ Cm})^2}{-5.821 \times 10^{13} \text{ s}^{-1}} + \frac{(3.584 \times 10^{-29} \text{ Cm})^2}{-1.0295 \times 10^{14} \text{ s}^{-1}} \right] = -2.142 \times 10^{-26} \text{ J}$$

and

$$\frac{\hbar\Delta}{k_B} = 1.55 \text{ mK}, \quad \frac{\hbar\Delta}{h} = 32.3 \text{ MHz}.$$

We see that these atoms experience a potential that is attractive (due to the negative sign), and deep enough to confine atoms whose kinetic energy is a few times less than a mK of temperature.

The second case is where we have an optical lattice. We may use the interference of counter-propagating light to create a standing wave, and thus a periodic potential. Let us assume that we have the same laser at our disposal.¹⁴



Figure 27: Laser sent to stimulate a cloud of atoms, then reflected by a mirror onto itself.

¹⁴I think "prism" is wrong in Fig. 27. But that's okay, just figure out what it is later.

The electric field of the retro-reflected beam is

$$\begin{aligned}\mathbf{E} &= E_0 \cos(kz - \omega t) \mathbf{e}_y + E_0 \cos(-kz - \omega t) \mathbf{e}_y \\ &= E_0 [\cos(kz) \cos(-\omega t) + \sin(kz) \sin(-\omega t) + \cos(-kz) \cos(-\omega t) + \sin(-kz) \sin(-\omega t)] \mathbf{e}_y \\ &= 2E_0 \cos(kz) \cos(\omega t) \mathbf{e}_y.\end{aligned}$$

Here, $\cos(kz)$ is the spatial interference term, while $\cos(\omega t)$ is our usual time-dependence term. We treat $2E_0 \cos(kz)$ as a spatially varying field amplitude, and use it as E_0 in the above formulae. Thus,

$$\begin{aligned}\hbar\Delta &= \frac{E^2}{4\hbar} \left[\frac{(2.537 \times 10^{-29} \text{ Cm})^2}{-5.821 \times 10^{13} \text{ s}^{-1}} + \frac{(3.584 \times 10^{-29} \text{ Cm})^2}{-1.0295 \times 10^{14} \text{ s}^{-1}} \right] \\ &= 4 \cos^2(kz) \frac{E_0^2}{4\hbar} \left[\frac{(2.537 \times 10^{-29} \text{ Cm})^2}{-5.821 \times 10^{13} \text{ s}^{-1}} + \frac{(3.584 \times 10^{-29} \text{ Cm})^2}{-1.0295 \times 10^{14} \text{ s}^{-1}} \right] \\ &= 4(-2.142 \times 10^{-26} \text{ J}) \cos^2(kz) \\ &= -8.568 \times 10^{-26} \text{ J} \cdot \cos^2(kz).\end{aligned}$$

Of course, we note that

$$\frac{\hbar\Delta}{k_B} = -6.21 \text{ mK} \cdot \cos^2(kz).$$

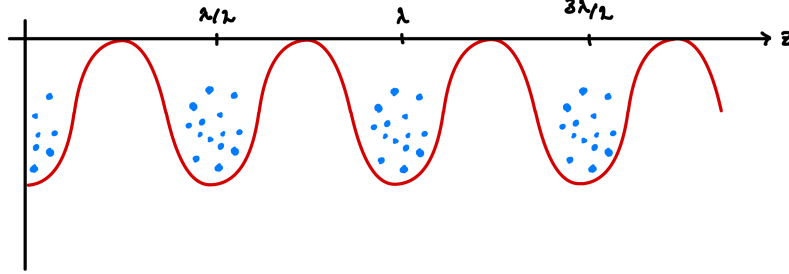


Figure 28: Periodically attractive potential for a cloud of atoms stimulated by a ground-state "light shift."

Observe that the ground-state "light shift" creates a periodically attractive potential for the atoms. This result has applications in "quantum simulation," where we may engineer atomic systems that look like crystal structures of electrons in a solid-state lattice.

And, to change the pattern of trapping, we need only to interfere light in multiple dimensions to change the pattern of trapping.

2.9 Transition Rates and the Einstein Equations

The semiclassical description also lets us consider non-monochromatic radiation. That is, light with a finite bandwidth. For our purposes, we will consider "quasi-mono-chromatic" light whose bandwidth is much less than the central/carrier frequency.

In our non-rotating frame, we have

$$\begin{aligned} i\dot{c}_1(t) &= \frac{\Omega_0}{2} e^{i\delta t} c_2(t) \\ i\dot{c}_2(t) &= \frac{\Omega_0}{2} e^{-i\delta t} c_1(t) \end{aligned}$$

and assume *weak excitation*: the population in the excited state remains small. So, we may assume that $c_1(t) \approx c_1(0) = 1$, to give

$$\begin{aligned} \dot{c}_2(t) &= -i \frac{\Omega_0}{2} e^{-i\delta t} \\ c_2(t) &= -\frac{\Omega_0}{2} \int_0^t dt' e^{-i\delta t'} \\ &= -\frac{i\Omega_0}{2} \left(\frac{1}{-i\delta} \right) e^{-i\delta t'} \Big|_0^t \\ &= \frac{\Omega_0}{2\delta} (e^{-i\delta t} - 1) \\ &= \frac{\Omega_0}{2\delta} e^{-i\delta t/2} (e^{-i\delta t/2} - e^{i\delta t/2}) \\ &= \frac{\Omega_0}{2\delta} e^{-i\delta t/2} (-2i) \sin\left(\frac{\delta t}{2}\right) \\ &= -\frac{i\Omega_0}{\delta} e^{-i\delta t/2} \sin\left(\frac{\delta t}{2}\right). \end{aligned}$$

The population of our excited states are

$$|c_2(t)|^2 = \frac{|\Omega_0|^2}{\delta^2} \sin^2\left(\frac{\delta t}{2}\right) = \frac{|\Omega_0|^2 t^2/4}{\delta^2 t^2/4} \sin^2\left(\frac{\delta t}{2}\right) = \frac{|\Omega_0|^2 t^2}{4} \text{sinc}^2\left(\frac{\delta t}{2}\right).$$

From our result, we can see that the probability of a transition peaks when $\delta = 0$, since $\text{sinc}(\delta t/2) = 0$. The probability also depends on the matrix-element-squared, namely $|\Omega_0|^2$.

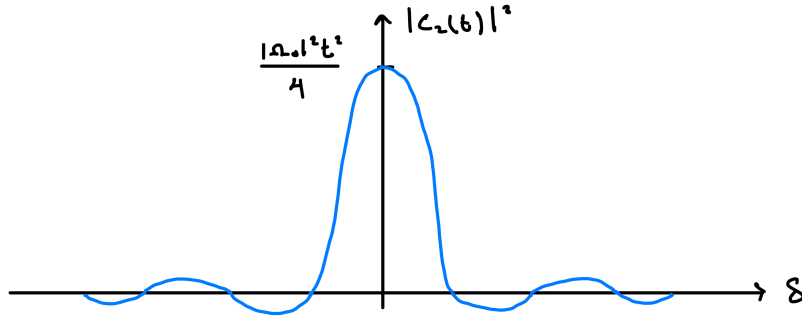


Figure 29: Probability population of the excited state as a function of δ . Observe the peak at $\delta = 0$.

The energy density of a classical EM field is given by

$$U = \int d\omega u(\omega) = \frac{\epsilon_0 E_0^2}{2}$$

which has units of energy per volume. We define U by the energy density per unit frequency, since we often know that instead (for example, in the blackbody problem).

Given the definition of Ω_0 , we have

$$|\Omega_0|^2 = \frac{e^2 E_0^2}{\hbar^2} |\langle \psi_2 | \mathbf{e}_{\text{rad}} \cdot \mathbf{r} | \psi_1 \rangle|^2 \equiv \frac{e^2 E_0^2 X_{12}}{\hbar^2}$$

where X_{12} is the matrix element $X_{12} = |\langle \psi_2 | \mathbf{e}_{\text{rad}} \cdot \mathbf{r} | \psi_1 \rangle|$ that holds all the information from atomic physics, and selection rules. So,

$$|\Omega_0|^2 = \frac{e^2}{\hbar^2} \left(\frac{2U}{\epsilon_0} \right) X_{12}^2 = \frac{2e^2 X_{12}^2}{\epsilon_0 \hbar^2} \int d\omega u(\omega).$$

Thus, the population of the excited state is

$$|c_2(t)|^2 = \frac{e^2 X_{12}^2}{2\epsilon_0 \hbar^2} \int_0^\infty d\omega t^2 u(\omega) \text{sinc}^2\left(\frac{\delta t}{2}\right) = \frac{e^2 X_{12}^2}{2\epsilon_0 \hbar^2} \int_0^\infty d\omega t^2 u(\omega) \text{sinc}^2\left[\frac{(\omega - \omega_0)t}{2}\right].$$

Using the relationship of the sinc function to the Dirac-delta function, we have

$$\delta(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{\pi x} \sin\left(\frac{x}{\epsilon}\right).$$

Observe that $\epsilon \rightarrow 1/t$ and $x \rightarrow (\omega - \omega_0)/2$, so

$$\lim_{t \rightarrow \infty} \frac{2}{\pi(\omega - \omega_0)} \sin\left[\frac{(\omega - \omega_0)t}{2}\right] = \lim_{t \rightarrow \infty} \frac{t}{\pi(\omega - \omega_0)(t/2)} \sin\left[\frac{(\omega - \omega_0)t}{2}\right] = \delta(\omega - \omega_0).$$

Inserting this into the above integral, we replace one sinc with our result, so

$$|c_2(t)|^2 = \frac{e^2 X_{12}^2}{2\epsilon_0 \hbar^2} \int_0^\infty d\omega t u(\omega) \text{sinc}\left[\frac{(\omega - \omega_0)t}{2}\right] \pi \delta(\omega - \omega_0) = \frac{\pi e^2 X_{12}^2}{2\epsilon_0 \hbar^2} t u(\omega_0) \text{sinc}(0).$$

The rate of population transfer is thus

$$\frac{d}{dt} |c_2(t)|^2 = \frac{\pi e^2 X_{12}^2}{2\epsilon_0 \hbar^2} u(\omega_0) = R_{21}.$$

Comparing this to the Einstein coefficient rate formulas, there are a few things we can analyze.¹⁵ To do so, we will consider three processes: (i) absorption (B_{12}); (ii) stimulated emission (B_{21}); and (iii) spontaneous emission (A_{21}). Note that (i) and (ii) depend on how much light is around, while (iii) does not.

We thus define the rate of these three as

- (i) Absorption: $B_{12}u(\omega)$
- (ii) Stimulated Emission: $B_{21}u(\omega)$
- (iii) Spontaneous Emission: A_{21} .

The populations follow by

$$N_2 = N |c_2(t)|^2, \quad N_1 = N |c_1(t)|^2.$$

¹⁵The "Einstein rate equations" and the "Einstein coefficients" are often used, especially in the theory of lasers, to describe the detailed balance of transitions in a 2-level atom.

The rate equations follow

$$\begin{aligned}\frac{\partial N_2}{\partial t} &= -N_2 B_{21} u(\omega) + N_1 B_{12} u(\omega) - N_2 A_{21} \\ \frac{\partial N_1}{\partial t} &= -\frac{\partial N_2}{\partial t} = N_2 B_{21} u(\omega) - N_1 B_{12} u(\omega) + N_2 A_{21}.\end{aligned}$$

Assuming $N_2 + N_1 = N$ is constant (no emitters are added or subtracted from the closed system), we have

$$\frac{\partial N_{1,2}}{\partial t} = N_2 B_{21} u(\omega) - N_1 B_{12} u(\omega) + N_2 A_{21} = 0 \implies u(\omega) = \frac{A_{21} N_2}{N_1 B_{12} - N_2 B_{21}}.$$

In steady-state, there is a constant exchange of energy between the field and the atoms: absorption takes energy from the field and adds it to an atom, while emission takes energy from the atom and adds it to the field. Steady-state is when these two things balance. This is also the condition for equilibrium in the *blackbody (BB) problem*. Under these conditions, there is a constant energy density in the field and constant populations in the 2-levels.

Next, we will apply the Maxwell-Boltzmann distribution function to the populations, so

$$\frac{N_2}{N_1} = \frac{e^{-E_2/k_B T}}{e^{-E_1/k_B T}} = e^{-(E_2-E_1)/k_B T}.$$

The resonant frequency of the light we consider is thus related to this energy difference, so

$$\omega = \frac{E_2 - E_1}{\hbar} \implies e^{-(E_2-E_1)/k_B T} = e^{-\hbar\omega/k_B T}$$

and the energy density is

$$u(\omega) = \frac{A_{21} N_1 e^{-\hbar\omega/k_B T}}{N_1 B_{12} - N_1 B_{21} e^{-\hbar\omega/k_B T}} = \frac{A_{21}}{B_{12} e^{\hbar\omega/k_B T} - B_{21}}.$$

Finally, since this really is just the BB problem, we will use those results (from many other books) to describe our physical situation, where

$$u_{BB}(\omega) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\hbar\omega/k_B T} - 1}$$

which arises due to Bose statistics. This has the same form as our expression so long as $B_{12} = B_{21}$ and the ratio

$$\frac{A_{21}}{B_{21}} = \frac{\hbar\omega^3}{\pi^2 c^3}$$

holds. Observe that the $\pi^3 c^3$ pre-factor comes from the counting of modes of the EM field in the BB problem.

Notice that the rates of absorption and stimulated emission are equal, indicating the analogous nature of these two processes, while the rates of stimulated emission and spontaneous emission differ only by constant, as they are intrinsically related processes.

Back to our earlier result for the rate of change in the populations, we have

$$\frac{\partial}{\partial t} |c_2(t)|^2 = \frac{\pi e^2 X_{12}^2}{\epsilon_0 \hbar^2} u(\omega) \equiv B_{12} u(\omega).$$

Using $X_{12}^2 = D_{12}^2/3$, where $\mathbf{D}_{12} = |\langle \psi_2 | \mathbf{r} | \psi_1 \rangle|$ (and averaging over all 3 directions and polarizations¹⁶), we find

$$B_{21} = \frac{\pi e^2 D_{12}^2}{3\epsilon_0 \hbar^2}, \quad A_{21} = \frac{\pi e^2 D_{12}^2 \hbar \omega_0^3}{3\epsilon_0 \hbar^2 \pi^2 c^3} = \frac{\omega_0^3}{3\pi\epsilon_0 \hbar c^3} (eD_{12})^2.$$

Observe that all these rates depend on D_{12}^2 , the matrix element squared for the transition. The larger the matrix element, the more likely the transition will occur. This scales the same way for the stimulated and spontaneous transitions. All three processes are driven by the same physical mechanism; the inducing of an electric dipole moment in the atom.

Spontaneous emission depends on ω like ω^3 , which arises from the counting of available modes in 3D space. The number of available modes is greater for higher frequency (thus lower wavelength light). Lower frequency transitions are less susceptible to spontaneous emission, and thus can be "long lived." This, for example, arises in hyperfine transitions in hydrogen and the alkali's.

Now, we will take a closer look at spontaneous emission. The rate of spontaneous emission is

$$\frac{\partial N_2}{\partial t} = -A_{21}N_2 \implies N_2(t) = N_2(0)e^{-A_{21}t} \equiv N_2(0)e^{-\Gamma_{21}t} \equiv N_2(0)e^{-t/\tau_{21}}.$$

The damping constant and the damping time are related to A_{21} by

$$\Gamma_{21} = A_{21} = \frac{1}{\tau_{21}}.$$

Since the lifetime is rather straightforwardly measured, we can use these results to get an experimental measure of the transition matrix element. Similarly, the transition linewidth is identically $\Gamma_{21} = A_{21}$, which controls the width of the transition in frequency space

$$\Gamma = \frac{\omega^3 (eD_{12})^2}{3\pi\epsilon_0 \hbar c^3} \implies (eD_{12})^2 = \frac{3\pi\epsilon_0 \hbar c^3}{\omega^3} \Gamma.$$

Thus,

$$B_{12} = \frac{\pi}{3\epsilon_0 \hbar^3} \frac{3\pi\epsilon_0 \hbar c^3}{\omega^3} \Gamma = \frac{\pi^2 c^3}{\hbar \omega^3} \Gamma = \frac{\lambda^3}{8\pi} \frac{\Gamma}{\hbar}$$

using $\omega/c = 2\pi/\lambda$. We can apply this result to the other quantities we have calculated,

$$|\Omega_0|^2 = \frac{|eD_{12}|^2}{\hbar^2} E_0^2 = \frac{3\pi\epsilon_0 c^3}{\hbar \omega^3} \Gamma E_0^2 = \frac{3\pi\epsilon_0}{\hbar \omega^3} \left(\frac{2I}{\epsilon_0 c} \right) \Gamma$$

where we have used $I = \epsilon_0 c E_0^2/2$. Thus,

$$|\Omega_0|^2 = \frac{6\pi c^2}{\hbar \omega^3} \Gamma I$$

showing that the Rabi frequency squared is proportional to intensity and linewidth. So, the Rabi frequency is the quantity that encapsulates all of the atomic information. It also depends on the field condition, that is, its amplitude and intensity.

For the light or ac-Stark shift, we have

$$\hbar \Delta = \frac{\hbar |\Omega|^2}{4\delta} = \frac{3\pi c^2}{2\omega^3} \frac{\Gamma}{\delta} I.$$

¹⁶See Foot chapter 7.

2.10 Optical Bloch Equations with Damping

In our previous examination of 2-level dynamics, our system conserved energy; now we will introduce damping through spontaneous emission, where

$$\begin{aligned} N_2 &= N_{\text{tot}} \tilde{\rho}_{22} = N_2 \tilde{c}_2^* \tilde{c}_2 \\ N_2(t) &= N_2(0) e^{-\Gamma t} \end{aligned}$$

so that

$$\begin{aligned} \tilde{\rho}_{22}(t) &= \tilde{\rho}_{22}(0) e^{-\Gamma t} \\ \dot{\tilde{\rho}}_{22} &= -\Gamma \tilde{\rho}_{22} \\ \dot{\tilde{\rho}}_{11} &= \Gamma \tilde{\rho}_{22}. \end{aligned}$$

Thus, we arrive at the population difference

$$\begin{aligned} w &= \tilde{\rho}_{11} - \tilde{\rho}_{22} \\ \dot{w} &= \dot{\tilde{\rho}}_{11} - \dot{\tilde{\rho}}_{22} = \Gamma \tilde{\rho}_{22} - (-\Gamma \tilde{\rho}_{22}) = 2\Gamma \tilde{\rho}_{22} \\ \dot{w} &= \Gamma(1 - w) = -\Gamma(w - 1). \end{aligned}$$

If we added this loss term "by hand" to the Hamiltonian dynamics of before, then we would have

$$\dot{w} = -\Omega_0 v - \Gamma(w - 1).$$

Similarly, the coherences are affected by this damping, but a bit differently. If an atom is in a state at t_1 , then

$$|\psi(t_1)\rangle = c_1(t_1) e^{-i\omega_1 t} |\psi_1\rangle + c_2(t) e^{-i\omega_2 t} |\psi_2\rangle$$

and undergoes spontaneous emission, after which at $t = t_2$,

$$|\psi(t_2)\rangle \rightarrow 1 e^{-i\omega_1 t} |\psi_1\rangle \implies c_1(t_2) = 1, \quad c_2(t_2) = 0.$$

So, coherence (superposition) is lost by spontaneous emission. We can argue that the effect of spontaneous emission is on the coefficient c_2 ($\tilde{c}_2$), not c_1 ($\tilde{c}_1$). We take

$$\tilde{c}_2(t) = \tilde{c}_2(0) e^{-\gamma t}$$

so that

$$\begin{aligned} \tilde{\rho}_{12} &= \tilde{c}_1 \tilde{c}_2^* \implies \tilde{\rho}_{12}(t) = \tilde{c}_1(0) \tilde{c}_2^*(0) e^{-\gamma t} \\ \tilde{\rho}_{21}(t) &= \tilde{c}_2 \tilde{c}_1^* \implies \tilde{\rho}_{21}(t) = \tilde{c}_2(0) \tilde{c}_1^*(0) e^{-\gamma t} \\ \tilde{\rho}_{22}(t) &= \tilde{c}_2 \tilde{c}_2^* \implies \tilde{\rho}_{22}(t) = \tilde{c}_2(0) \tilde{c}_2^*(0) e^{-2\gamma t}. \end{aligned}$$

so, we find

$$2\gamma = \Gamma \implies \gamma = \frac{\Gamma}{2}.$$

Thus, on the coherences, we have

$$\begin{aligned} u &= \tilde{\rho}_{12} + \tilde{\rho}_{21} \\ \dot{u} &= \dot{\tilde{\rho}}_{12} + \dot{\tilde{\rho}}_{21} + -\gamma u + (\text{other terms}) = -\frac{\Gamma}{2} u + (\text{other terms}) \\ \dot{v} &= -i(\tilde{\rho}_{12} - \tilde{\rho}_{21}) = -\frac{\Gamma}{2} v + (\text{other terms}) \end{aligned}$$

which leaves us with the *Optical Bloch Equations* (OBE).

$$\begin{aligned}\dot{u} &= \delta v - \frac{\Gamma}{2}u \\ \dot{v} &= -\delta v + \Omega_0 w - \frac{\Gamma}{2}v \\ \dot{w} &= -\Omega_0 v - \Gamma(w - 1).\end{aligned}$$

We can use these equations to study various steady state solutions, modelling the behaviour of a 2-level system subject to spontaneous emission. We will set $\dot{u} = \dot{v} = \dot{w} = 0$ so that

$$\begin{aligned}\delta v - \frac{\Gamma}{2}u &= 0 \\ -\delta u + \Omega_0 w - \frac{\Gamma}{2}v &= 0 \\ -\Omega_0 v - \Gamma(w - 1) &= 0\end{aligned}$$

and we substitute $u = 2\delta v/\Gamma$ so that

$$\begin{aligned}-\frac{2\delta^2 v}{\Gamma} + \Omega_0 w - \frac{\Gamma}{2}v &= 0 \\ v \frac{\Gamma}{2} \left[\frac{\delta^2}{(\Gamma/2)^2} + 1 \right] &= \Omega_0 w\end{aligned}$$

and

$$\begin{aligned}-\Omega_0 v - \Gamma(w - 1) &= 0 \\ -\frac{\Omega_0^2 w}{\Gamma[1 + \delta^2/(\Gamma/2)^2]/2} - \Gamma(w - 1) &= 0 \\ w \left\{ \frac{\Omega_0^2}{\Gamma[1 + \delta^2/(\Gamma/2)^2]/2} + \frac{\Gamma(\Gamma/2)[1 + \delta^2/(\Gamma/2)^2]}{(\Gamma/2)[1 + \delta^2/(\Gamma/2)^2]} \right\} &= \Gamma.\end{aligned}$$

Thus,

$$\begin{aligned}w &= \frac{(\Gamma^2/2)[1 + \delta^2/(\Gamma/2)^2]/2}{(\Omega_0^2 + \Gamma^2/2 + 2\delta^2)/2} = \frac{(\Gamma/2)^2 + \delta^2}{\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2} \\ v &= \frac{\Omega_0}{(\Gamma/2)[1 + \delta^2/(\Gamma/2)^2]} \frac{(\Gamma/2)^2 + \delta^2}{\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2} = \frac{\Omega_0(\Gamma/2)}{\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2} \\ u &= \frac{\delta\Omega_0}{\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2}.\end{aligned}$$

We will now consider the limiting cases. The first is a strong, resonant driving field, where $\Omega_0 \gg \Gamma$, $\delta = 0$. Here, we have

$$\begin{aligned}w &= \frac{(\Gamma/2)^2}{\Omega_0^2/2 + (\Gamma/2)^2} \longrightarrow \tilde{\rho}_{11} - \tilde{\rho}_{22} = 0 \\ u &= 0 \\ v &= \frac{\Omega_0(\Gamma/2)}{\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2} \longrightarrow 0.\end{aligned}$$

Observe that since $w = 0$ there are equal populations in both states, and since $u = v = 0$, there is no coherence and thus no superposition between states. The system is a "mixed" state, and we say that this transition is "saturated."

Now let us consider a weaker drive field, where $\Omega_0 \lesssim \Gamma$. We want to ask what the amount of light absorbed in the system is. To consider absorption, we will think about light transmission through a collection of opaque absorbers.

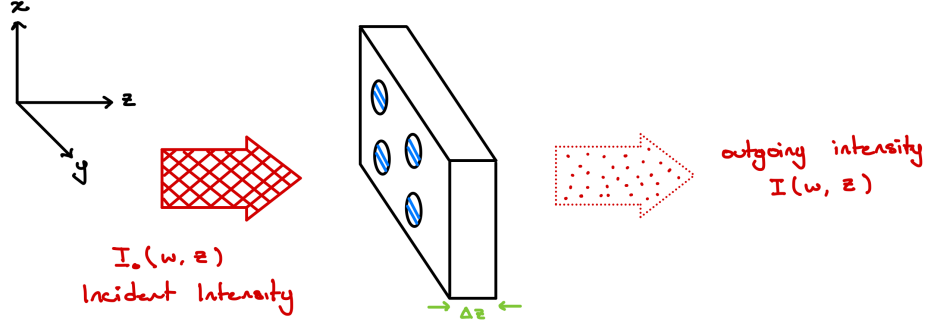


Figure 30: Light transmission through a collection of opaque absorbers. The incident intensity is given by $I_0(\omega, z)$ and the outgoing intensity is given by $I(\omega, z)$.

We can define the following relationship

$$\frac{I(\omega, z) - I_0(\omega, z)}{I_0(\omega, z)} = -N\sigma\Delta z$$

where the LHS is the probability of absorption, N is the number of absorbers per volume, σ is the scattering cross-section (in cm), and Δz is the thickness. So,

$$\frac{\Delta I(\omega, z)}{I} \rightarrow \frac{dI}{I} = -N\sigma dz \implies \frac{dI}{dz} = -N\sigma I.$$

Thus,

$$I(\omega, z) = I_0(\omega, z)e^{-N\sigma z}.$$

For 2-level systems, stimulated emission can add intensity to the output,

$$\frac{dI}{I} = -N_1\sigma dz + N_2\sigma dz = -(N_1 - N_2)\sigma dz$$

giving

$$\frac{dI}{dz} = -(N_1 - N_2)\sigma I \quad \text{and} \quad I(\omega, z) = I_0(\omega, z)e^{-(N_1 - N_2)\sigma z}.$$

This is the rate of energy per unit volume per time leaving the beam due to more absorption than stimulated emission, assuming $N_1 > N_2$. In considering the conservation of energy, the spontaneously emitted photons carry away the extra energy, so

$$\frac{dI}{dt} = -N_2 A_{21} \hbar \omega.$$

Thus,

$$\begin{aligned} -\frac{(N_1 - N_2)}{N}\sigma I &= -\frac{N_2}{N}A_{21}\hbar\omega \\ (\rho_{11} - \rho_{22})\sigma I &= \rho_{22}\Gamma\hbar\omega \\ \sigma &= \frac{\rho_{22}}{w}\frac{\Gamma\hbar\omega}{I} \end{aligned}$$

which is the scattering cross-section. From our previous results, we have $\rho_{22} = (1 - w)/2$, so

$$\begin{aligned} w &= \frac{(\Gamma/2)^2 + \delta^2}{\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2} \\ \frac{1}{2}(1 - w) &= \frac{1}{2}\left[\frac{\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2 - (\Gamma/2)^2 - \delta^2}{\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2}\right] \\ \frac{\rho_{22}}{w} &= \frac{\Omega_0^2/4}{(\Gamma/2)^2 + \delta^2} \end{aligned}$$

and, using the definition $|\Omega_0|^2 = 6\pi c^2 \Gamma I / \hbar \omega^3$ gives the scattering cross-section

$$\sigma(\omega) = \frac{(1/4)(6\pi c^2 / \hbar \omega^3) \Gamma I \hbar \omega}{I[(\Gamma/2)^2 + \delta^2]} = \frac{3\pi c^2}{2\omega^2} \frac{\Gamma^2}{(\Gamma/2)^2 + \delta^2}.$$

On resonance ($\delta = 0$), this simplifies to

$$\sigma(\omega_0) = \frac{3\pi c^2}{2\omega_0^2} \frac{\Gamma^2}{\Gamma^2/4} = \frac{6\pi c^2}{\omega_0^2} = 6\pi \frac{\lambda^2}{(2\pi)^2} = \frac{3\lambda^2}{2\pi}.$$

Resonant scattering depends only on the wavelength or frequency (and thus the energy) of the transition, which is a consequence of spontaneous emission being due to the number of modes into which to scatter.

For incoming light that is not necessarily on resonance, we have

$$K(\omega) = (N_1 - N_2)\sigma(\omega)$$

can be defined as an absorption coefficient,

$$I(z) = I_0 e^{-K(\omega)z}.$$

So, we can write

$$K(\omega) = N \frac{[\delta^2 + (\Gamma/2)^2]}{[\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2]} \frac{3\pi c^2}{2\omega^2} \frac{\Gamma^2}{[\delta^2 + (\Gamma/2)^2]} = N\sigma_0 \left[\frac{(\Gamma/2)^2}{\Omega_0^2/2 + (\Gamma/2)^2 + \delta^2} \right].$$

Here, N is the total number and the term $3\pi c^2/2\omega^2$ is close to resonance, so $\omega \rightarrow \omega_0$. In the low Rabi frequency limit (or low-intensity limit), we have $\Omega_0^2/2 \ll (\Gamma/2)^2$

$$K(\omega) = N\sigma_0 \frac{(\Gamma/2)^2}{\delta^2 + (\Gamma/2)^2} = \frac{N\sigma_0}{1 + \delta^2/(\Gamma/2)^2}.$$

This is a "Lorentzian" line shape, and is often characterized by the *full-width half-maximum*, so $K(\omega_{\text{FWHM}}) = N\sigma_0/2$. So,

$$\begin{aligned} \frac{1}{2} &= \frac{1}{1 + \delta^2/(\Gamma/2)^2} \\ 1 + \frac{\delta^2}{(\Gamma/2)^2} &= 2 \\ \frac{\delta^2}{(\Gamma/2)^2} &= 1 \\ \delta &= \pm \frac{\Gamma}{2}. \end{aligned}$$

Thus the full-width half-maximum is

$$\omega_{\text{FWHM}} = \omega_0 \pm \frac{\Gamma}{2}.$$

For high-intensity, it may be useful to rewrite the expression in terms of a "saturation intensity,"

$$K(\omega) = N\sigma_0 \frac{(\Gamma/2)^2}{\delta^2 + (1 + I/I_{\text{sat}})(\Gamma/2)^2}$$

which is also a Lorentzian, but with an increased width and a reduced peak amplitude:

$$K(\omega_0) = \frac{N\sigma_0}{1 + I/I_{\text{sat}}}$$

and a width characterized by

$$\Delta\omega_{\text{FWHM}} = \Gamma \left(1 + \frac{I}{I_{\text{sat}}} \right)^{1/2}$$

as $I \gg I_{\text{sat}}$ and $\kappa(\omega_0) \rightarrow 0$.

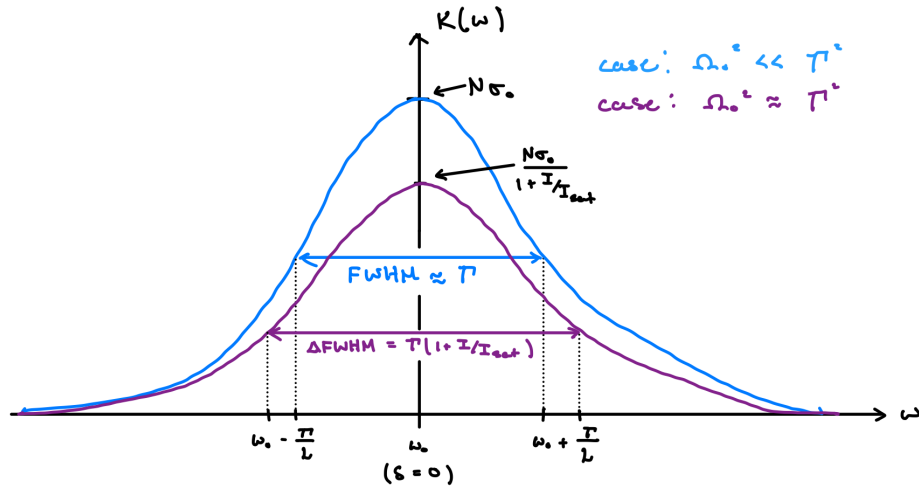


Figure 31: High and low-intensity limits for the absorption coefficient $K(\omega)$ with the full-width half-maximum labelled.

2.11 Application: Laser Cooling

Laser cooling uses the fact that there is a force associated with absorbing and emitting light.¹⁷ We will imagine some moving 2-level atoms (and consider just a single dimension for now). We subject the atoms to a light of frequency ω and wavevector $\mathbf{k}$, where the two beams are counter-propagating.

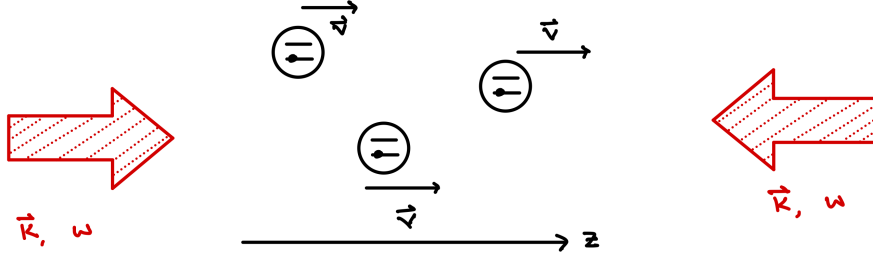


Figure 32: Counter-propagating beams with wavevector $\mathbf{k}$ and frequency ω impacting 2-level atoms travelling at $\mathbf{v}$ in the z -direction.

The light "seen" by the atoms depends on the velocity in the frame of the atom. We may write

$$\omega_{\text{at}} > \omega_{\text{lab}} - \mathbf{k} \cdot \mathbf{v}$$

where ω_{lab} is the frequency in the lab frame. This implies

$$\begin{aligned} \delta &= \omega - \omega_0 \\ \delta_{\text{at}} &= \omega_{\text{lab}} - \mathbf{k} \cdot \mathbf{v} - \omega_0 \\ &= \delta_{\text{lab}} - \mathbf{k} \cdot \mathbf{v}. \end{aligned}$$

Additionally, there is a force acting on the atoms due to the momentum transfer of the absorption process, so

$$\hbar \mathbf{k} + m \mathbf{v}_0 = m \mathbf{v}_{\text{max}} \implies m \Delta \mathbf{v} = \hbar \mathbf{k} \implies \Delta v = v_r = \frac{\hbar k}{m}$$

assuming $\mathbf{v}_0 = 0$.

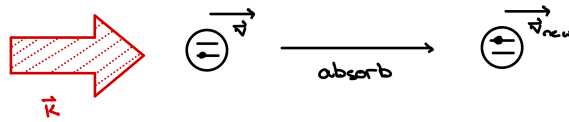


Figure 33: Beam of light with wavevector $\mathbf{k}$ exciting an atom travelling at $\mathbf{v}$, inducing a new velocity $\mathbf{v}_{\text{new}}$.

Now, we will consider the force,

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = \frac{1}{2} \Gamma_{\text{sc}} \cdot \Delta \mathbf{p}$$

¹⁷This is due to the photon's momentum and the need to conserve momentum overall.

where Γ_{sc} is the scattering rate, depending on absorption conditions, and $\Delta\mathbf{p}$ is the momentum transfer per event. We add a $1/2$ factor for the randomness of the re-emitted photons since, on average, the change in momentum is reduced by this same number.

From our recent results, we can see that

$$\Gamma_{\text{sc}} = \Gamma_{\rho_{22}} = \Gamma \left[\frac{\Omega_0^2/4}{\delta^2 + (\Gamma/2)^2} \right] = \frac{\Omega_0^2/\Gamma}{1 + \delta^2/(\Gamma/2)^2}$$

assuming the low-intensity limit so $\Omega_0 \ll \Gamma$. If we include the Doppler effect due to the counter-propagating beams, then

$$\begin{aligned} F_{\text{tot}} &= |\mathbf{F}_{\text{left}} + \mathbf{F}_{\text{right}}| \\ &= \left| \frac{\hbar k(\Omega_0^2/\Gamma)/2}{1 + (\delta_0 - kv)^2/(\Gamma/2)^2} + \frac{(-\hbar k)(\Omega_0^2/\Gamma)/2}{1 + (\delta_0 + kv)^2/(\Gamma/2)^2} \right| \\ &= \frac{\hbar k\Omega_0^2}{2\Gamma} \left[\frac{1}{1 + (\delta_0 - kv)^2/(\Gamma/2)^2} - \frac{1}{1 + (\delta_0 + kv)^2/(\Gamma/2)^2} \right] \end{aligned}$$

where the $\delta \pm kv$ terms arise from Doppler shift from the left or right side. Specifically, we have a minus sign for a shift from the left, and a plus sign for a shift from the right.

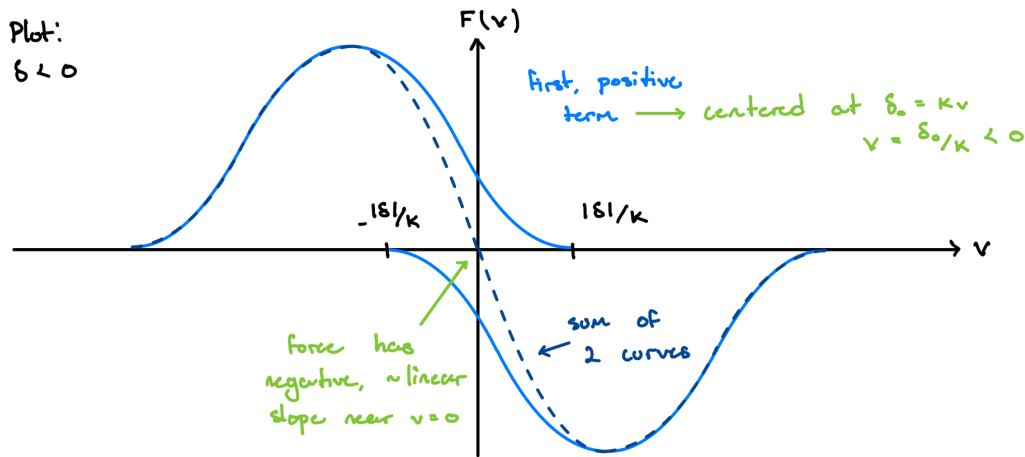


Figure 34: Plot of the force as experienced by the atom due to velocity. Observe that the force is a sum of two curves, centred at $\pm|\delta|/k$.

We can see explicitly from both Fig. 34 and by directly analyzing F_{tot} that near $v = 0$ (consider small displacements from $v = 0$) that $F_{\text{tot}} \propto -v$. That is, it is a damping force.

The velocity range between peaks defines the "capture range" of such a cooling scheme. We can use magnetic fields ("magneto-optical trapping") to broaden this range significantly.

2.12 Application: Magneto-Optical Trapping

The effectiveness of laser cooling is greatly enhanced in the presence of a magnetic field gradient. We can use this to shift atoms of different speeds into resonance with light at different points. Let

us consider a simple 1D version first, called the *Zeeman slower*. The goal of this setup is to slow an atomic beam by sending single frequency light along a tapered solenoid.

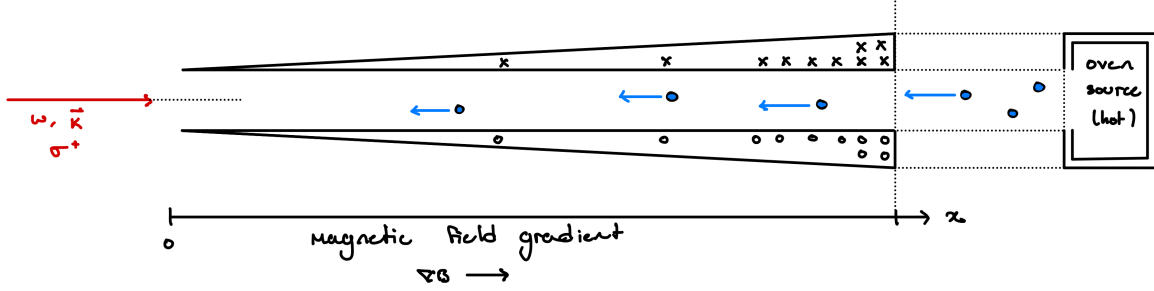


Figure 35: Single frequency light along a magnetic field gradient, with a source of atoms travelling into it.

We define the magnetic field to be

$$\mathbf{B}(x) = bx\hat{x}$$

where the resonance condition for moving atoms depends on their position (the local B -field) and their speed. Suppose we are interested in the $J = 0 \rightarrow J' = 1$ transition.

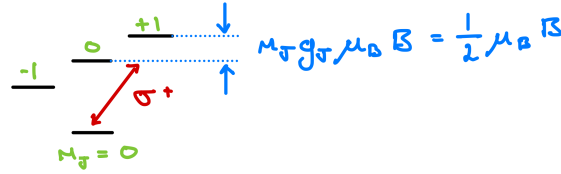


Figure 36: The $J = 0 \rightarrow J' = 1$ transition.

The resonance condition for this transition is

$$\omega'_0 = \omega_0(B = 0) + \frac{1}{2}\mu_B B.$$

The detuning "seen" by such a moving atom is given by

$$\delta = \omega_{\text{at}} - \omega'_0 = \omega_{\text{lab}} - kv - \left(\omega_0 + \frac{1}{2}\mu_B B \right).$$

For a B -field gradient as above, we have thus

$$\delta = \omega_{\text{lab}} - kv - \omega_0 - \frac{1}{2}\mu_B bx$$

and when $\delta = 0$, the light is absorbed and provides a momentum "kick" opposing motion. So,

$$\delta = 0 \implies \omega_{\text{lab}} - \omega_0 = kv + \frac{1}{2}\mu_B bx.$$

Setting $\omega_{\text{lab}} - \omega_0 = 0$, we find the resonant velocity to be

$$v = -\frac{\mu_B b x}{2k}.$$

This shows fast moving atoms scattering at large x , so they're slowed down and resonant at smaller x . This process repeats over and over until eventually the atoms stop.

Let us now look at this process in 3D. We will instead use a quadrupole magnetic field to produce a B -field gradient in all directions; then, we will counteract them all with a counter-propagating laser beam.

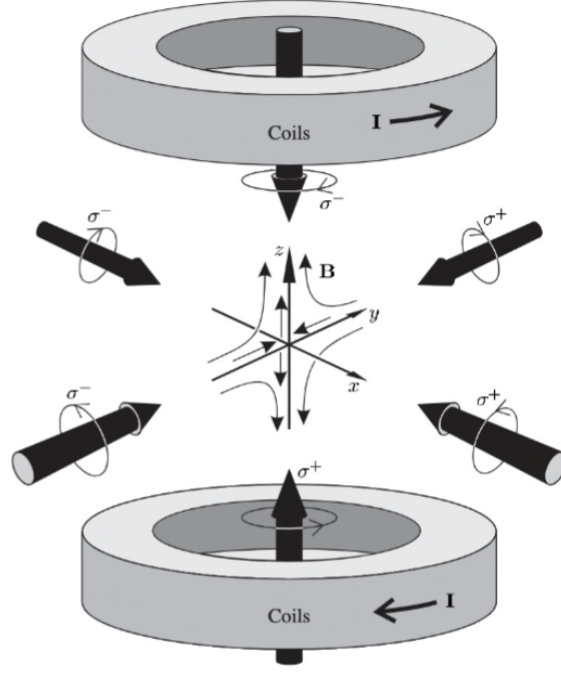


Figure 37: Quadrupole magnetic field setup to create an anti-Helmholtz configuration (optical-trap).

The scattering force in a magneto-optical trap (MOT) is given by

$$F_{\text{MOT}} = F_{\text{scat.}}^{\sigma^+}[\omega - k - (\omega_0 + bx)] - F_{\text{scat.}}^{\sigma^-}[\omega + kv - (\omega_0 - bx)]$$

where $\omega_0 \pm bx$ are polarization definitions with respect to the local field and laser directions. Here,

$$F_{\text{scat.}}(\delta) = \frac{\hbar k (\Omega_0^2 / \Gamma) / 2}{1 + \delta^2 / (\Gamma/2)^2}.$$

To see the force's effect, we will write this as an approximation using the Taylor expansion around $\omega - \omega_0 \approx 0$ for small $kv \ll \Gamma$ and $bx \ll \Gamma$. So, we have

$$\begin{aligned} F_{\text{scat.}}^{\sigma^+} &\approx F_{\text{scat.}}^{\sigma^+}(\delta = \omega - \omega_0) + \frac{\partial F}{\partial \omega} + \frac{\partial F}{\partial \omega} \\ F_{\text{scat.}}^{\sigma^-} &\approx F_{\text{scat.}}^{\sigma^-}(\delta = \omega - \omega_0) + \frac{\partial F}{\partial \omega} + \frac{\partial F}{\partial \omega}. \end{aligned}$$

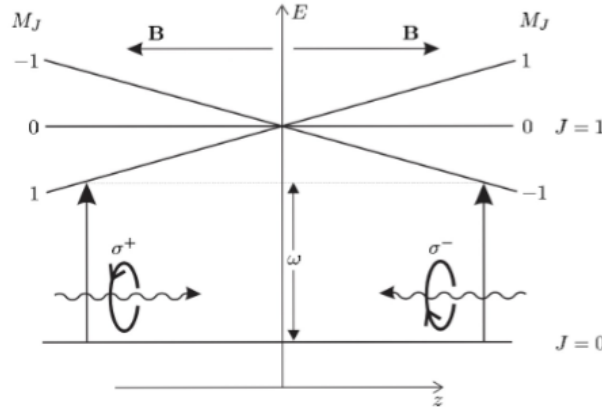


Figure 38: MOT setup as viewed from y -axis, with J -lines shown.

Subtracting these two equations, we find

$$F_{\text{MOT}} \approx -2 \frac{\partial F}{\partial \omega} kv - 2 \frac{\partial F}{\partial \omega} bx = -\alpha v - \alpha bx$$

where $-\alpha v$ is a damping force and αbx is a restoring force. So, the restoring force acts in simple harmonic motion (SHM) and is thus a trap.

2.13 Application: Optical Clocks

While the current frequency standard is the hyperfine transition in cesium, there has been tremendous progress in building clocks at optical frequencies. The long progression of time standards has a history of ever-increasing frequencies: from the daily rotation of the earth, to a pendulum clock, to quartz oscillators, to atomic transitions. With optical transitions, we gain orders of magnitude over microwave transitions.

Optical clock transitions are chosen with accessibility and precision in mind. They are generally "forbidden" electric quadrupole transitions. Take, for example, the world leading "SrI" clock at JILA.¹⁸

We see some of the relevant transitions in Sr (using NIST Atomic Spectral Database). The "clock transition" is very much forbidden, where $^1S_0 \rightarrow ^3P_0$ (this corresponds to $S = L = J = 0 \rightarrow S = L = 1, J = 0$). This transition is disallowed since, for multi-electron atoms, $\Delta S = 0$.

However, for the listed transitions, A (which is Γ) leads us to the matrix elements. To isolate atoms from each other and avoid systematic uncertainties due to collisions between atoms (mean-field shifts, etc.), optical lattices are often used to make many tiny traps of atoms that are all interrogated.

However, we have discussed optical lattices in the context of 2-level systems, which in reality are multi-level; states experience shifts due to off-resonant couplings to all levels. In this case, if we consider the two clock states, they are effectively uncoupled (super-narrow linewidth) and the lattice trap is due to coupling to other levels.

We will do an estimate of the wavelength we should use that shifts both levels the same amount. This is known as the *magic wavelength*. Our previous result, including the RWA (not

¹⁸See Metrologia 56, 065004 (2019).

rigorously valid), is

$$\hbar\Delta = \frac{\hbar\Omega_0^2}{4\delta}.$$

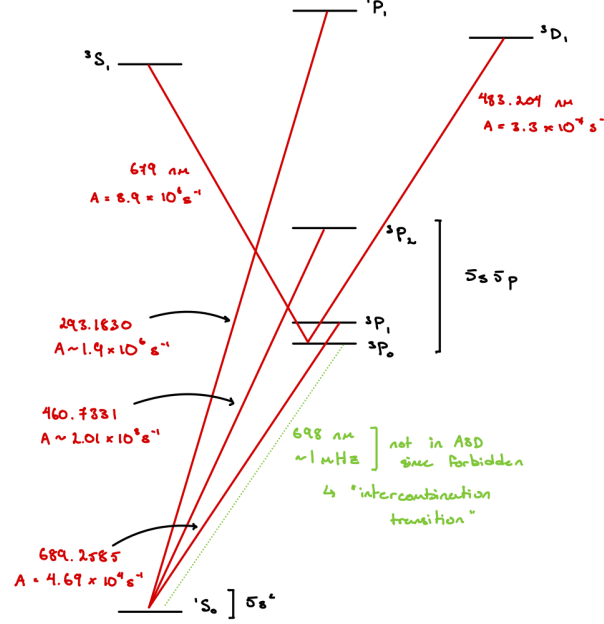


Figure 39: Relevant transitions in Sr using the NIST Atomic Spectral Database.

For the ground state, the light shift is

$$\hbar\Delta_{\text{tot}} \approx \hbar\Delta_1 + \hbar\Delta_2 + \hbar\Delta_3 + \dots$$

where

$$\hbar\Delta_i = \frac{\hbar|\Omega_i|^2}{4\delta} = \frac{3\pi c^2 \Gamma_i}{2\omega_i^3 \delta_i} I$$

and ω_i is the transition frequency, Γ_i is the transition linewidth, and $\delta_i = \omega_{\text{magic}} - \omega_i$. The relevant transitions (for larger Γ and smaller δ) that we will consider are

$$\begin{aligned} 5^1S_0 - 5^3P_1 : \lambda_1 &= 689.2585 \text{ nm}, \Gamma_1 = 4.69 \times 10^4 \text{ s}^{-1} \\ 5^1S_0 - 5^1P_1 : \lambda_2 &= 460.7221 \text{ nm}, \Gamma_2 = 2.01 \times 10^8 \text{ s}^{-1} \\ 5^1S_0 - 6^1P_1 : \lambda_3 &= 289.1380 \text{ nm}, \Gamma_3 = 6.9 \times 10^6 \text{ s}^{-1} \\ 5^1S_0 - 7^1P_1 : \lambda_4 &= 256.9469 \text{ nm}, \Gamma_4 = 5.3 \times 10^6 \text{ s}^{-1} \\ &\vdots \end{aligned}$$

We can do the same for the excited state, where we find

$$\begin{aligned}
 5^3P_0 - 6^3S_1 : \lambda_1 &= 679.1022 \text{ nm}, \Gamma_1 = 8.9 \times 10^6 \text{ s}^{-1} \\
 5^3P_0 - 7^3S_1 : \lambda_2 &= 432.6445 \text{ nm}, \Gamma_2 = 3.1 \times 10^6 \text{ s}^{-1} \\
 5^3P_0 - 8^3S_1 : \lambda_3 &= 378.0527 \text{ nm}, \Gamma_3 = \text{N/A} \\
 5^3P_0 - 5^3D_1 : \lambda_4 &= 483.2043 \text{ nm}, \Gamma_4 = 3.3 \times 10^7 \text{ s}^{-1} \\
 &\vdots
 \end{aligned}$$

We can write a quick script to determine at what magic wavelength(s) this cancels. Observe from Fig. 39 that the energy shifts of the two strongest transitions are $5s^21S_0 \rightarrow 5s5p^1P_1$ (blue, $\lambda_0 = 460.7 \text{ nm}$) and $5s5p^3P_0 \rightarrow 5s6p^3S_1$ (green, $\lambda_0 = 679.1 \text{ nm}$).

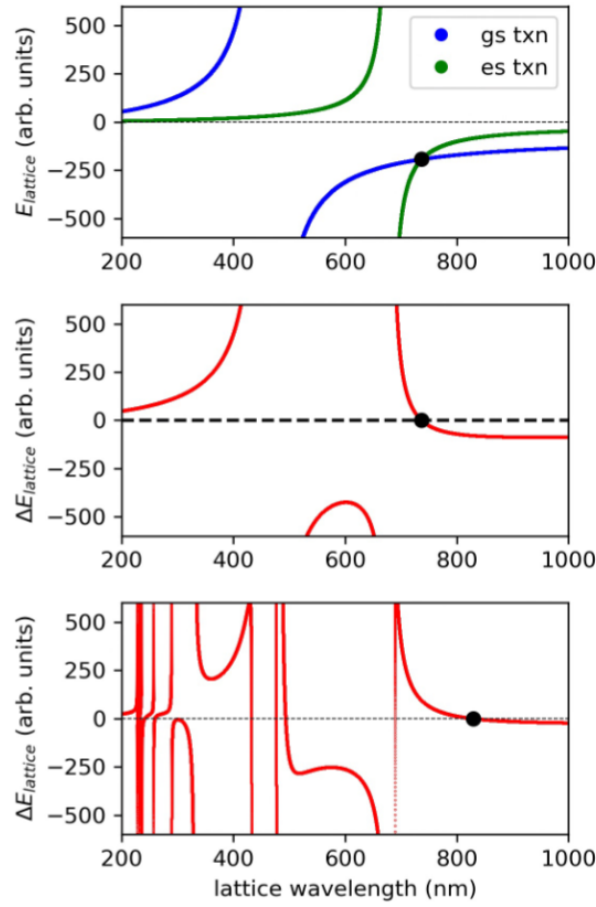


Figure 40: Energy shifts as a function of the lattice wavelength.

The difference between these two shifts highlight where the difference is zero gives us where we have equal ground state and excited state shifts.

We can include all transitions listed above (may be omitting a few medium-important shifts, also uses RWA which we said before is not rigorously appropriate), we can see how the energy shift evolves over the lattice wavelength.

Using our parameters, we find that there is a shallow cancellation at 830.230 nm. However, full calculations show that we have an optical lattice wavelength at about 813 nm, leading to the most accurate measurements of time.

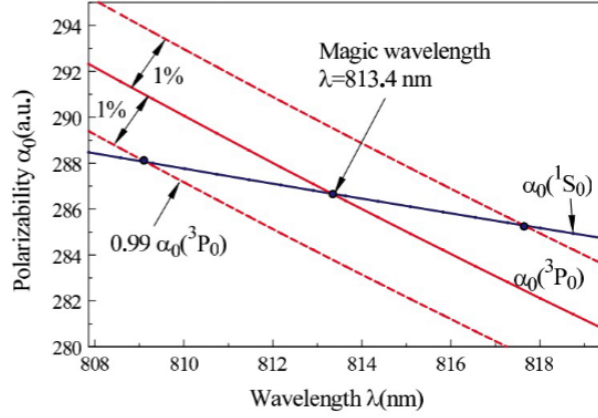


Figure 41: Magic wavelength after extrapolating calculations of the polarizability against the wavelength.

3 Introductory Quantum Optics

Following this section, we will quantize the EM field, and study both the nature of quantum light itself. In doing so, we shall study the interactions between this light and quantum matter.

There are many approaches to quantizing the EM field; in our analysis, we will follow one of them, which is the approach from the Loudon text.

3.1 Classical EM Review

We begin with some classical review. Consider the vector potential,

$$\begin{aligned}\mathbf{E}(\mathbf{r}) &= -\frac{\partial \mathbf{A}(\mathbf{r})}{\partial t} - \nabla \phi(\mathbf{r}) \\ \mathbf{B}(\mathbf{r}) &= \nabla \times \mathbf{A}(\mathbf{r})\end{aligned}$$

where the vector and scalar potentials $\{\mathbf{A}(\mathbf{r}), \phi(\mathbf{r})\}$ are not unique, but subject to the choice of gauge. Here, we choose the Coulomb gauge:

$$\nabla \cdot \mathbf{A}(\mathbf{r}) = 0.$$

With this choice, Maxwell's equations, in terms of $\mathbf{A}(\mathbf{r})$, become

$$\nabla \times \mathbf{B} = \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J}$$

so that

$$\begin{aligned}\nabla \times \mathbf{B}(\mathbf{r}) &= \nabla \times (\nabla \times \mathbf{A}) \\ &= \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \\ &= -\nabla^2 \mathbf{A}(\mathbf{r}) \\ \frac{\partial \mathbf{E}(\mathbf{r})}{\partial t} &= -\frac{\partial^2 \mathbf{A}(\mathbf{r})}{\partial t^2} - \frac{\partial}{\partial t} \nabla \phi.\end{aligned}$$

Thus, we find

$$-\nabla^2 \mathbf{A} = -\frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} - \frac{1}{c^2} \frac{\partial}{\partial t} (\nabla \phi) + \mu_0 \mathbf{J}.$$

And,

$$\nabla \cdot \mathbf{E} = -\frac{\partial}{\partial t} (\nabla \cdot \mathbf{A}) - \nabla \cdot (\nabla \phi) = \frac{\rho}{\epsilon_0} = -\nabla^2 \phi.$$

Next we consider the current density, $\mathbf{J}$. Since any vector field can be split into a part with no curl and a part with no divergence, we have

$$\mathbf{J} = \mathbf{J}_L + \mathbf{J}_T$$

where $\mathbf{J}_L$ is the longitudinal current density (which has no curl) and $\mathbf{J}_T$ is the transverse current density (which has no divergence). Indeed, we may write

$$\nabla \cdot \mathbf{J}_T = 0, \quad \nabla \times \mathbf{J}_L = 0.$$

Thus, we arrive at

$$\begin{aligned}\mu_0(\mathbf{J}_T + \mathbf{J}_L) &= -\nabla^2 \mathbf{A} + \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} + \frac{1}{c^2} \frac{\partial}{\partial t}(\nabla \phi) \\ \mu_0 \mathbf{J}_T &= -\nabla^2 \mathbf{A} + \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} \\ \mu_0 \mathbf{J}_L &= \frac{1}{c^2} \frac{\partial}{\partial t}(\nabla \phi).\end{aligned}$$

Likewise, we may write the transverse and longitudinal electric field

$$\mathbf{E} = \mathbf{E}_L + \mathbf{E}_T$$

so that the transverse equations are

$$\nabla \cdot \mathbf{E}_T = 0, \quad \nabla \times \mathbf{B} = \frac{1}{c^2} \frac{\partial \mathbf{E}_T}{\partial t} + \mathbf{J}_T, \quad \nabla \cdot \mathbf{B} = 0$$

and the longitudinal equations are

$$\nabla \cdot \mathbf{E}_L = \frac{\sigma}{\epsilon_0}, \quad 0 = \frac{1}{c^2} \frac{\partial \mathbf{E}_L}{\partial t} + \mathbf{J}_L, \quad \nabla \times \mathbf{E}_L = -\frac{\partial \mathbf{B}}{\partial t}.$$

Observe that the only transverse $\mathbf{J}_T$ is relevant and not the free charge σ in the transverse equations; we expect the transverse equations to describe EM waves. However, the longitudinal equations describe charge distributions.

In free space, where there are no charges or currents, we have

$$\mathbf{J}_T = 0 \implies -\nabla^2 \mathbf{A} + \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0.$$

We can find a wave equation describing the EM field (in terms of any of $\mathbf{E}$, $\mathbf{B}$, or $\mathbf{A}$).

3.2 Quantizing the EM Field

To quantize the EM field, we shall make the following assumptions:

- (i) we will "promote" $\mathbf{A}$ from a field to an operator, so $\mathbf{A} \rightarrow \hat{\mathbf{A}}$;
- (ii) we will quantize the field in a particular region of space, and later expand this region to all space.

Consider a cubic region of space (no real boundaries) known as a *quantization cavity*. In the cavity the EM field has running-wave solutions with periodic boundary conditions. The cavity has an infinite number of modes, where each mode has a wavevector $\mathbf{k} = (k_x, k_y, k_z)$ with

$$k_x = \frac{2\pi\nu_x}{L}, \quad k_y = \frac{2\pi\nu_y}{L}, \quad k_z = \frac{2\pi\nu_z}{L},$$

a frequency,

$$\nu_x, \nu_y, \nu_z = 0, \pm 1, \pm 2, \dots$$

and a polarization $\mathbf{e}_{\mathbf{k}\lambda}$ (*unit polarization vector*). We note that the polarization is perpendicular to propagation in free space, so

$$\mathbf{e}_{\mathbf{k}\lambda} \cdot \mathbf{k} = 0$$

and that each mode may have one of two polarizations, where the mode is indexed by $\lambda = \{1, 2\}$, thus

$$\mathbf{e}_{\mathbf{k}1} \cdot \mathbf{e}_{\mathbf{k}2} = 0 \quad \text{for every } \mathbf{k}, \quad \mathbf{e}_{\mathbf{k}\lambda} \cdot \mathbf{e}_{\mathbf{k}\lambda'} = \delta_{\lambda\lambda'} \quad (\text{normalized}).$$

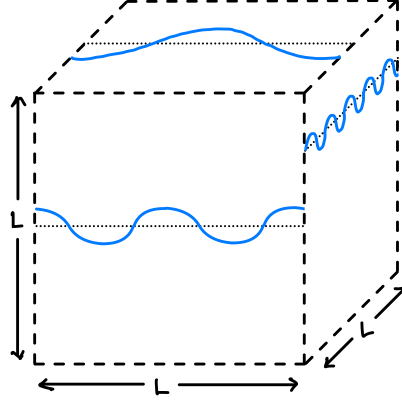


Figure 42: Waves of different λ along 3 dimensions on a cube of length L .

The vector potential can be written as a sum over all modes of the cavity

$$\mathbf{A}(\mathbf{r}, t) = \sum_{\lambda=1,2} \sum_{\mathbf{k}} \mathbf{e}_{\mathbf{k}\lambda} \mathbf{A}_{\mathbf{k}\lambda}(\mathbf{r}, t)$$

where we note that $\mathbf{A}$ is in the same direction as $\mathbf{E}$. We also have the classical vector potential

$$\mathbf{A}_{\mathbf{k}\lambda}(\mathbf{r}, t) = A_{\mathbf{k}\lambda}(t)e^{i\mathbf{k}\cdot\mathbf{r}} + A_{\mathbf{k}\lambda}^*(t)e^{-i\mathbf{k}\cdot\mathbf{r}}.$$

Here, $A_{\mathbf{k}\lambda}(t)$ is a time-dependence in scalar coefficients and $\mathbf{k} \cdot \mathbf{r}$ is a periodic bias encapsulated in $\mathbf{k}$. Every mode is independent of the others, so all modes individually satisfy the wave equation:

$$k^2 A_{\mathbf{k}\lambda}(t) + \frac{1}{c^2} \frac{\partial^2 A_{\mathbf{k}\lambda}(t)}{\partial t^2} = 0, \quad k = |\mathbf{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2}$$

which has solutions

$$A_{\mathbf{k}\lambda}(t) = A_{\mathbf{k}\lambda} e^{-ickt} = A_{\mathbf{k}\lambda} e^{-i\omega_k t}$$

where $\omega_k = ck$ is an equally good index for each mode. Thus, we find

$$\mathbf{A}_{\mathbf{k}\lambda}(\mathbf{r}, t) = A_{\mathbf{k}\lambda} e^{-i\omega_k t + i\mathbf{k}\cdot\mathbf{r}} + A_{\mathbf{k}\lambda}^* e^{i\omega_k t - i\mathbf{k}\cdot\mathbf{r}}.$$

From this, we can calculate the associated electric and magnetic fields

$$\begin{aligned} \mathbf{E}_T(\mathbf{r}, t) &= -\frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t} \\ &= \sum_{\lambda=1,2} \sum_{\mathbf{k}} \mathbf{e}_{\mathbf{k}\lambda} E_{\mathbf{k}\lambda}(\mathbf{r}, t) \\ E_{\mathbf{k}\lambda}(\mathbf{r}, t) &= i\omega_k \left(A_{\mathbf{k}\lambda} e^{-i\omega_k t + i\mathbf{k}\cdot\mathbf{r}} - A_{\mathbf{k}\lambda}^* e^{i\omega_k t - i\mathbf{k}\cdot\mathbf{r}} \right) \end{aligned}$$

and

$$\begin{aligned}\mathbf{B}(\mathbf{r}, t) &= \nabla \times \mathbf{A}(\mathbf{r}, t) \\ &= \sum_{\lambda} \sum_{\mathbf{k}} \frac{\mathbf{k} \times \mathbf{e}_{\mathbf{k}\lambda}}{k} B_{\mathbf{k}\lambda}(\mathbf{r}, t) \\ B_{\mathbf{k}\lambda}(\mathbf{r}, t) &= ik \left(A_{\mathbf{k}\lambda} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k} \cdot \mathbf{r}} - A_{\mathbf{k}\lambda}^* e^{i\omega_{\mathbf{k}}t - i\mathbf{k} \cdot \mathbf{r}} \right).\end{aligned}$$

Next, we find the energy of this field with an eye towards determining a (quantum) Hamiltonian that describes the field. Classically, the energy density of an EM field is

$$\begin{aligned}u &= \frac{1}{2} \epsilon_0 |E|^2 + \frac{1}{2\mu_0} \epsilon_0 |B|^2 \\ U_{\text{EM}} &= \int d^3\mathbf{r} u.\end{aligned}$$

For our collection of modes, we have

$$U_{\text{EM}} = \frac{1}{2} \int d^3\mathbf{r} \left(\epsilon_0 \mathbf{E}_t \cdot \mathbf{E}_T + \frac{1}{\mu_0} \mathbf{B} \cdot \mathbf{B} \right).$$

Evaluating these explicitly, we have

$$\begin{aligned}\mathbf{E}_T \cdot \mathbf{E}_T &= \sum_{\lambda\lambda'} \sum_{\mathbf{k}\mathbf{k}'} \mathbf{e}_{\mathbf{k}\lambda} \cdot \mathbf{e}_{\mathbf{k}'\lambda'} \\ &\quad \times \left[-\omega_{\mathbf{k}}\omega_{\mathbf{k}'} A_{\mathbf{k}\lambda} A_{\mathbf{k}'\lambda'} e^{-i(\omega_{\mathbf{k}}+\omega_{\mathbf{k}'})t + i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{r}} \right. \\ &\quad - \omega_{\mathbf{k}}\omega_{\mathbf{k}'} A_{\mathbf{k}\lambda}^* A_{\mathbf{k}'\lambda'}^* e^{i(\omega_{\mathbf{k}}+\omega_{\mathbf{k}'})t - i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{r}} \\ &\quad + \omega_{\mathbf{k}}\omega_{\mathbf{k}'} A_{\mathbf{k}\lambda} A_{\mathbf{k}'\lambda'}^* e^{i(\omega_{\mathbf{k}}+\omega_{\mathbf{k}'})t - i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{r}} \\ &\quad \left. + \omega_{\mathbf{k}}\omega_{\mathbf{k}'} A_{\mathbf{k}\lambda}^* A_{\mathbf{k}'\lambda'} e^{-i(\omega_{\mathbf{k}}+\omega_{\mathbf{k}'})t + i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{r}} \right].\end{aligned}$$

The phase factors are non-zero only if $\mathbf{k} \pm \mathbf{k}' = 0$, and the polarization vectors are non-zero for $\lambda = \lambda'$, so

$$\mathbf{e}_{\mathbf{k}\lambda} \cdot \mathbf{e}_{\mathbf{k}'\lambda'} = \delta_{\lambda\lambda'} \mathbf{e}_{\mathbf{k}} \cdot \mathbf{e}_{\mathbf{k}'} = \begin{cases} 1, & \mathbf{k} = \mathbf{k}' \\ 1 & = \mathbf{k} = -\mathbf{k}', \lambda = \lambda_1 \\ -1, & \mathbf{k} = -\mathbf{k}', \lambda = \lambda_2 \end{cases}$$

due to cyclic permutations of the indices. Thus, we find

$$\mathbf{E}_T \cdot \mathbf{E}_T = \sum_{\mathbf{k}} \sum_{\lambda} \left[\omega_{\mathbf{k}}^2 A_{\mathbf{k}\lambda} A_{\mathbf{k}\lambda}^* - \omega_{\mathbf{k}}^2 A_{\mathbf{k}\lambda} A_{(-\mathbf{k})\lambda} (\pm 1) e^{-2i\omega_{\mathbf{k}}t} - \omega_{\mathbf{k}}^2 A_{\mathbf{k}\lambda}^* A_{(-\mathbf{k})\lambda}^* (1) e^{2i\omega_{\mathbf{k}}t} \right].$$

Similarly, for $\mathbf{B}(\mathbf{r}, t)$, we have

$$\begin{aligned}\mathbf{B} \cdot \mathbf{B} &= \sum_{\lambda\lambda'} \sum_{\mathbf{k}\mathbf{k}'} \left(\frac{\mathbf{k} \times \mathbf{e}_{\mathbf{k}\lambda}}{k} \right) \cdot \left(\frac{\mathbf{k}' \times \mathbf{e}_{\mathbf{k}'\lambda'}}{k'} \right) \\ &\quad \times \left[-\mathbf{k}\mathbf{k}' A_{\mathbf{k}\lambda} A_{\mathbf{k}'\lambda'} e^{-i(\omega_{\mathbf{k}}+\omega_{\mathbf{k}'})t + i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{r}} \right. \\ &\quad - \mathbf{k}\mathbf{k}' A_{\mathbf{k}\lambda}^* A_{\mathbf{k}'\lambda'}^* e^{-i(\omega_{\mathbf{k}}+\omega_{\mathbf{k}'})t - i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{r}} \\ &\quad + \mathbf{k}\mathbf{k}' A_{\mathbf{k}\lambda} A_{\mathbf{k}'\lambda'}^* e^{-i(\omega_{\mathbf{k}}+\omega_{\mathbf{k}'})t - i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{r}} \\ &\quad \left. + \mathbf{k}\mathbf{k}' A_{\mathbf{k}\lambda}^* A_{\mathbf{k}'\lambda'} e^{-i(\omega_{\mathbf{k}}+\omega_{\mathbf{k}'})t + i(\mathbf{k}+\mathbf{k}') \cdot \mathbf{r}} \right].\end{aligned}$$

Using the same arguments as above for the phase factors, but for the polarizations instead, we find

$$\mathbf{B} \cdot \mathbf{B} = \sum_{\lambda} \sum_{\mathbf{k}} \left[\pm \mathbf{k} A_{\mathbf{k}\lambda} A_{(-\mathbf{k})\lambda} e^{-2i\omega_{\mathbf{k}}t} \pm k^2 A_{\mathbf{k}\lambda}^* A_{(-\mathbf{k})\lambda}^* e^{2i\omega_{\mathbf{k}}t} + k^2 A_{\mathbf{k}\lambda} A_{\mathbf{k}\lambda}^* + k^2 A_{\mathbf{k}\lambda}^* A_{\mathbf{k}\lambda} \right].$$

Thus, we assemble the total energy

$$U_{EM} = \frac{1}{2} \int d^3\mathbf{r} \sum_{\lambda} \sum_{\mathbf{k}} \left[\epsilon_0 \omega_{\mathbf{k}}^2 \left(A_{\mathbf{k}\lambda} A_{\mathbf{k}\lambda}^* + A_{\mathbf{k}\lambda}^* A_{\mathbf{k}\lambda} \mp A_{\mathbf{k}\lambda} A_{(-\mathbf{k})\lambda} e^{-2i\omega_{\mathbf{k}}t} \mp A_{\mathbf{k}\lambda}^* A_{(-\mathbf{k})\lambda}^* e^{2i\omega_{\mathbf{k}}t} \right) \right. \\ \left. + \frac{1}{2} \int d^3\mathbf{r} \sum_{\lambda} \sum_{\mathbf{k}} \left[\frac{1}{\mu_0} k^2 \left(A_{\mathbf{k}\lambda} A_{\mathbf{k}\lambda}^* + A_{\mathbf{k}\lambda}^* A_{\mathbf{k}\lambda} \pm A_{\mathbf{k}\lambda} A_{(-\mathbf{k})\lambda} e^{-2i\omega_{\mathbf{k}}t} \pm A_{\mathbf{k}\lambda}^* A_{(-\mathbf{k})\lambda}^* e^{2i\omega_{\mathbf{k}}t} \right) \right] \right].$$

Noting that $\omega_{\mathbf{k}} = ck$, $\epsilon_0 \mu_0 = 1/c^2$ so that $\epsilon_0 \omega_{\mathbf{k}}^2 = c^2 k^2 / \mu_0 c^2$, we find

$$U_{EM} = \frac{1}{2} \epsilon_0 \int_V d^3\mathbf{r} \sum_{\lambda} \sum_{\mathbf{k}} 2\omega_{\mathbf{k}}^2 [A_{\mathbf{k}\lambda} A_{\mathbf{k}\lambda}^* + A_{\mathbf{k}\lambda}^* A_{\mathbf{k}\lambda}] = \epsilon_0 V [A_{\mathbf{k}\lambda} A_{\mathbf{k}\lambda}^* + A_{\mathbf{k}\lambda}^* A_{\mathbf{k}\lambda}].$$

To quantize the EM field, we associate with each mode labelled and defined by $\{\mathbf{k}, \lambda\}$ a quantum harmonic oscillator (QHM). This is well-justified since the energy in each mode is quadratic in the $\mathbf{E}$, $\mathbf{B}$ fields, just like a harmonic oscillator's energy is quadratic in $\mathbf{x}$ and $\mathbf{p}$. Let us define the following operators

$$\hat{a}_{\mathbf{k}\lambda} |n_{\mathbf{k},\lambda}\rangle = \sqrt{n_{\mathbf{k},\lambda}} |n_{\mathbf{k},\lambda} - 1\rangle \quad (\text{lowering operator}) \quad (\text{lowering operator})$$

$$\hat{a}_{\mathbf{k}\lambda}^\dagger |n_{\mathbf{k},\lambda}\rangle = \sqrt{n_{\mathbf{k},\lambda} + 1} |n_{\mathbf{k},\lambda} + 1\rangle. \quad (\text{raising operator})$$

Observe that $n_{\mathbf{k}\lambda}$ is the number of excitations in the mode. So, we can define

$$\hat{n}_{\mathbf{k}\lambda} = \hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda} \quad (\text{number operator})$$

$$[\hat{a}_{\mathbf{k}\lambda}, \hat{a}_{\mathbf{k}\lambda}^\dagger] = \delta_{\mathbf{k}\mathbf{k}'} \delta_{\lambda\lambda'} \quad (\text{commutation})$$

In quantizing the EM field, we proceed with the following interpretation:

- each mode of the EM field is represented by a QHM, whose occupation and energy act like a harmonic oscillator with frequency ω_k ;
- the number of excitations per mode is labelled as $n_{\mathbf{k}\lambda}$, with a single excitation being called a *photon*;
- the energy of one mode is

$$\hat{U}_{\mathbf{k}\lambda} = \hbar \omega_k \left(\hat{n}_{\mathbf{k}\lambda} + \frac{1}{2} \right) = \frac{\hbar \omega_k}{2} \left(\hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda}^\dagger \right);$$

- the sum of energy over all modes is

$$\hat{U}_{EM} = \frac{1}{2} \sum_{\mathbf{k}, \lambda} \hbar \omega_k \left(\hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda}^\dagger \right);$$

- by comparing to our previous expression, we can identify

$$\begin{aligned} \left(\frac{\hbar\omega_k}{2}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda} &= \sqrt{\epsilon_0 V \omega_k} A_{\mathbf{k}\lambda} \implies \hat{A}_{\mathbf{k}\lambda} = \left(\frac{\hbar}{2\epsilon_0 \omega_k V}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda} \\ \left(\frac{\hbar\omega_k}{2}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda} &= \sqrt{\epsilon_0 V \omega_k} A_{\mathbf{k}\lambda}^* \implies \hat{A}_{\mathbf{k}\lambda}^\dagger = \left(\frac{\hbar}{2\epsilon_0 \omega_k V}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda}^\dagger. \end{aligned}$$

The final expression for the vector field is thus

$$\hat{\mathbf{A}}(\mathbf{r}, t) = \sum_{\mathbf{k}\lambda} e_{\mathbf{k}\lambda} \hat{A}_{\mathbf{k}\lambda}(\mathbf{r}, t)$$

where

$$\hat{A}_{\mathbf{k}\lambda}(\mathbf{r}, t) = \hat{A}_{\mathbf{k}\lambda} e^{-i\omega_k t + i\mathbf{k}\cdot\mathbf{r}} + \hat{A}_{\mathbf{k}\lambda}^\dagger e^{i\omega_k t - i\mathbf{k}\cdot\mathbf{r}} = \left(\frac{\hbar}{2\epsilon_0 \omega_k V}\right)^{1/2} \left(\hat{a}_{\mathbf{k}\lambda} e^{-i\omega_k t + i\mathbf{k}\cdot\mathbf{r}} + \hat{a}_{\mathbf{k}\lambda}^\dagger e^{i\omega_k t - i\mathbf{k}\cdot\mathbf{r}}\right).$$

Using this result, we can also formulate the quantum field operators for the electric and magnetic fields. To simplify the notation, we can define a generic phase variable

$$\chi_{\mathbf{k}}(\mathbf{r}, t) = \omega_k t - \mathbf{k} \cdot \mathbf{r} - \frac{\pi}{2}$$

where the $\pi/2$ term is added for convenience. The electric field can be written in terms of a "positive frequency" and a "negative frequency part":

$$\begin{aligned} \hat{\mathbf{E}}_T &= \hat{\mathbf{E}}_T^+(\mathbf{r}, t) + \hat{\mathbf{E}}_T^-(\mathbf{r}, t) \\ &= \sum_{\mathbf{k}, \lambda} \mathbf{e}_{\mathbf{k}\lambda} \left(\frac{\hbar\omega_k}{2\epsilon_0 V}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda} e^{-i\chi_{\mathbf{k}}(\mathbf{r}, t)} + \sum_{\mathbf{k}, \lambda} \mathbf{e}_{\mathbf{k}\lambda} \left(\frac{\hbar\omega_k}{2\epsilon_0 V}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda}^\dagger e^{-i\chi_{\mathbf{k}}(\mathbf{r}, t)} \end{aligned}$$

as can the magnetic field

$$\begin{aligned} \hat{\mathbf{B}}(\mathbf{r}, t) &= \hat{\mathbf{B}}^+(\mathbf{r}, t) + \hat{\mathbf{B}}^-(\mathbf{r}, t) \\ &= \sum_{\mathbf{k}, \lambda} (\mathbf{k} \times \mathbf{e}_{\mathbf{k}\lambda}) \left(\frac{\hbar}{2\epsilon_0 V \omega_k}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda} e^{-i\chi_{\mathbf{k}}(\mathbf{r}, t)} + \sum_{\mathbf{k}, \lambda} (\mathbf{k} \times \mathbf{e}_{\mathbf{k}\lambda}) \left(\frac{\hbar}{2\epsilon_0 V \omega_k}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda}^\dagger e^{-i\chi_{\mathbf{k}}(\mathbf{r}, t)}. \end{aligned}$$

We can, as a units-check, consider the ratio of electric to magnetic fields,

$$\frac{\langle |\hat{\mathbf{E}}| \rangle}{\langle |\hat{\mathbf{B}}| \rangle} = \frac{(\hbar\omega_k/2\epsilon_0 V)^{1/2}}{k(\hbar/2\epsilon_0 V \omega_k)^{1/2}} = \frac{(\omega_k^2)^{1/2}}{k} = \frac{\omega_k}{k} = c$$

which is as expected in SI units. Notice that the vector potential differs in phase by a quarter period from the electric and magnetic fields:

$$\begin{aligned} |\hat{\mathbf{A}}_{\mathbf{k}\lambda}^+(\mathbf{r}, t)| &= \left(\frac{\hbar}{2\epsilon_0 V \omega_k}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda} e^{-i(\omega_k t - \mathbf{k}\cdot\mathbf{r})} \\ |\hat{\mathbf{E}}_{\mathbf{k}\lambda}^+(\mathbf{r}, t)| &= \left(\frac{\hbar\omega_k}{2\epsilon_0 V}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda} e^{-i(\omega_k t - \mathbf{k}\cdot\mathbf{r} - \pi/2)} \\ |\hat{\mathbf{B}}_{\mathbf{k}\lambda}^+(\mathbf{r}, t)| &= \left(\frac{\hbar k^2}{2\epsilon_0 V \omega_k}\right)^{1/2} \hat{a}_{\mathbf{k}\lambda} e^{-i(\omega_k t - \mathbf{k}\cdot\mathbf{r} - \pi/2)}. \end{aligned}$$

This result implies that the vector potential is conjugate to the $\mathbf{E}$ and $\mathbf{B}$ fields, and is, in fact, the generator of these fields; this is not surprising in the classical sense, since it is related through derivatives, akin to momentum, the "generator of translations" relationship to position through a derivative.

To make the connection between our approach to quantizing the field here, and other approaches, we complete our definitions by considering *quadrature operators*. We will separate the $\mathbf{E}$ and $\mathbf{B}$ fields into a part that oscillates "like a cosine" and a part that oscillates "like a sine," which we shall call the *in-phase* and *out-of-phase* components. This is a way to characterize a final phase with respect to a reference (input) phase.

For the electric field, we have

$$\begin{aligned}\hat{\mathbf{E}}_{\mathbf{k}\lambda}(\mathbf{r}, t) &= \left(\frac{\hbar\omega_k}{2\epsilon_0 V}\right)^{1/2} \left[\hat{a}_{\mathbf{k}\lambda} e^{-i\chi_{\mathbf{k}}(\mathbf{r}, t)} + \hat{a}_{\mathbf{k}\lambda}^\dagger e^{i\chi_{\mathbf{k}}(\mathbf{r}, t)} \right] \\ &= \left(\frac{\hbar\omega_k}{2\epsilon_0 V}\right)^{1/2} \left[\hat{a}_{\mathbf{k}\lambda} (\cos \chi_k - i \sin \chi_k) + \hat{a}_{\mathbf{k}\lambda}^\dagger (\cos \chi_k + i \sin \chi_k) \right] \\ &= \left(\frac{\hbar\omega_k}{2\epsilon_0 V}\right)^{1/2} \left[\cos \chi_k (\hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda}^\dagger) + i \sin \chi_k (\hat{a}_{\mathbf{k}\lambda}^\dagger - \hat{a}_{\mathbf{k}\lambda}) \right].\end{aligned}$$

We define two new operators

$$\hat{X}_{\mathbf{k}\lambda} = \frac{1}{2} (\hat{a}_{\mathbf{k}\lambda}^\dagger + \hat{a}_{\mathbf{k}\lambda}), \quad \hat{Y}_{\mathbf{k}\lambda} = \frac{i}{2} (\hat{a}_{\mathbf{k}\lambda}^\dagger - \hat{a}_{\mathbf{k}\lambda}).$$

which have the commutation property

$$[\hat{X}_{\mathbf{k}\lambda}, \hat{Y}_{\mathbf{k}'\lambda'}] = \frac{i}{2} \delta_{\mathbf{k}\mathbf{k}'} \delta_{\lambda\lambda'}.$$

This lets us write the field operators

$$\begin{aligned}\hat{\mathbf{E}}(\mathbf{r}, t) &= \sum_{\mathbf{k}\lambda} \mathbf{e}_{\mathbf{k}\lambda} \left(\frac{2\hbar\omega_k}{\epsilon_0 V}\right)^{1/2} \left\{ \hat{X}_{\mathbf{k}\lambda} \cos[\chi_{\mathbf{k}}(\mathbf{r}, t)] + \hat{Y}_{\mathbf{k}\lambda} \sin[\chi_{\mathbf{k}}(\mathbf{r}, t)] \right\} \\ \hat{\mathbf{B}}(\mathbf{r}, t) &= \sum_{\mathbf{k}\lambda} (\mathbf{k} \times \mathbf{e}_{\mathbf{k}\lambda}) \left(\frac{2\hbar}{\epsilon_0 V \omega_k}\right)^{1/2} \left\{ \hat{X}_{\mathbf{k}\lambda} \cos[\chi_{\mathbf{k}}(\mathbf{r}, t)] + \hat{Y}_{\mathbf{k}\lambda} \sin[\chi_{\mathbf{k}}(\mathbf{r}, t)] \right\}\end{aligned}$$

while

$$\hat{\mathbf{A}}(\mathbf{r}, t) = \sum_{\mathbf{k}\lambda} \mathbf{e}_{\mathbf{k}\lambda} \left(\frac{2\hbar}{\epsilon_0 V \omega_k}\right)^{1/2} \left\{ \hat{X}_{\mathbf{k}\lambda} \sin[\chi_{\mathbf{k}}(\mathbf{r}, t)] - \hat{Y}_{\mathbf{k}\lambda} \cos[\chi_{\mathbf{k}}(\mathbf{r}, t)] \right\}.$$

This gives us (at least) 3 ways to express the Hamiltonian, where $U_{EM} \rightarrow \hat{H}$, so

$$\begin{aligned}\hat{H}_{EM} &= \sum_{\mathbf{k}\lambda} \frac{\hbar\omega_k}{2} (\hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda}^\dagger) \\ &= \sum_{\mathbf{k}\lambda} \hbar\omega_k \left(n_{\mathbf{k}\lambda} + \frac{1}{2} \right) \\ &= \sum_{\mathbf{k}\lambda} \hbar\omega_k (X_{\mathbf{k}\lambda}^2 + Y_{\mathbf{k}\lambda}^2) \\ &= \sum_{\mathbf{k}\lambda} \int_{\text{cavity}} d^3\mathbf{r} \left[\epsilon_0 \hat{\mathbf{E}}_0(\mathbf{r}, t) \cdot \hat{\mathbf{E}}_0(\mathbf{r}, t) + \frac{1}{\mu_0} \hat{\mathbf{B}}(\mathbf{r}, t) \cdot \hat{\mathbf{B}}(\mathbf{r}, t) \right].\end{aligned}$$

There are a few things we can say about this result. The first is that each mode has its own energy, independent of the others. The second is that the eigenstates of the *photon-number operator* $\hat{n}_{\mathbf{k}\lambda} = \hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda}$ are the eigenstates of the Hamiltonian, and so the energy of each mode depends on this operator. And, in each mode $(\mathbf{k}, \lambda)$, the eigenvalues are

$$n_{\mathbf{k}\lambda} = 0, 1, 2, \dots$$

with eigenstates (labelled by $n_{\mathbf{k}\lambda}$)

$$\hat{n}_{\mathbf{k}\lambda} |n_{\mathbf{k}\lambda}\rangle = n_{\mathbf{k}\lambda} |n_{\mathbf{k}\lambda}\rangle$$

where

$$|n_{\mathbf{k}\lambda}\rangle = \{|0\rangle, |1\rangle, |2\rangle, \dots\}$$

which are the photon-number states. This implies that there can be zero photons (excitations) of a mode, but there is still energy, given by

$$\langle 0 | \hat{H}_{\mathbf{k}\lambda} | 0 \rangle = \frac{1}{2} \hbar \omega_k.$$

So, every mode of the field carries $\hbar \omega_k / 2$ of energy (each polarization), which is actually the energy of the electromagnetic vacuum

$$E_0 = \sum_k \sum_\lambda \frac{1}{2} \hbar \omega_k = \sum_k \hbar \omega_k.$$

The *vacuum field* is the set of states with no photons. That is, $\hat{n}_{\mathbf{k}\lambda} = 0$ for all $\{\mathbf{k}, \lambda\}$. A very formal definition of the photon follows:

”A photon is an excitation of a mode of the electromagnetic field.”

Consider now a simple case, where we imagine just one mode of the EM field with $(\mathbf{k}, \lambda = 1)$.

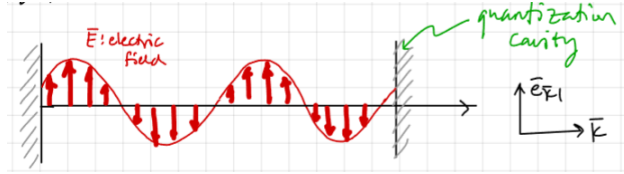


Figure 43: A single mode of the EM field within a quantization cavity.

Here, the wavevector under a cavity length L and a 2π rotation is given by

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{L/2} = \frac{4\pi}{L} \implies \nu = 2, \quad \omega = ck = \frac{4\pi c}{L}.$$

The number operator follows and is $\hat{n}_{\mathbf{k}\lambda} = \hat{n}_{\mathbf{k}1} = \hat{a}_{\mathbf{k}1}^\dagger \hat{a}_{\mathbf{k}1}$, giving an expectation value $\langle \hat{n}_{\mathbf{k}\lambda} \rangle = \langle \hat{a}_{\mathbf{k}1}^\dagger \hat{a}_{\mathbf{k}1} \rangle$. Thus, we expect the total energy

$$\langle \hat{H}_{k1} \rangle = \hbar \omega_k \left(\langle \hat{n}_{\mathbf{k}1} \rangle + \frac{1}{2} \right).$$

For eigenstates of the Hamiltonian, we will consider the electric field operator's expectation values

$$\begin{aligned}\langle \hat{E}_T \rangle &= \langle \hat{n}_{\mathbf{k}\lambda} | \left(\frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0 V} \right)^{1/2} \left(\hat{a}_{\mathbf{k}\lambda} e^{-i\chi_{\mathbf{k}}} + \hat{a}_{\mathbf{k}\lambda}^\dagger e^{i\chi_{\mathbf{k}}} \right) | n_{\mathbf{k}\lambda} \rangle \\ &= \left(\frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0 V} \right)^{1/2} \left[\langle n_{\mathbf{k}\lambda} | \hat{a}_{\mathbf{k}\lambda} | n_{\mathbf{k}\lambda} \rangle e^{-i\chi_{\mathbf{k}}} + \langle n_{\mathbf{k}\lambda} | \hat{a}_{\mathbf{k}\lambda}^\dagger | n_{\mathbf{k}\lambda} \rangle e^{i\chi_{\mathbf{k}}} \right] \\ &= 0\end{aligned}$$

since both inner products equate to zero. Similarly,

$$\begin{aligned}\langle \hat{E}_T^2 \rangle &= \frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0 V} \langle n_{\mathbf{k}\lambda} | \left(\hat{a}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda} e^{-2i\chi_{\mathbf{k}}} + \hat{a}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda}^\dagger + \hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda}^\dagger e^{2i\chi_{\mathbf{k}}} \right) | n_{\mathbf{k}\lambda} \rangle \\ &= \frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0 V} \left[\langle n_{\mathbf{k}\lambda} + 1 | (n_{\mathbf{k}\lambda} + 1)^{1/2} (n_{\mathbf{k}\lambda} + 1)^{1/2} | n_{\mathbf{k}\lambda} + 1 \rangle + \langle n_{\mathbf{k}\lambda} - 1 | n_{\mathbf{k}\lambda}^{1/2} n_{\mathbf{k}\lambda}^{1/2} | n_{\mathbf{k}\lambda} - 1 \rangle \right] \\ &= \frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0 V} (n_{\mathbf{k}\lambda} + 1 + n_{\mathbf{k}\lambda}) \\ &= \frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0 V} \left(n_{\mathbf{k}\lambda} + \frac{1}{2} \right)\end{aligned}$$

which shows that the RMS electric field amplitude in mode $\mathbf{k}\lambda$ $E_{\text{rms}} \sim \sqrt{\langle \hat{E}_T^2 \rangle}$ is discretized for the eigenstates.

For our schematic of an electric field in the quantization cavity, let's consider $E_{\text{rms}} = \sqrt{\langle \hat{E}_T^2 \rangle}$ via some "snapshots" in time and space:

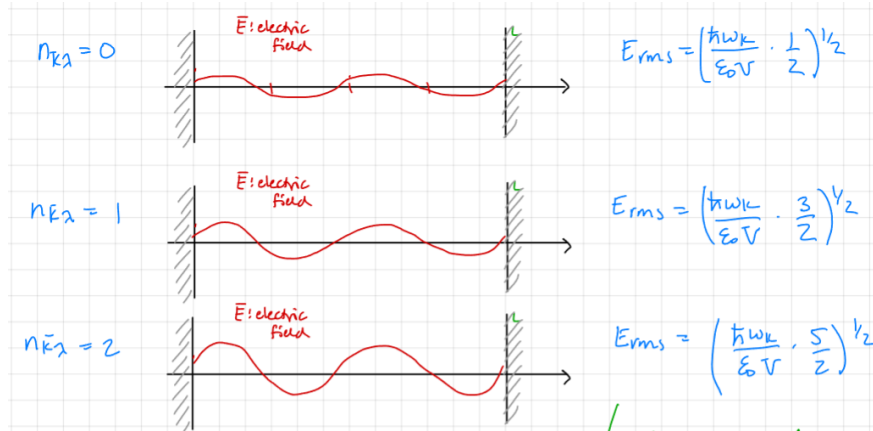


Figure 44: Root-mean-square electric field for different harmonics $n_{\mathbf{k}\lambda} = 0, 1, 2$.

Observe that the electric field amplitude grows at $\sqrt{n_{\mathbf{k}\lambda} + 1/2}$. We make a note about the modes of our electric fields. We can see that the number of photons in a mode determines the field amplitude of that mode. That is, if a state is not an eigenstate, i.e., a superposition state, then a mode may have a non-integer expectation value of the expectation value of the number of photons in a mode. This, in itself, is not a big deal.

The spectrum of energies is of the form of the QHM. However, we must not confuse the spatial wavefunctions of the QHM with the spatial field distributions of the EM field. These are not the same.

Before moving on, we will take a look at the quadrature operators' expectation values for these eigenstates of the Hamiltonian. We have

$$\langle X_{\mathbf{k}\lambda} \rangle = \langle n_{\mathbf{k}\lambda} | \frac{1}{2} (\hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda}^\dagger) | n_{\mathbf{k}\lambda} \rangle = 0 = \langle Y_{\mathbf{k}\lambda} \rangle.$$

However, for the squares,

$$\begin{aligned} \langle X_{\mathbf{k}\lambda}^2 \rangle &= \langle n_{\mathbf{k}\lambda} | \frac{1}{4} (\hat{a}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda}^\dagger + \hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda}^\dagger) | n_{\mathbf{k}\lambda} \rangle \\ &= \frac{1}{4} [\langle n_{\mathbf{k}\lambda} | \hat{a}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda}^\dagger + \hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}\lambda} | n_{\mathbf{k}\lambda} \rangle] \\ &= \frac{1}{4} (n_{\mathbf{k}\lambda} + 1 + n_{\mathbf{k}\lambda}) \\ &= \frac{1}{2} \left(n_{\mathbf{k}\lambda} + \frac{1}{2} \right) \\ \langle Y_{\mathbf{k}\lambda}^2 \rangle &= \frac{i^2}{4} \langle n_{\mathbf{k}\lambda} | (\hat{a}_{\mathbf{k}\lambda}^\dagger - \hat{a}_{\mathbf{k}\lambda}) (\hat{a}_{\mathbf{k}\lambda}^\dagger - \hat{a}_{\mathbf{k}\lambda}) | n_{\mathbf{k}\lambda} \rangle \\ &= -\frac{1}{4} [-(n_{\mathbf{k}\lambda} + 1) - n_{\mathbf{k}\lambda}] \\ &= \frac{1}{2} \left(n_{\mathbf{k}\lambda} + \frac{1}{2} \right). \end{aligned}$$

Notice that both $\langle X_{\mathbf{k}\lambda} \rangle$ and $\langle Y_{\mathbf{k}\lambda} \rangle$ have the same dependence on $n_{\mathbf{k}\lambda}$ as the field-squared operators, and we observe that they are the same as each other.

In fact, if we recall the electric field definition in terms of these (for a single mode), we find

$$\hat{E}_{\mathbf{k}\lambda}(\mathbf{r}, t) = \left(\frac{2\hbar\omega_k}{\epsilon_0 V} \right)^{1/2} \left\{ \hat{X}_{\mathbf{k}\lambda} \cos[\chi_{\mathbf{k}}(\mathbf{r}, t)] + \hat{Y}_{\mathbf{k}\lambda} \sin[\chi_{\mathbf{k}}(\mathbf{r}, t)] \right\}$$

so the X and Y have equal rms amplitudes for eigenstates of the Hamiltonian. That is, the sum of the amplitudes is a constant.

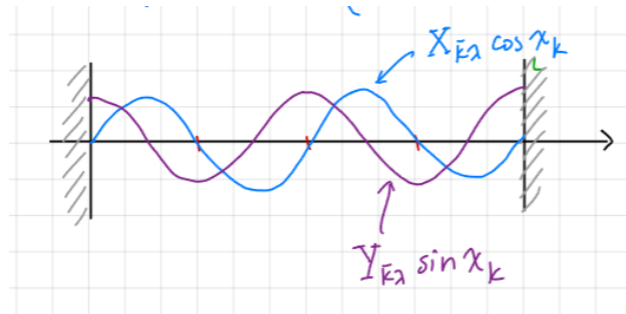


Figure 45: $X_{\mathbf{k}\lambda} \cos \chi_{\mathbf{k}}$ and $Y_{\mathbf{k}\lambda} \sin \chi_{\mathbf{k}}$ plotted within a quantization cavity.

Indeed, we can use a phasor picture of the electric field operator that captures this character of conjugate variables:

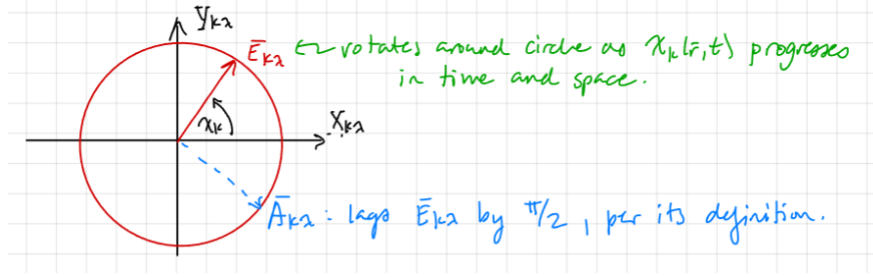


Figure 46: Phasor diagram of the electric field operator rotating with angle $\chi_{\mathbf{k}}$.

3.3 States of the EM Field

Let us consider different states of the EM field, including number states, coherent states, thermal states, and squeezed states. Specifically, we will consider states of a *single mode*: one $\{\mathbf{k}, \lambda\}$ (and we drop the $\mathbf{k}, \lambda$ subscripts for clarity).

3.3.1 Number States

Number states are the eigenstates of the EM Hamiltonian that we started discussing above. These number states have an exactly defined photon number, and they are also commonly called *Fock States*. We may write the number state of a single mode as

$$|\psi\rangle = |n\rangle, \quad \hat{n} = \hat{a}^\dagger \hat{a}$$

where there is no uncertainty in the photon number

$$\Delta n \equiv (\langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2)^{1/2} = \langle n | \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} | n \rangle - \left(\langle n | \hat{a}^\dagger \hat{a} | n \rangle \right)^2 = n^2 - n^2 = 0.$$

As eigenstates of the Hamiltonian, they are also eigenstates of the quadratures operator, so

$$\begin{aligned} \langle n | \hat{X}^2 + \hat{Y}^2 | n \rangle &= \langle n | \hat{X}^2 | n \rangle + \langle n | \hat{Y}^2 | n \rangle \\ &= \langle n | \frac{1}{4} (\hat{a}^\dagger \hat{a}^\dagger + \hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger + \hat{a} \hat{a}) | n \rangle + \frac{i^2}{4} \langle n | \hat{a}^\dagger \hat{a}^\dagger - \hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger + \hat{a} \hat{a} | n \rangle \\ &= \frac{1}{2} (\langle n | \hat{a}^\dagger \hat{a} | n \rangle + \langle n | \hat{a} \hat{a}^\dagger | n \rangle) \\ &= \frac{1}{2} [(\sqrt{n})^2 + (\sqrt{n+1})^2] \\ &= n + \frac{1}{2}. \end{aligned}$$

In a phasor diagram, this state sits on the circle of radius $r = \sqrt{n + 1/2}$ with angle χ , where the state $|n\rangle$ lies somewhere along that circle. However, there is nothing distinguishing any angle χ for this state.

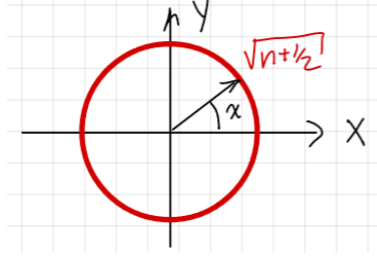


Figure 47: Circle of radius $r = \sqrt{n + 1/2}$, where the circle designates the state $|n\rangle$

Consider now the expectation value of the electric field. As above, we find $\langle \hat{E}_T \rangle = 0$, and we calculate

$$\begin{aligned} \langle \hat{E}_T^2 \rangle &= \left\langle \frac{2\hbar\omega_k}{\epsilon_0 V} \left(\hat{X} \cos \chi + \hat{Y} \sin \chi \right)^2 \right\rangle \\ &= \frac{2\hbar\omega_k}{\epsilon_0 V} \langle \hat{X}^2 \cos^2 \chi + \hat{X}\hat{Y} \cos \chi \sin \chi + \hat{Y}\hat{X} \sin \chi \cos \chi + \hat{Y}^2 \sin^2 \chi \rangle \\ &= \frac{2\hbar\omega_k}{\epsilon_0 V} \left(\langle \hat{X}^2 \rangle \cos^2 \chi + \langle \hat{X}\hat{Y} \rangle \cos \chi \sin \chi + \langle \hat{Y}\hat{X} \rangle \cos \chi \sin \chi + \langle \hat{Y}^2 \rangle \sin^2 \chi \right). \end{aligned}$$

Observe that

$$\langle \hat{X}\hat{Y} + \hat{Y}\hat{X} \rangle = \langle n | \frac{i}{4} (\hat{a}^\dagger \hat{a}^\dagger - \hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger - \hat{a} \hat{a} + \hat{a}^\dagger \hat{a}^\dagger + \hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger - \hat{a} \hat{a}) | n \rangle = 0$$

so that by recalling our results for $\langle \hat{X}^2 \rangle$ and $\langle \hat{Y}^2 \rangle$, we find

$$\langle \hat{E}_T^2 \rangle = \frac{2\hbar\omega_k}{\epsilon_0 V} \frac{1}{2} \left(n + \frac{1}{2} \right) = \frac{\hbar\omega_k}{\epsilon_0 V} \left(n + \frac{1}{2} \right).$$

Thus, the variance of the electric field is

$$\Delta E_T \equiv \sqrt{\langle \hat{E}_T^2 \rangle - \langle \hat{E}_T \rangle^2} = \left(\frac{\hbar\omega_k}{\epsilon_0 V} \right)^{1/2} \sqrt{n + \frac{1}{2}}.$$

Note that for the vacuum state, we have $|\psi\rangle = |0\rangle$, so there is the special case

$$\Delta E_T^{\text{vac}} = \left(\frac{\hbar\omega_k}{\epsilon_0 V} \right)^{1/2}$$

which is a measure of the amplitude of the EM vacuum field.

To picture the field of a number state, we must think of a field that has a perfectly well-defined amplitude but no defined phase: we will picture a single amplitude (no variance or uncertainty) at all possible phases.

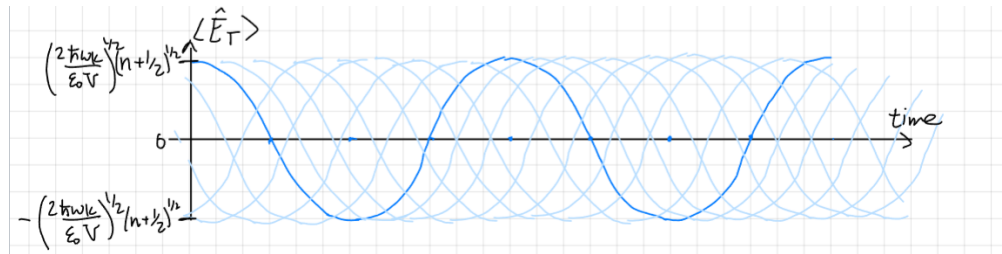


Figure 48: A number state with amplitude $\pm(2\hbar\omega_k/\epsilon_0 V)^{1/2}(n + 1/2)^{1/2}$ existing at all phases.

3.3.2 Coherent State

A *coherent state* corresponds to the classical monochromatic EM field, i.e., it has a well-defined amplitude and phase. Thus, it is coherent. The above-threshold lasers we have discussed thus far generate coherent states.

We will label the coherent state as

$$|\psi\rangle = |\alpha\rangle$$

and define an operator $\hat{\alpha} = \hat{X} + i\hat{Y}$. The coherent state is a linear superposition of number states

$$|\alpha\rangle = \exp\left(-\frac{1}{2}|\alpha|^2\right) \sum_{n=0}^{\infty} \frac{\alpha^n}{(n!)^{1/2}} |n\rangle$$

and has the following properties:

- (1) the coherent state is normalized

$$\begin{aligned} \langle\alpha|\alpha\rangle &= \exp(-|\alpha|^2) \sum_{n=0}^{\infty} \sum_{n'=0}^{\infty} \frac{(\alpha^*)^n (\alpha)^{n'}}{(n'!n!)^{1/2}} \langle n'|n\rangle \\ &= \exp(-|\alpha|^2) \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} \\ &= \exp(-|\alpha|^2) \exp(|\alpha|^2) \\ &= 1; \end{aligned}$$

- (2) the coherent states are not orthonormal

$$\begin{aligned} \langle\alpha|\beta\rangle &= \exp\left(-\frac{1}{2}|\alpha|^2 - \frac{1}{2}|\beta|^2\right) \sum_{n=0}^{\infty} \frac{(\alpha^*)^n \beta^n}{n!} \\ &= \exp\left(-\frac{1}{2}|\alpha|^2 - \frac{1}{2}|\beta|^2 - \alpha^* \beta\right) \\ &= \exp\left[-\frac{1}{2}\left(|\alpha|^2 + |\beta|^2 + 2\alpha^* \beta\right)\right] \\ &= \exp\left[-\frac{1}{2}(\alpha - \beta)^2\right] \end{aligned}$$

where the set of all coherent states is overcomplete. So, the similarity of states decreases as α, β differ from each other. These states are considered approximately orthogonal for $|\alpha - \beta| \gg 1$;

(3) the coherent states are eigenstates of the lowering and destruction operator

$$\begin{aligned}
\hat{a} |\alpha\rangle &= \exp\left(-\frac{1}{2}|\alpha|^2\right) \sum_{n=0}^{\infty} \frac{\alpha^n}{(n!)^{1/2}} \hat{a} |n\rangle \\
&= \exp\left(-\frac{1}{2}|\alpha|^2\right) \sum_{n=1}^{\infty} \frac{\alpha^n}{(n!)^{1/2}} \sqrt{n} |n-1\rangle \\
&= \exp\left(-\frac{1}{2}|\alpha|^2\right) \sum_{n=0}^{\infty} \frac{\alpha^{n+1}}{[(n+1)!]^{1/2}} \sqrt{n+1} |n\rangle \\
&= \alpha \exp\left(-\frac{1}{2}|\alpha|^2\right) \sum_{n=0}^{\infty} \frac{\alpha^n}{n!} |n\rangle \\
\hat{a} |\alpha\rangle &= \alpha |\alpha\rangle
\end{aligned}$$

which implies also that $\langle\alpha| \hat{a}^\dagger = \alpha^* \langle\alpha|$. In a way these are the definition of a coherent state.

Suppose we want to calculate the average and the variances associated with the coherent state. Here, the photon number $\hat{n} = \hat{a}^\dagger \hat{a}$ shows that

$$\begin{aligned}
\langle\hat{n}\rangle &= \langle\alpha| \hat{a}^\dagger \hat{a} |\alpha\rangle = \alpha^* \alpha = |\alpha|^2 \\
\langle\hat{n}^2\rangle &= \langle\alpha| \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} |\alpha\rangle = \langle\alpha| \hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a} + \hat{a}^\dagger \hat{a} |\alpha\rangle = |\alpha|^4 + |\alpha|^2 \\
\Delta n &\equiv \sqrt{\langle\hat{n}^2\rangle - \langle\hat{n}\rangle^2} = \sqrt{|\alpha|^4 + |\alpha|^2 - |\alpha|^4} = |\alpha|^2 = \sqrt{\langle\hat{n}\rangle}.
\end{aligned}$$

Notice that the number of excitations (number of photons) in the mode is not exact. Indeed, there is a variance that scales as the square root of the number of excitations. Specifically, a coherent state has an average photon number $\langle\hat{n}\rangle = |\alpha|^2 \equiv n$, but this number has an associated uncertainty of $\sqrt{\langle\hat{n}\rangle}$. We can determine the fractional uncertainty to be

$$\frac{\Delta n}{n} = \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{\langle\hat{n}\rangle}}$$

which is the form of Poissonian "counting" statistics

The probability of a particular eigenstate being measured in a coherent state (e.g., through a photon-counting detector) is the square of the coefficient in the sum

$$P(n) = |\langle n|\alpha\rangle|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!} = e^{-\langle\hat{n}\rangle} \frac{\langle\hat{n}\rangle^n}{n!} = e^{-\langle\hat{n}\rangle} \frac{\langle\hat{n}\rangle^n}{n!}$$

which is the Poisson distribution. For example, if we had a coherent state with one photon on average (so $|\alpha|^2 = 1$), we have

$$\begin{aligned}
P(0) &= \frac{e^{-1} \times 1^0}{0!} = e^{-1} = 0.368 \\
P(1) &= \frac{e^{-1} \times 1^1}{1!} = e^{-1} = 0.368 \\
P(2) &= \frac{e^{-1} \times 1^2}{2!} = \frac{e^{-1}}{2} = 0.184 \\
P(3) &= \frac{e^{-1} \times 1^3}{3!} = \frac{e^{-1}}{6} = 0.0613 \\
&\vdots
\end{aligned}$$

and so forth. Observe that for $\langle n \rangle = 1$, there is a significant vacuum component and a significant $n \geq 2$ component.

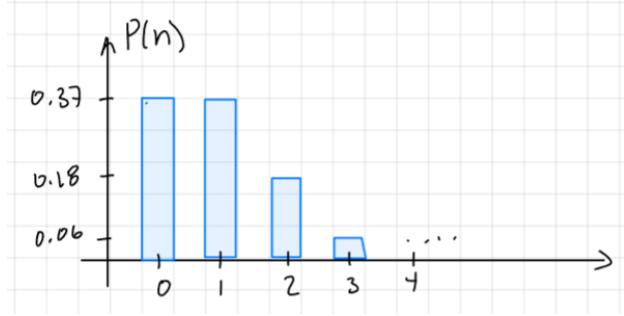


Figure 49: Probability of a coherent state with a single photon on average as the number of photons increases.

Next, we will consider the quadratures and associated fields:

$$\begin{aligned}\langle \hat{X} \rangle &= \langle \alpha | \hat{X} | \alpha \rangle = \frac{1}{2} \langle \alpha | \hat{a} + \hat{a}^\dagger | \alpha \rangle = \frac{1}{2} \langle \alpha | \hat{a} | \alpha \rangle + \frac{1}{2} \langle \alpha | \hat{a}^\dagger | \alpha \rangle = \frac{1}{2} (\alpha + \alpha^*) = \text{Re}(\alpha) = |\alpha| \cos \phi \\ \langle \hat{Y} \rangle &= \langle \alpha | \hat{Y} | \alpha \rangle = \frac{i}{2} \langle \alpha | \hat{a}^\dagger - \hat{a} | \alpha \rangle = \frac{i}{2} (\alpha^* - \alpha) = \text{Im}(\alpha) = |\alpha| \sin \phi.\end{aligned}$$

We have also

$$\begin{aligned}\langle \hat{X}^2 \rangle &= \frac{1}{4} \langle \alpha | \hat{a}^\dagger \hat{a}^\dagger + \hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger + \hat{a} \hat{a} | \alpha \rangle \\ &= \frac{1}{4} [(\alpha^*)^2 + |\alpha|^2 + |\alpha|^2 + 1 + \alpha^2] \\ &= \frac{1}{4} [(\alpha^*)^2 + \alpha^2 + 2|\alpha|^2 + 1] \\ \langle \hat{Y}^2 \rangle &= -\frac{1}{4} \langle \alpha | \hat{a}^\dagger \hat{a}^\dagger - \hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger + \hat{a} \hat{a} | \alpha \rangle \\ &= -\frac{1}{4} [(\alpha^*)^2 + \alpha^2 - 2|\alpha|^2 - 1].\end{aligned}$$

The variances follow

$$\begin{aligned}\Delta X &= \sqrt{\langle \hat{X}^2 \rangle - \langle \hat{X} \rangle^2} \\ &= \frac{1}{2} \left\{ [(\alpha^*)^2 + \alpha^2 + 2|\alpha|^2 + 1] - (\alpha + \alpha^*)^2 \right\}^{1/2} \\ &= \frac{1}{2} [(\alpha^*)^2 + \alpha^2 + 2|\alpha|^2 + 1 - \alpha^2 - 2|\alpha|^2 - (\alpha^*)^2] = \frac{1}{2} \\ \Delta Y &= \sqrt{\langle \hat{Y}^2 \rangle - \langle \hat{Y} \rangle^2} \\ &= \left\{ -\frac{1}{4} [(\alpha^*)^2 + \alpha^2 - 2|\alpha|^2 - 1] + \frac{1}{4} (\alpha^* - \alpha)^2 \right\}^{1/2} = \frac{1}{2}.\end{aligned}$$

We see that the variances are independent of α (and n) and are the same as a vacuum state. So, we can say that coherent states are the *minimum uncertainty states*.

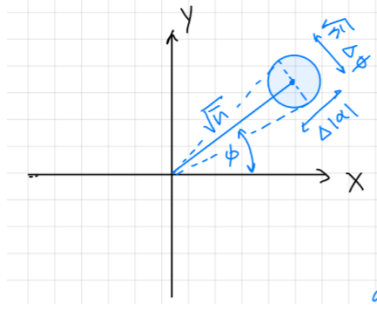


Figure 50: A "displaced vacuum state," displaced from the origin by $|\alpha| = \sqrt{n}$.

Observe from Fig. 50 that the size (radius) of the disk, which represents the square rooted variance, is the same everywhere in the xy -plane. The uncertainty in the length and phase can be related to Δn by

$$\Delta|\alpha| = \frac{1}{2}, \quad \sqrt{n}\Delta\phi = \frac{1}{2}, \quad \sqrt{n} = \Delta n$$

so

$$\Delta n \Delta\phi = \left(\frac{1}{2}\right)^2$$

which is the *number/phase uncertainty product*. In contrast to a number state, where n is well-defined but phase is unknown, here the uncertainty is distributed between phase and number of photons.

To picture the propagating field, we will plot the electric field,

$$\begin{aligned} \langle \hat{E}_T \rangle &= \left(\frac{2\hbar\omega}{\epsilon_0 V} \right)^{1/2} \left[\langle \hat{X} \cos \chi_{\mathbf{k}}(\mathbf{r}, t) + \hat{Y} \sin \chi_{\mathbf{k}}(\mathbf{r}, t) \right] \\ &= \left(\frac{2\hbar\omega}{\epsilon_0 V} \right)^{1/2} \left[\langle \hat{X} \rangle \cos \chi(\mathbf{r}, t) + \langle \hat{Y} \rangle \sin \chi(\mathbf{r}, t) \right] \\ &= \left(\frac{2\hbar\omega}{\epsilon_0 V} \right)^{1/2} [|\alpha| \cos \phi \cos \chi + |\alpha| \sin \phi \sin \chi] \\ &= \left(\frac{2\hbar\omega}{\epsilon_0 V} \right)^{1/2} |\alpha| [\cos(\phi - \chi)] \\ &= \sqrt{n} \left(\frac{2\hbar\omega}{\epsilon_0 V} \right)^{1/2} \cos(\phi - \chi) \end{aligned}$$

and

$$\langle \hat{E}_T^2 \rangle = \left(\frac{2\hbar\omega}{\epsilon_0 V} \right) \left[\langle \hat{X}^2 \rangle \cos^2 \chi + \langle \hat{X} \hat{Y} + \hat{Y} \hat{X} \rangle \cos \chi \sin \chi + \langle \hat{Y}^2 \rangle \sin^2 \chi \right].$$

Evaluating the expectation values explicitly, we use the relations

$$\hat{a}^\dagger \hat{a}^\dagger |\alpha\rangle = \alpha^2 |\alpha\rangle, \quad \langle \alpha | \hat{a} \hat{a} = (\alpha^*)^2 \langle \alpha |$$

so that

$$\langle \hat{X} \hat{Y} + \hat{Y} \hat{X} \rangle = \frac{i}{2} [\alpha^2 - (\alpha^*)^2] = \frac{i}{2} |\alpha|^2 [e^{2i\phi} - e^{-2i\phi}] = |\alpha|^2 \sin 2\phi$$

and

$$\begin{aligned}
 \langle \hat{X}^2 \rangle &= \frac{1}{4} [(\alpha^*)^2 + |\alpha|^2 + |\alpha|^2 + 1 + (\alpha)^2] \\
 &= \frac{1}{4} [2|\alpha|^2 + 1 + (\alpha^*)^2 + (\alpha)^2] \\
 &= \frac{1}{4} [2|\alpha|^2 + 1 + 2|\alpha|^2 \cos(2\phi)] \\
 \langle \hat{Y}^2 \rangle &= -\frac{1}{4} [(\alpha^*)^2 - |\alpha|^2 - |\alpha|^2 - 1 + (\alpha)^2] \\
 &= -14 [-2|\alpha|^2 - 1 + 2|\alpha|^2 \cos^2(2\phi)].
 \end{aligned}$$

Thus,

$$\begin{aligned}
 \langle \hat{E}_T^2 \rangle &= \left(\frac{2\hbar\omega}{\epsilon_0 V} \right) \left(\frac{|\alpha|^2}{2} + \frac{1}{4} \right) \cos^2 \chi + \left(\frac{2\hbar\omega}{\epsilon_0 V} \right) \left(\frac{|\alpha|^2}{2} + \frac{1}{4} \right) \sin^2 \chi \\
 &\quad + \left(\frac{2\hbar\omega}{\epsilon_0 V} \right) \left[\frac{|\alpha|^2}{2} \cos 2\phi \cos^2 \chi - \frac{|\alpha|^2}{2} \cos 2\phi \sin^2 \chi + |\alpha| \sin 2\phi \sin \chi \cos \chi \right] \\
 &= \left(\frac{2\hbar\omega}{\epsilon_0 V} \right) \left[\frac{|\alpha|^2}{2} (1 + \cos 2\phi \cos 2\chi + \sin 2\phi \sin 2\chi) + \frac{1}{4} \right] \\
 &= \left(\frac{2\hbar\omega}{\epsilon_0 V} \right) \left\{ \frac{|\alpha|^2}{2} [1 + \cos(2\phi - 2\chi)] + \frac{1}{4} \right\} \\
 &= \left(\frac{2\hbar\omega}{\epsilon_0 V} \right) \left[|\alpha|^2 \cos^2(\phi - \chi) + \frac{1}{4} \right].
 \end{aligned}$$

Finally, the variance of the electric field is

$$\Delta E_T = [\langle \hat{E}_T^2 \rangle - \langle \hat{E}_T \rangle^2]^{1/2} = \left[\left(\frac{2\hbar\omega}{\epsilon_0 V} \right) |\alpha|^2 \cos^2(\phi - \chi) + \left(\frac{\hbar\omega}{2\epsilon_0 V} \right) - \left(\frac{2\hbar\omega}{\epsilon_0 V} \right) |\alpha|^2 \cos^2(\phi - \chi) \right].$$

So,

$$\Delta E_T = \left(\frac{\hbar\omega}{2\epsilon_0 V} \right)^{1/2}$$

which shows that the electric field variance is independent of α and is equal to that of the vacuum. This is the minimum uncertainty state.

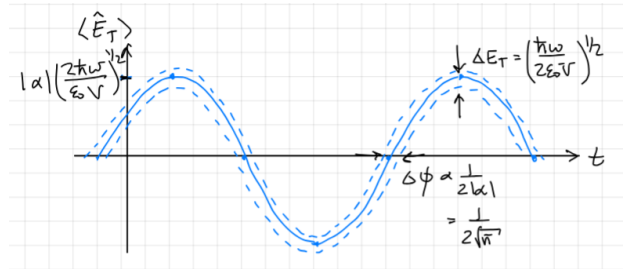


Figure 51: The electric field plotted with the "minimum uncertainty" width of the wave.

This field has a (fairly) well-defined amplitude and phase, and certainly much better resembles what we think of a classical field. Observe that the phase, ϕ , can be interpreted as an initial phase offset of the field's evolution.

3.3.3 Squeezed States

Currently, the state we have looked at had the same variances in the X and Y quadratures. Though, for instance, the coherent state is the minimum uncertainty state, it does not mean that these quadratures must equally share the uncertainty. *Squeezed states* trade fluctuations along quadratures one to the other. Let us consider the vacuum state:

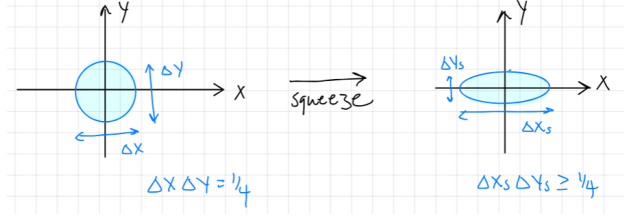


Figure 52: Squeeze vacuum state from a circle of uncertainty $\Delta X \Delta Y = 1/4$ to an ellipse of uncertainty $\Delta X_s \Delta Y_s \geq 1/4$.

The *squeezing operator* generates such a quadrature-squeeze state

$$\hat{S}(\zeta) = \exp \left[\frac{1}{2} \zeta^* \hat{a}^2 - \frac{1}{2} \zeta (\hat{a}^\dagger)^2 \right], \quad \zeta = s e^{i\nu}$$

where ζ is the *squeezing parameter*. We can apply this squeezing operator to the vacuum state, so

$$\hat{S}(\zeta) |0\rangle = (\text{sech } s)^{1/2} \sum_{n=0}^{\infty} \frac{[(2n)!]^{1/2}}{n!} \left[-\frac{1}{2} e^{i\nu} \tanh s \right]^n |2n\rangle$$

which shows that the squeeze vacuum is analogous to a coherent state with only even-photon-number states. We state, without proof, the following relationships

$$\begin{aligned} \langle n \rangle &= \sinh^2 s \\ \langle n^2 \rangle &= 3 \sinh^4 s + 2 \sinh^2 s \\ (\Delta n)^2 &= 2 \langle n \rangle (\langle n \rangle + 1) \\ (\Delta X)^2 &= \frac{1}{4} \left[e^{2s} \sin^2 \left(\frac{\nu}{2} \right) + e^{-2s} \cos^2 \left(\frac{\nu}{2} \right) \right] \\ (\Delta Y)^2 &= \frac{1}{4} \left[e^{2s} \cos^2 \left(\frac{\nu}{2} \right) + e^{-2s} \sin^2 \left(\frac{\nu}{2} \right) \right] \\ [\Delta E(x)]^2 &= \frac{1}{4} \left[e^{2s} \sin^2 \left(\chi - \frac{\nu}{2} \right) + e^{-2s} \cos^2 \left(\chi - \frac{\nu}{2} \right) \right]. \end{aligned}$$

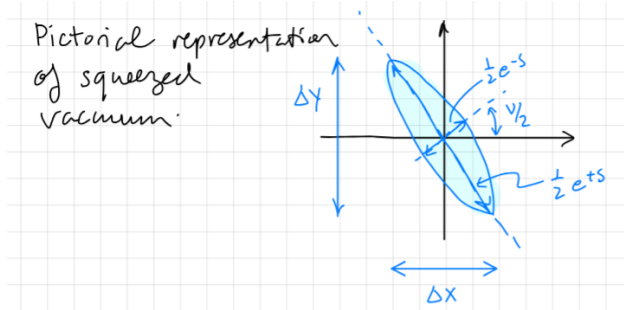


Figure 53: Pictorial representation of a squeeze vacuum.

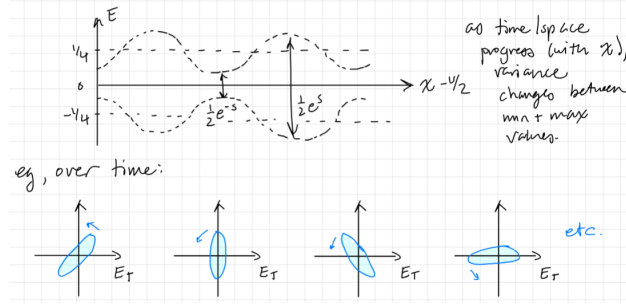


Figure 54: Variance plotted as a function of $\chi - \nu/2$, with the rotation of the squeezed state reflecting the plot.

3.3.4 Squeeze Coherent States

An alternate way to think about a coherent state is as a "displaced vacuum." That is, we take the vacuum state and shift it in the xy -plane to a non-zero ($\langle \hat{X} \rangle, \langle \hat{Y} \rangle$) position, where

$$\langle \hat{X} \rangle = |\alpha| \cos \phi, \quad \langle \hat{Y} \rangle = |\alpha| \sin \phi.$$

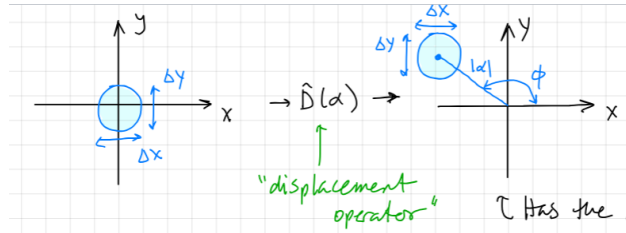


Figure 55: Vacuum state affected by a displacement operator $\hat{D}(\alpha)$.

Observe that this new displaced state has the same variance, but is no longer at the origin. We see that a *displacement operator* shifts the vacuum state to non-zero centroid coordinates, leaving the variances unchanged:

$$\hat{D}(\alpha) = \exp[\alpha \hat{a}^\dagger - \alpha^* \hat{a}].$$

This operator is equivalent to a creation operator for the entire coherent state, so we may write

$$\hat{D}(\alpha) |0\rangle = |\alpha\rangle$$

and, since it is unitary ($\hat{D}^\dagger(\alpha)\hat{D}(\alpha) = \hat{D}(\alpha)\hat{D}^\dagger(\alpha) = 1$), the effect on raising and lowering operators is

$$\hat{D}^\dagger(\alpha) \hat{a} \hat{D}(\alpha) = e^{-\alpha \hat{a}^\dagger + \alpha^* \hat{a}} \hat{a} e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} = e^{-\alpha \hat{a}^\dagger + \alpha^* \hat{a}} \left(e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} \hat{a} + [\hat{a}, e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}}] \right).$$

We use the relationship $e^{\hat{A} + \hat{B} + \frac{1}{2}[\hat{A}, \hat{B}]} = e^{\hat{A}} e^{\hat{B}}$ to find

$$\begin{aligned} e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a} - \frac{1}{2}[\alpha \hat{a}^\dagger, \alpha^* \hat{a}]} &= e^{\alpha \hat{a}^\dagger} e^{\alpha^* \hat{a}} \\ e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a} - \frac{1}{2}|\alpha|^2 [\hat{a}^\dagger, \hat{a}]} &= e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a} + \frac{1}{2}|\alpha|^2} \end{aligned}$$

so

$$e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} = e^{\alpha \hat{a}^\dagger} e^{\alpha^* \hat{a}} e^{-\frac{1}{2}|\alpha|^2}.$$

Then,

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = e^0\hat{a} + e^{-\alpha\hat{a}^\dagger + \alpha^*\hat{a}} \left[\hat{a}, e^{\alpha\hat{a}^\dagger} e^{\alpha^*\hat{a}} e^{-\frac{1}{2}|\alpha|^2} \right].$$

We use the commutation relationship $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$ to find that

$$\left[\hat{a}, e^{\alpha\hat{a}^\dagger} e^{\alpha^*\hat{a}} \right] = \left[\hat{a}, e^{\alpha\hat{a}^\dagger} \right] e^{\alpha^*\hat{a}} + e^{\alpha\hat{a}^\dagger} \left[\hat{a}, e^{\alpha^*\hat{a}} \right] = \left[\hat{a}, e^{\alpha\hat{a}^\dagger} \right] e^{\alpha^*\hat{a}}$$

since the last term commutes. We state also (without proof¹⁹) that $[\hat{a}, e^{\beta\hat{a}^\dagger}] = \beta e^{\beta\hat{a}^\dagger}$, so

$$\begin{aligned} \hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) &= e^0\hat{a} + e^{-\alpha\hat{a}^\dagger + \alpha^*\hat{a}}(\alpha)e^{\alpha\hat{a}^\dagger}e^{\alpha^*\hat{a}}e^{-\frac{1}{2}|\alpha|^2} \\ &= \hat{a} + \alpha e^{-\alpha\hat{a}^\dagger + \alpha^*\hat{a}}e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}} \\ \hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) &= \hat{a} + \alpha. \end{aligned}$$

Thus, we find that the displacement operator has the following properties:

$$\hat{D}(\alpha)\hat{a}\hat{D}^\dagger(\alpha) = \hat{a} + \alpha, \quad \hat{D}(\alpha)\hat{a}^\dagger\hat{D}^\dagger(\alpha) = \hat{a}^\dagger + \alpha^*.$$

Now, the same displacement can be applied to the squeezed vacuum, so

$$|\alpha, \zeta\rangle = \hat{D}(\alpha)\hat{S}(\zeta)|0\rangle$$

where we first squeeze with $\hat{S}(\zeta)$, the displace with $\hat{D}(\alpha)$. Observe that this squeezed coherent state maintains the noise characteristics of the squeeze vacuum state. Also, the same properties of the displacement operator apply, where

$$\begin{aligned} \hat{S}^\dagger(\zeta)\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha)\hat{S}(\zeta) &= \hat{a} \cosh(s) - \hat{a}^\dagger e^{-i\nu} \sinh(s) + \alpha \\ \hat{S}^\dagger(\zeta)\hat{D}^\dagger(\alpha)\hat{a}^\dagger\hat{D}(\alpha)\hat{S}(\zeta) &= \hat{a}^\dagger \cosh(s) - \hat{a} e^{-i\nu} \sinh(s) + \alpha^*. \end{aligned}$$

We thus find

$$\langle n \rangle = |\alpha|^2 + \sinh^2(s)$$

where $|\alpha|^2$ comes from the coherent state, and $\sinh^2(s)$ is from the squeeze vacuum. The $\langle \hat{X} \rangle$ and $\langle \hat{Y} \rangle$ parameters are the same as the coherent state, while the $(\Delta X)^2$ and $(\Delta Y)^2$ parameters are the same as the squeeze vacuum state.

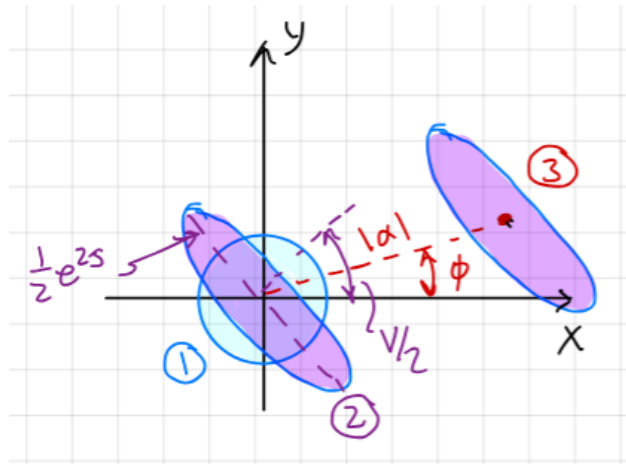


Figure 56: Pictorial representation of applying the squeeze operator then the displacement operator of a vacuum state.

¹⁹Maybe stick the proof in an appendix later.

We thus have three steps:

- (1) Begin with the vacuum state.
- (2) Apply $\hat{S}(\zeta)$ to get the squeeze vacuum with squeezing parameter $\zeta = se^{i\nu}$.
- (3) Displace squeezed vacuum to get the squeeze coherent state $\hat{D}(\alpha)$ with $\alpha = |\alpha|e^{i\phi}$.

The angle ν determines whether we have amplitude or phase-squeezing. Specifically, we can consider the time evolution of the electric field amplitude with $\chi(\mathbf{r}, t)$,

$$\langle \hat{E}_T(\chi) \rangle = |\alpha| \cos(\chi - \phi)$$

where the measured field is the projection along the horizontal axis. For amplitude squeeze, the uncertainty is minimum at the maximum amplitude of the field.

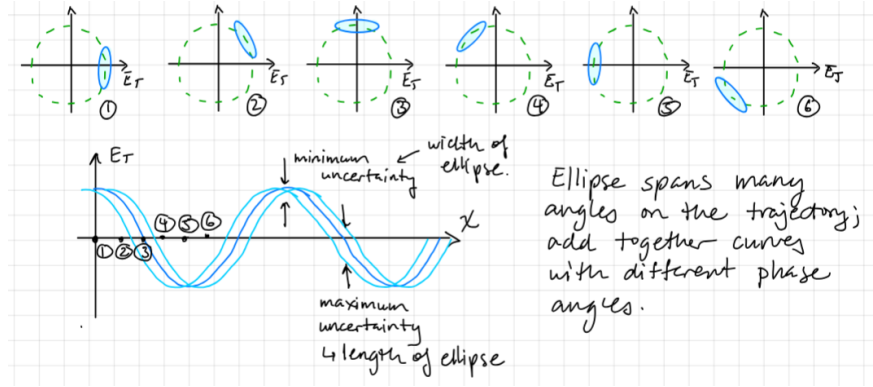


Figure 57: Amplitude squeezing of the ellipse showing minimum uncertainty at the peak of the field.

On the other hand, we may have phase-squeezing, where the uncertainty is minimum in phase, that is at zero-crossings.

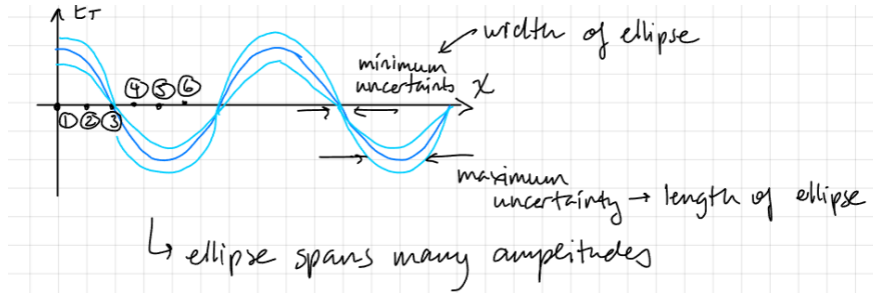


Figure 58: Phase-squeezing of the ellipse showing minimum uncertainty at the zero-crossing of the field.

The squeezing can also happen along other axes, but these are the most useful. Now we might ask: how can we make a squeezed state? One of the first and most common ways to generate light is through a *parametric down conversion*.

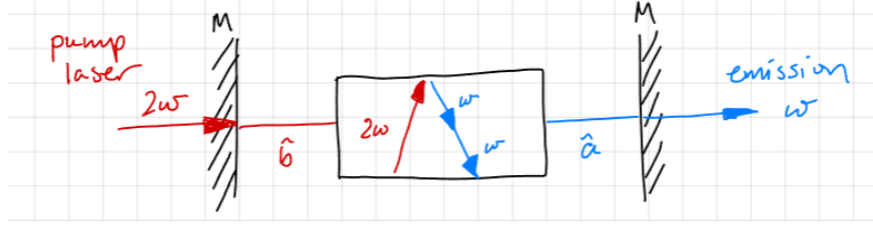


Figure 59: Parametric down converter, squeezing light with frequency 2ω into light with two instances of frequency ω .

We begin with a squeezed state $|\zeta, \alpha\rangle = \hat{S}(\zeta)|\alpha\rangle$ with operators $(\hat{a}^\dagger, \hat{a})$ for the mode with ω and $(\hat{b}^\dagger, \hat{b})$ for the mode with 2ω . Now, consider the interaction Hamiltonian

$$\hat{H}_I = i \frac{\hbar R}{2T} ((\hat{a}^\dagger)^2 \hat{b} - \hat{a}^2 \hat{b}^\dagger)$$

where the first $(\hat{a}^\dagger)^2$ term is from simultaneous emission or absorption of 2-photons simultaneously. This Hamiltonian is much like the argument of the squeeze operator

$$e^{-i\hat{H}_I T/\hbar} |\alpha\rangle = \hat{S}(\zeta) |\alpha\rangle.$$

We apply this Hamiltonian to the state

$$e^{-iH_I T/\hbar} |0\rangle = e^{\frac{R}{2}[(\hat{a}^\dagger)^2 \hat{b} - \hat{a}^2 \hat{b}^\dagger]} |0\rangle.$$

Now, recalling the form of the squeezing operator, we have

$$\zeta = \frac{R}{2} \hat{b}^\dagger, \quad \nu = 0 \longrightarrow s = \frac{R}{2}$$

which shows that the degree of squeezing is related to the rate of pair creation/destruction. Looking specifically at the expectation values and variances, we have

$$\langle \hat{X} \rangle = \langle 0 | e^{\frac{R}{2}[\hat{a}^2 \hat{b}^\dagger - (\hat{a}^\dagger)^2 \hat{b}]} \frac{1}{2} (\hat{a}^\dagger + \hat{a}) e^{\frac{R}{2}[(\hat{a}^\dagger)^2 \hat{b} - \hat{a}^2 \hat{b}^\dagger]} |0\rangle.$$

3.4 Beamsplitters, Detectors, and Coherence

While quantum optics descriptions of the fields are complete, they do not (yet) account for the non-uniform space of a practical setup (i.e., the presence of mirrors, beamsplitters, etc.), nor the method of detection, i.e., beamsplitters, which involve measuring field intensities rather than amplitudes.

3.4.1 Photodetectors

Photodetectors range from the eye to photodiodes to CCD cameras; they generally measure something proportional to intensity, like the square of the field amplitude. Typically, we can discuss photodetectors as classical light interacting with a quantum detector (semiclassical description). We can describe a photocurrent as

$$ip(\mathbf{r}, t) = \alpha \mathbf{E}^-(\mathbf{r}, t) \mathbf{E}^+(\mathbf{r}, t) = E_0^2 e^{-i((\mathbf{k} \cdot \mathbf{r} - \omega t))} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}.$$

We can imagine a detector that can record individual events (which we call "clicks"), which gives a probability per time of recording a click (e.g., via ionization of a detector particle):

$$w(\mathbf{r}, t) = \beta \mathbf{E}^-(\mathbf{r}, t) \mathbf{E}^+(\mathbf{r}, t)$$

where the probability of detection between t and $t + dt$ is $w(\mathbf{r}, t) dt$. For quantum light, it turns out²⁰ that the intensity operator is the analog

$$\hat{I} = 2\epsilon_0 c \hat{\mathbf{E}}_T^-(\mathbf{r}, t) \hat{\mathbf{E}}_T^+(\mathbf{r}, t) = 2\epsilon_0 c \sum_{k\lambda} \left(\frac{\hbar\omega_k}{2\epsilon_0 V} \right)^{1/2} \hat{a}_{k\lambda}^+ \hat{a}_{k\lambda}.$$

We will thus treat photodetectors as though they measure something proportional to the number operator.

3.4.2 Beamsplitters

Classically, assuming no losses or absorption, we use the following picture and mathematics:

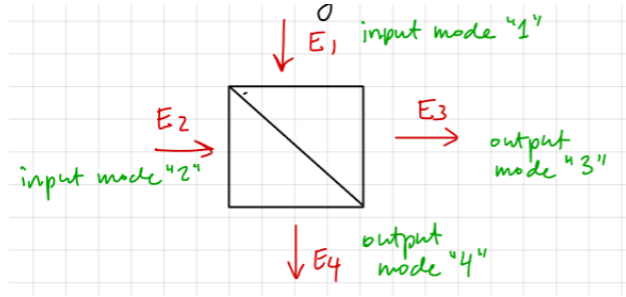


Figure 60: Beamsplitter with input and outputs modes labelled.

where we have specifically

$$E_3 = r_{13}E_1 + t_{23}E_2$$

$$E_4 = t_{14}E_1 + r_{23}E_2$$

so that

$$\begin{bmatrix} E_3 \\ E_4 \end{bmatrix} = \begin{bmatrix} r_{13} & t_{23} \\ t_{14} & r_{23} \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \end{bmatrix}.$$

Here, r and t are complex numbers that define the reflection or transmission of the beamsplitter. Energy into the beamsplitter must equal the output, so

$$|E_1|^2 + |E_2|^2 = |E_3|^2 + |E_4|^2$$

²⁰It can be shown starting from

$$\hat{I}(\mathbf{r}, t) = \epsilon_0 c^2 \left[\mathbf{E}_T^-(\mathbf{r}, t) \times \mathbf{B}^+(\mathbf{r}, t) - \hat{\mathbf{B}}^-(\mathbf{r}, t) \times \hat{\mathbf{E}}_T^+(\mathbf{r}, t) \right]$$

(Loudon problem #4.14).

which gives

$$\begin{aligned} |r_{13}|^2 + |t_{14}|^2 &= 1 \\ |t_{23}|^2 + |r_{24}|^2 &= 1 \\ (r_{13}^* t_{23} + t_{14}^* r_{24}) &= 0 \\ (r_{13} t_{23}^* + t_{14} r_{24}^*) &= 0. \end{aligned}$$

The phase change is not simply π upon reflection because of the angle of reflection. We are free to choose between conventions, so long as these relationships are satisfied. Taking these complex numbers, we have $r = |r|e^{i\phi_r}$ and $t = |t|e^{i\phi_t}$, suggesting that

$$|r_{13}||t_{23}|e^{-i\phi_{r13}}e^{i\phi_{t23}} + |t_{14}||r_{24}|e^{-i\phi_{t14}}e^{i\phi_{r24}} = 0.$$

So,

$$|r_{13}||t_{23}| = |t_{14}||r_{24}|$$

and

$$e^{-i\phi_{r23}+i\phi_{t23}} = -e^{-i\phi_{t14}+i\phi_{r24}} = e^{\pm i\pi - i\phi_{t14} + i\phi_{r24}}.$$

Also,

$$-\phi_{r23} + \phi_{t23} = \pm\pi - \phi_{t14} + i\phi_{r24} \implies \phi_{t23} + \phi_{t14} - \phi_{r23} - \phi_{r24} = \pm\pi.$$

We will consider the case of a 50/50 beamsplitter, so

$$|r_{13}| = |r_{24}| = |t_{14}| = |t_{23}| = \frac{1}{\sqrt{2}}$$

and a phase convention

$$\phi_{t23} = \phi_{t14} = \phi_{r23} = 0, \quad \phi_{r24} = \pi.$$

Then, we have the matrix equation

$$\begin{bmatrix} E_3 \\ E_4 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \end{bmatrix} \longrightarrow \begin{cases} E_3 = \frac{1}{\sqrt{2}}(E_1 + E_2) \\ E_4 = \frac{1}{\sqrt{2}}(E_1 - E_2). \end{cases}$$

For quantum fields, we will ultimately be measuring the intensities with photodetectors, so it is appropriate to consider the plus component of the field and how it propagates,

$$\hat{\mathbf{E}}^+(\mathbf{r}, t) = \sum_{\mathbf{k}\lambda} \mathbf{e}_{\mathbf{k}\lambda} \left(\frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0 V} \right)^{1/2} \hat{a}_{\mathbf{k}\lambda} e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t - \pi/2)}.$$

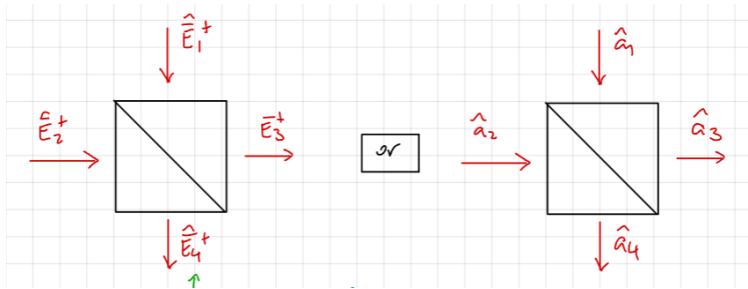


Figure 61: Beamsplitter configuration with the plus component propagation of the electric field.

We note that subscripts define single modes; in particular, we assume perfect transfer from input to output. Thus, we have the following

$$\begin{aligned} \hat{\mathbf{E}}_3^+ &= \frac{1}{\sqrt{2}}(\hat{\mathbf{E}}_1^+ + \hat{\mathbf{E}}_2^+) & \hat{a}_3 &= \frac{1}{\sqrt{2}}(\hat{a}_1 + \hat{a}_2) \\ \hat{\mathbf{E}}_4^+ &= \frac{1}{\sqrt{2}}(\hat{\mathbf{E}}_1^+ - \hat{\mathbf{E}}_2^+) & \hat{a}_4 &= \frac{1}{\sqrt{2}}(\hat{a}_1 - \hat{a}_2). \end{aligned}$$

3.5 Coherences

Coherences can be measured classically and quantum mechanically, and they can tell us about the nature of light we are dealing with. There are two particular cases: first-order coherence and second-order coherence.

3.5.1 First-Order Coherence

First-order coherence is the coherence of amplitude. We pose a question: are two sources of light able to form interference fringes? The two sources can be independent sources or a single source split and recombined over time or space.

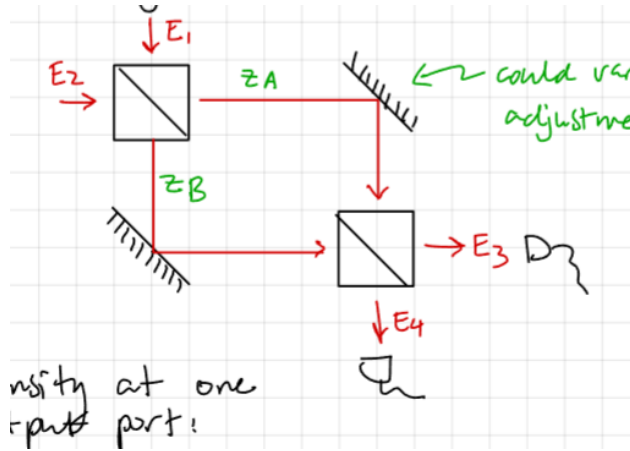


Figure 62: Mach-Zender interferometer setup for first-order coherence.

We can measure, for example, with a Mach-Zender interferometer first-order coherence. The travel time is given by

$$t_A = t - \frac{z_A}{c}, \quad t_B = t - \frac{z_B}{c}$$

and the intensity at one output port is

$$I_4 = \frac{\epsilon_0 c}{2} |I_4|^2 = \frac{\epsilon_0 c}{2} |t[rE_1(t_1) + tE_2(t_1)] - r[tE_1(t_2) - rE_2(t_2)]|^2.$$

For a single input, we can determine self-coherence

$$\begin{aligned}
 I_4 &= \frac{\epsilon_0 c}{2} |tr E_1(t_1) - rt E_1(t_2)|^2 \\
 &= \frac{\epsilon_0 c}{2} |rt|^2 \left[|E_1(t_2)|^2 + |E_1(t_2)|^2 - E_1^*(t_1) E_1(t_2) + E_1(t_1) E_1^*(t_2) \right] \\
 &= \frac{\epsilon_0 c}{2} |rt|^2 \left[|E_1(t_1)|^2 + |E_1(t_2)|^2 - E_{10}^2 e^{-i\omega t_1} e^{i\omega t_2} + E_{10}^2 e^{i\omega t_1} e^{-i\omega t_2} \right] \\
 &= \frac{\epsilon_0 c}{2} |rt|^2 E_{10}^2 \{2 - 2 \cos[\omega(t_1 - t_2)]\} \\
 &= \frac{\epsilon_0 c}{2} |rt|^2 E_0^2 \sin^2 \left[\frac{\omega(t_1 - t_2)}{2} \right].
 \end{aligned}$$

Quantum mechanically, we have

$$\langle \hat{I}_4 \rangle = \frac{\epsilon_0 c}{2} \langle \hat{a}_4^\dagger \hat{a}_4 \rangle = \frac{\epsilon_0 c}{2} \left\langle \left[t(r\hat{a}_1^\dagger + t\hat{a}_2^\dagger) - r(t\hat{a}_1^\dagger - r\hat{a}_2^\dagger) \right] [t(r\hat{a}_1 + t\hat{a}_2) - r(t\hat{a}_1 - r\hat{a}_2)] \right\rangle.$$

Again, if one input is a vacuum, then $|\psi_{in}\rangle = |\psi\rangle_1 |0\rangle_2$, with $\hat{a}_2 |\psi_{in}\rangle = \langle \psi_{in} | \hat{a}_2^\dagger = 0$. So,

$$\begin{aligned}
 \langle \hat{I}_4 \rangle &= \frac{\epsilon_0 c}{2} \left\langle \left(tr\hat{a}_1^\dagger e^{-i\chi_1} - rt\hat{a}_1^\dagger e^{-i\chi_2} \right) (tr\hat{a}_1 e^{i\chi_1} - rt\hat{a}_1 e^{i\chi_2}) \right\rangle \\
 &= \frac{\epsilon_0 c}{2} (tr)^2 \langle \hat{a}_1^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_1 e^{-i(\chi_1 - \chi_2)} - \hat{a}_1^\dagger \hat{a}_1 e^{i(\chi_1 - \chi_2)} + \hat{a}_1^\dagger \hat{a}_1 \rangle \\
 &= \frac{\epsilon_0 c}{2} (rt)^2 [2\langle \hat{n} \rangle - 2\langle \hat{n} \rangle \cos(\chi_1 - \chi_2)] \\
 &= \frac{\epsilon_0 c}{2} (rt)^2 \langle \hat{n} \rangle \sin^2 \left[\frac{1}{2}(\chi_1 - \chi_2) \right]
 \end{aligned}$$

where we note $\chi_i \propto \omega_i t_i$ for $i = \{1, 2\}$. Observe that this result is completely analogous to the classical case. The first-order correlator is the *g-one*, given

$$g^{(1)}(\tau) = \frac{\langle E^*(t) E(t + \tau) \rangle}{\langle E^*(t) E^*(t) \rangle}$$

where the angle brackets are time average brackets. The *g-one* quantity measures whether there is a stable, well-defined phase of the light over time. Specifically, if $g^{(1)}(\tau) \rightarrow 0$, then the light is incoherent, but if $g^{(1)}(\tau) \rightarrow 1$, the light is coherent.

3.5.2 Second-Order Coherence

We can determine second-order coherence by the Brown-Twiss interferometer, after their measurement of the effect.

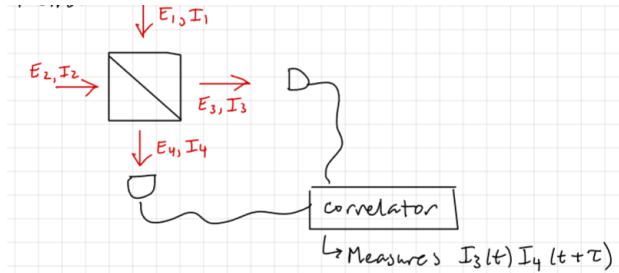


Figure 63: Brown-Twiss interferometer setup.

Using measurement to calculate second order g -two,

$$g^{(2)}(\tau) = \frac{\langle I_3(t)I_4(t+\tau) \rangle}{\langle I_3(t) \rangle \langle I_4(t+\tau) \rangle}.$$

For classical light, we have $g^{(2)}(0) \geq 1$.²¹ The utility of this is to measure fluctuations: if the light is coherent from the two parts, the fluctuations will add and give extra amplitude; randomness will average and $g^{(2)}(\tau) \rightarrow 1$.

For quantum states of light, we have the alternative setup:

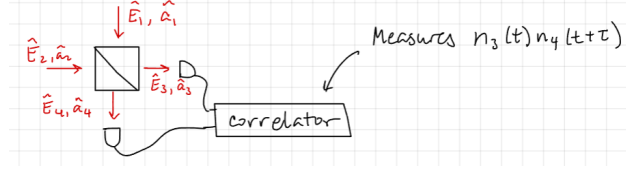


Figure 64: Alternative Brown-Twiss setup to measure $n_3(t)$.

²¹See Loudon Ch. 3.

References